ReThink Summary

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01. Gambaran Umum Produk

1.1 Identitas & Posisi Produk

Atribut	Definisi
Nama produk	ReThink
Kategori	Platform Pemulihan Pemahaman Konseptual Matematika Berbantuan AI
Domain	Pendidikan matematika
Pengguna utama	Siswa SMA yang mengalami kesulitan pada topik matematika tertentu; profil ini masih perlu divalidasi
Mode belajar	Belajar mandiri dengan dukungan terarah
Fokus	Diagnosis berbasis bukti → pembelajaran terbimbing → validasi pemahaman
Pendekatan AI	Interpretasi bukti berbantuan AI dengan otonomi yang dibatasi

1.1.2 Posisi produk

ReThink merupakan platform pemulihan pemahaman konseptual matematika berbantuan AI. Produk ini membantu siswa menelusuri kemungkinan hambatan konsep prasyarat berdasarkan bukti respons matematika, mempelajari kembali konsep yang relevan melalui intervensi terkurasi, dan menilai penerapannya secara mandiri melalui pengerjaan ulang serta soal transfer. Alur tersebut menyatukan diagnosis, pembelajaran, dan validasi dalam satu pengalaman yang terarah.

Berbeda dari platform belajar matematika yang mengutamakan cakupan materi luas atau pemberian langkah penyelesaian secara umum, ReThink memprioritaskan kedalaman pemulihan pada konsep yang benar-benar didukung sistem. AI membantu interpretasi bukti dalam batas pengetahuan dan pemeriksaan matematika yang terkurasi; diagnosis tidak menganggap setiap kesulitan sebagai conceptual blocker, dan keberhasilan pada konsep tertentu tidak disamakan dengan penguasaan seluruh topik. Struktur konsep memungkinkan perluasan bertahap tanpa mengklaim dukungan yang belum tersedia.

1.2 Visi Produk

Menjadikan kesulitan belajar matematika lebih mudah dipahami dan diatasi dengan membantu siswa menemukan kemungkinan kesenjangan konsep dasar yang relevan, memulihkan pemahamannya melalui pembelajaran terarah, dan menunjukkan kemampuan menerapkan konsep tersebut secara mandiri—tanpa menggantikan proses bernalar siswa.

1.3 Deskripsi Produk

Siswa memulai dari katalog topik matematika dan memilih topik yang mendukung diagnosis. Sistem mengevaluasi bukti dari respons matematika untuk mempertimbangkan apakah hambatan konsep yang didukung dapat dikonfirmasi, ditolak, atau masih belum pasti. Ketika bukti memadai, sistem merekomendasikan paket pembelajaran terarah berupa penjelasan visual, contoh beralasan, dan latihan terbimbing parsial. Hasil latihan mandiri dan soal transfer kemudian menjadi dasar status belajar pada konsep yang diperiksa.

Peta Belajar Hibrida menampilkan relasi topik–konsep untuk eksplorasi, fokus yang didukung diagnosis, serta perkembangan yang didukung bukti validasi. Jika diagnosis tidak mengonfirmasi hambatan, siswa tetap dapat mengakses ulasan konsep opsional atau kembali mengeksplorasi topik, tanpa sistem menyatakan adanya kelemahan yang belum terbukti.

1.4 Nilai Utama Produk

Nilai ReThink terletak pada keterhubungan tiga aktivitas yang sering ditemui terpisah: diagnosis berbasis bukti, pemulihan pemahaman yang relevan, dan penilaian penerapan mandiri.

Dimensi	Nilai bagi pengguna
Kejelasan — mengetahui apa yang perlu ditangani	Menjelaskan kemungkinan hambatan konsep berdasarkan bukti respons, bukan sekadar hasil benar atau salah.
Fokus — memulihkan pemahaman yang relevan	Mengarahkan perhatian pada satu konsep yang didukung dan menyediakan pembelajaran terstruktur tanpa mengharuskan peninjauan materi yang terlalu luas.
Verifikasi — mengetahui apa yang dapat diterapkan mandiri	Menunjukkan bukti penerapan konsep tanpa bantuan, termasuk pada representasi soal yang berbeda.

Manfaat tersebut tetap bergantung pada cakupan diagnosis, intervensi, dan validasi yang benar-benar tersedia. Jika bukti tidak cukup, sistem menyampaikan ketidakpastian secara transparan.

1.5 Diferensiasi Produk

Pembeda ReThink bukan klaim bahwa setiap teknik yang dipakai sepenuhnya baru, melainkan fokus pada pemulihan satu hambatan konsep tertentu dan integrasi diagnosis, intervensi, serta validasi pemahaman. Pendekatan ini adalah posisi produk yang dituju; keunggulannya atas alternatif lain belum dibuktikan secara empiris.

Aspek pembanding	Pendekatan ReThink
Titik awal	Mencari hambatan konsep yang secara matematis relevan terhadap kesulitan siswa
Dasar rekomendasi	Sintesis bukti beberapa respons dengan kesimpulan yang dibatasi dan dapat tidak pasti
Bentuk bantuan	Paket konsep terkurasi: visual → contoh beralasan → latihan terbimbing parsial
Cara menilai pemahaman	Independent Retry dan Transfer Problem, bukan penyelesaian konten saja
Cakupan	Satu jalur utuh yang terverifikasi lebih diutamakan daripada banyak topik yang belum didukung

1.6 Prinsip Utama Produk

- Bukti sebelum inferensi: Kesimpulan diagnosis dan status belajar harus bersandar pada bukti yang teramati; ketidakpastian adalah hasil yang sah.

- Pemulihan di atas keluasan konten: Kualitas dukungan konsep lebih penting daripada banyaknya topik yang terlihat.
- Penalaran aktif di atas pemberian jawaban: Bantuan tidak boleh menggantikan proses bernalar matematika siswa.
- Validasi sebelum klaim pemahaman: Menyelesaikan pembelajaran tidak otomatis berarti memahami konsep.
- Otonomi AI yang dibatasi: Interpretasi AI dibatasi pengetahuan terkurasi, pemeriksaan matematika, dan pengaman keputusan; pembagian otoritas final masih OI-01.
- Struktur dapat diperluas, cakupan tetap terkendali: Konsep yang direpresentasikan tidak otomatis tersedia untuk diagnosis, intervensi, atau validasi.

02. Pernyataan Masalah & Pengguna Sasaran

2.1 Latar Belakang Masalah & Bukti

Ringkasan: Kesulitan matematika dapat berkaitan dengan konsep prasyarat, tetapi tidak semuakesalahan berasal dari hambatan konsep. Literatur memberi dasar konseptual; survei eksploratif 12 siswa memberi indikasi awal berbasis persepsi, bukan pembuktian kausal.

2.1.1 Kesulitan matematika & pengetahuan prasyarat

Konsep matematika saling berkaitan. Kesulitan pada suatu topik dapat berasal dari keterbatasan konsep sebelumnya, bukan semata dari topik yang sedang dipelajari. Dari 12 responden survei eksploratif, 10 mengaku kadang atau sering kesulitan belajar matematika; 6 mengaitkan sebagian kesulitan dengan materi terdahulu yang belum dipahami. Temuan ini bersifat perseptual.

2.1.2 Kesulitan mengatribusikan kesalahan & memahami konsep

Jawaban keliru dapat memiliki beragam penyebab dan jawaban benar belum tentu menunjukkan penalaran konseptual. Dalam survei awal, 8/12 kesulitan memahami makna rumus atau konsep, 7/12 kesulitan saat contoh berbeda dari soal, dan 5/12 dapat mengikuti contoh tetapi kesulitan mengerjakan mandiri. Perbedaan kompleksitas atau keakraban soal tetap merupakan penjelasan alternatif.

2.1.3 Batas bukti

Survei berskala kecil, tidak representatif, dan menggunakan laporan diri—terutama siswa kelas XI–XII. Hasilnya tidak menetapkan prevalensi pada populasi SMA, membuktikan conceptual blocker secara objektif, atau menunjukkan efektivitas ReThink. Wawancara dan observasi pengerjaan soal diperlukan untuk memperkuat pemahaman penyebab kesulitan.

2.2 Rumusan Masalah Inti

Ringkasan: Ketika siswa kesulitan pada suatu topik, mereka mungkin tidak tahu apakah kesulitan tersebut terkait kesenjangan konsep, materi mana yang perlu dipulihkan, dan apakah mereka sudah mampu menerapkannya secara mandiri.

Dimensi masalah	Penjelasan
Sumber kesulitan belum jelas	Siswa dapat menyadari kesulitan tanpa mengetahui kontribusi konsep prasyarat atau faktor lain.
Arah pemulihan belum terarah	Bantuan yang tersedia dapat menjelaskan prosedur tanpa menjangkau pemahaman yang relevan.
Pemahaman mandiri belum pasti	Mengikuti contoh tidak membuktikan penerapan konsep tanpa bantuan atau di bentuk soal berbeda.

2.3 Pengguna Sasaran & Segmentasi

Ringkasan: Sasaran awal adalah siswa SMA yang kesulitan memahami atau menerapkan matematika, terutama untuk konteks belajar mandiri. Definisi pengguna dan segmentasi masih bersifat sementara dan membutuhkan validasi.

2.3.1 Pengguna utama

Siswa SMA yang mengalami kesulitan pada topik matematika tertentu dan berpotensi terbantu oleh dukungan pemulihan konsep secara mandiri. Mereka tidak harus lebih dulu memahami penyebab kesulitan tersebut.

2.3.2 Fokus riset awal

Riset awal mencakup kelas X–XII dengan perhatian pada XI–XII; 11 dari 12 responden awal berasal dari kelas XI–XII. Aljabar dipilih sebagai domain pengembangan MVP, bukan karena terbukti menjadi kesulitan paling umum.

2.3.3 Dimensi segmentasi sementara

Dimensi	Peran
Kesulitan belajar	Frekuensi dan karakter tantangan memahami konsep atau menerapkan matematika.
Perilaku mencari bantuan	Cara mencari, memakai, atau menghindari bantuan saat kesulitan.
Kesadaran masalah (atribut tambahan)	Tingkat kekhususan siswa mengenali dan menjelaskan kesulitannya.

Dimensi tersebut belum membentuk segmen yang telah tervalidasi atau eksklusif satu sama lain.

2.4 Konteks Penggunaan & Perilaku Saat Ini

Ringkasan: Konteks potensial mencakup tertinggal pelajaran maupun menemui soal yang sulit. Perilaku mencari bantuan beragam dan tidak menunjukkan secara otomatis preferensi terhadap diagnosis berbantuan AI.

2.4.1 Situasi belajar dan pemicu

Konteks awal meliputi kesulitan mengikuti pelajaran kelas dan kesulitan menyelesaikan soal spesifik. Pemicu potensial mencakup tidak memahami penjelasan, tugas matematika, persiapan asesmen, atau menyadari konsep terdahulu belum kuat; tingkat kepentingan relatifnya belum diketahui.

2.4.2 Perilaku bantuan belajar

Dalam survei 12 siswa, 10 melaporkan bantuan guru/teman/keluarga, 8 materi tertulis atau video, 7 asisten AI, 6 mencoba secara mandiri, sedangkan les dan melewati soal masing-masing 2. Penggunaan AI tidak membuktikan kepercayaan terhadap diagnosis AI atau kemauan menggunakan ReThink.

2.4.3 Kondisi belajar mandiri

Enam responden melaporkan sekitar 15–30 menit belajar mandiri; tiga tidak memiliki periode tetap. Data ini belum menetapkan durasi sesi ReThink. Belajar mandiri berarti mengambil inisiatif, bukan belajar tanpa guru atau bantuan lain.

2.5 Kebutuhan yang Belum Terpenuhi & Peluang

Ringkasan: Ada indikasi kebutuhan kejelasan arah belajar, bantuan yang relevan secara konseptual, dan bukti penerapan mandiri. Belum ada bukti yang cukup mengenai permintaan terhadap ReThink atau keunggulannya dibanding alternatif.

Potensi kebutuhan	Batas interpretasi
Kejelasan kesenjangan belajar	Siswa mungkin membutuhkan penjelasan konsep mana yang relevan; penyebab kesulitan tidak dapat disimpulkan hanya dari persepsi.
Bantuan yang relevan secara konseptual	Bantuan perlu menjelaskan alasan matematika; kesulitan dapat pula disebabkan kompleksitas atau konteks soal.
Bukti penerapan mandiri	Siswa mungkin perlu tahu kemampuannya tanpa bantuan; belum terbukti bahwa mereka bersedia melakukan asesmen tambahan.

Peluang produk adalah memberi pengalaman pemulihan yang lebih terarah saat konsep prasyarat memang relevan. Peluang tersebut masih hipotesis produk, bukan bukti demand atau efektivitas.

2.6 Asumsi Kritis & Kesenjangan Riset

Ringkasan: Enam asumsi A1–A6 memandu riset lanjutan; semuanya perlu dibaca sesuai kualitas bukti, bukan sebagai karakteristik pengguna atau hasil produk yang telah terverifikasi.

ID	Asumsi	Status bukti
A1	Kesenjangan prasyarat berkontribusi pada sebagian kesulitan matematika.	Didukung literatur dan indikasi laporan diri; konfirmasi objektif tertunda.
A2	Sebagian siswa sulit mengenali sumber masalah dan arah belajar.	Indikasi survei terbatas; evaluasi objektif tertunda.
A3	Sebagian kesulitan belum terselesaikan setelah memakai bantuan yang tersedia.	Indikasi friksi; persistensi dan kecukupan alternatif belum terbukti.
A4	Siswa memahami dan menerima pengalaman diagnosis terarah.	Belum diuji langsung.
A5	Pemulihan terarah dapat meningkatkan penerapan konsep secara mandiri.	Hipotesis pedagogis; efektivitas ReThink belum terbukti.
A6	Siswa menilai bukti penerapan mandiri sebagai informasi yang berguna.	Indikasi tidak langsung; penerimaan validasi belum diuji.

Prioritas riset: observasi kesulitan matematika, kebiasaan mencari bantuan, pengujian pemahaman/penerimaan pengalaman produk, dan evaluasi hasil belajar secara terpisah. Temuan riset tidak otomatis mengubah keputusan produk tanpa peninjauan.

03. Tujuan Produk & Ukuran Keberhasilan

3.1 Tujuan Produk & Hasil yang Diharapkan

Tujuan ReThink adalah membantu siswa memperoleh arah pemulihan berbasis bukti, mempelajari konsep secara terarah, dan memahami apa yang dapat diterapkan tanpa bantuan. Tujuan di bawah adalah hasil yang dituju, bukan bukti efektivitas yang sudah dicapai.

Tujuan	Penjelasan	Hasil yang diharapkan
G1 — Arah pemulihan berbasis bukti	Menerjemahkan bukti respons menjadi informasi hambatan konsep yang terukur, termasuk hasil belum pasti.	Siswa memahami kemungkinan kesulitan yang relevan dan pilihan kelanjutan yang tersedia.
G2 — Pemulihan konsep terarah	Memberikan pembelajaran konsep yang fokus sambil mempertahankan penalaran aktif.	Siswa memperoleh kesempatan memperkuat pemahaman dan menerapkan konsep mandiri.
G3 — Evaluasi pemahaman berbasis bukti	Memeriksa penerapan tanpa bantuan dan pengenalan struktur pada representasi lain.	Siswa memperoleh bukti pemahaman dalam cakupan asesmen, termasuk kebutuhan penguatan.
G4 — Pengalaman belajar mandiri yang dapat dipahami	Memastikan tujuan diagnosis, alur belajar, dan batas arti hasil bisa dipahami.	Siswa dapat menavigasi pengalaman dan menafsirkan hasil tanpa pendampingan terus-menerus.

3.2 Definisi Keberhasilan MVP

MVP menargetkan satu jalur pemulihan utuh: topik SPL dengan Distributive Property sebagai blocker yang didukung. Keberhasilan fungsional dan validasi pengalaman pengguna dinilai secara terpisah.

Kelompok	Kondisi	Makna penerimaan
Fungsional	SC1 — Jalur Pemulihan Lengkap	Pemilihan topik → diagnosis → pembelajaran terarah → validasi mandiri → hasil tingkat blocker berjalan end-to-end pada skenario yang didukung.
Fungsional	SC2 — Integritas Keputusan Berbasis Bukti	Diagnosis dan validasi sesuai bukti matematika; tidak mengonfirmasi blocker atau pemahaman di luar dukungan.
Fungsional	SC3 — Kelanjutan Sesuai Hasil	Confirmed/Rejected/Uncertain ditangani secara berbeda; completion belajar tidak disamakan dengan pemahaman.
Pengalaman pengguna	SC4 — Pengalaman Pemulihan yang Dapat Dipahami	Pengguna memahami diagnosis, alur, hasil, dan keterbatasannya; dinilai melalui pengujian pengguna terpisah.

Interpretasi: SC1–SC3 baru dapat disebut terverifikasi bila semua skenario fungsional wajib di cakupan MVP telah dijalankan dan memenuhi kriterianya. Skenario gagal, belum diuji, atau belum memiliki expected behavior tidak dapat dianggap lulus. SC4 dilaporkan terpisah; keberhasilan fungsional tidak membuktikan usability atau peningkatan hasil belajar.

3.3 Dimensi & Metrik Keberhasilan

Dimensi	Metrik utama	Indikator pendukung / batas
D1 — Kelengkapan fungsional	Tingkat kelulusan skenario wajib yang sudah dijalankan	Catat jumlah lulus, gagal, belum diuji; tingkat kelulusan tidak boleh menutupi coverage yang belum lengkap.
D2 — Integritas keputusan	Kesesuaian keputusan dengan kasus acuan yang ditinjau	Insiden konfirmasi blocker tanpa bukti, penanganan uncertain, ketepatan status validasi; bukan akurasi pada populasi umum.
D3 — Usability & interpretabilitas	Keberhasilan tugas inti dan pemahaman hasil	Laporkan terpisah navigasi, kesalahan tafsir diagnosis/status, serta isu usability kritis.
D4 — Bukti hasil belajar	Penerapan mandiri dan performa transfer	Skor cakupan validasi; perubahan pre/post hanya eksploratif dengan desain yang tepat dan tidak otomatis kausal.

Pengukuran harus menyatakan cakupan soal, kasus, dan peserta; kegagalan kritis tidak disembunyikan dalam rata-rata. Ambang numerik, metode rinci, dan protokol pengujian tidak ditetapkan tanpa justifikasi.

3.4 Batas Keberhasilan & Non-Tujuan

Status Confirmed menunjukkan dukungan bukti atas hambatan konsep yang diperiksa, bukan sebab pasti seluruh kesulitan. Understood menunjukkan terpenuhinya kondisi validasi pada tingkat blocker, bukan penguasaan SPL, retensi jangka panjang, atau seluruh matematika. Hasil fungsional, usability, dan efektivitas belajar memerlukan bukti berbeda.

Tidak menjadi syarat MVP	Penjelasan
Cakupan kurikulum menyeluruh	Tidak semua konsep memiliki diagnosis atau paket pemulihan.
Penguasaan penuh topik SPL	Validasi dibatasi Distributive Property.
Pemulihan adaptif bertingkat	Tidak ada tracing beberapa lapis atau diagnosis per kompetensi.
Bukti retensi longitudinal	Tidak ada kewajiban mengukur pemahaman jangka panjang.
Peningkatan matematika umum	Tidak ada klaim dampak lintas topik/populasi.

Efektivitas pemulihan dalam konsep yang didukung tetap merupakan tujuan riset; pembuktiannya terpisah dari kelulusan fungsi MVP.

04. Cakupan Produk & Definisi MVP

4.1 Pernyataan Cakupan & Batas Produk

MVP ReThink menyediakan satu pengalaman lengkap pemulihan konseptual matematika berbasis bukti. Satu topik yang dapat memulai diagnosis adalah Sistem Persamaan Linear (SPL) dalam Aljabar; satu hambatan konseptual yang didukung secara utuh adalah Distributive Property, termasuk penerapannya secara langsung dan dalam SPL.

Alur: eksplorasi katalog/peta → asesmen diagnostik terarah → hasil diagnosis → pembelajaran terbimbing terkurasi jika terkonfirmasi (atau ulasan opsional tanpa klaim diagnosis) → validasi tingkat blocker untuk jalur confirmed yang memenuhi prasyarat. Keberadaan konsep di katalog atau peta tidak berarti tersedia diagnosis dan validasi.

Prinsip batas produk	Implikasi
Cakupan kemampuan terkendali	Satu jalur pemulihan lengkap, bukan semua topik aktif.
Hasil dibatasi bukti	Tidak menetapkan kelemahan atau pemahaman konsep yang belum diperiksa.
Siap diperluas secara konseptual	Relasi konsep reusable tidak mengaktifkan kemampuan di luar MVP.

4.2 Cakupan Matematika & Diagnosis

4.2.1 Materi yang didukung

Distributive Property diperiksa pada (a) manipulasi ekspresi aljabar secara langsung dan (b) transformasi dalam soal SPL. Keduanya menambah bukti mengenai satu konsep yang sama, bukan diagnosis blocker tambahan.

4.2.2 Rancangan asesmen diagnostik

Terdapat tiga soal inti dan paling banyak satu probe kondisional, sehingga jumlah keseluruhan tidak melebihi empat soal. Contoh di bawah hanya ilustrasi rancangan, bukan butir final yang telah disetujui atau rubrik OPS-04.

Soal	Peran dan format	Contoh ilustratif	Batas interpretasi
Q1	Performa target SPL; pilihan ganda	Pilih langkah yang tepat untuk menyelesaikan sistem dengan 2(x+3) dalam salah satu persamaan.	Jawaban benar/salah sendiri tidak cukup untuk menentukan blocker.
Q2	Aplikasi prasyarat terisolasi; jawaban matematika singkat	Uraikan 3(x + 2) menjadi bentuk ekuivalen.	Menunjukkan transformasi Distributive Property.
Q3	Alasan konseptual; penjelasan bebas	Jelaskan mengapa 2(x + 3) = 2x + 6 dan bukan 2x + 3.	Memberi bukti eksplanatoris, bukan sekadar ketepatan hasil.

Q4 (opsional)	Satu procedural trace atau conceptual probe	Tunjukkan letak transformasi pertama yang tidak setara pada rangkaian langkah yang disediakan.	Hanya jika terdapat kekosongan bukti yang perlu diklarifikasi.

Kategori bukti yang dapat digunakan meliputi hasil jawaban, transformasi matematika yang terlihat, dan penjelasan siswa; tidak seluruh kategori harus muncul di setiap soal. AI dapat membantu menafsirkan dan menyintesiskan bukti, tetapi pembagian otoritas keputusan final masih OI-01.

4.2.3 Hasil diagnosis dan batasnya

Hasil	Makna
Confirmed	Bukti cukup mendukung adanya hambatan Distributive Property sesuai aturan diagnosis.
Rejected	Bukti cukup tidak mendukung atau bertentangan dengan hambatan yang diperiksa.
Uncertain	Bukti kurang, ambigu, atau tidak konsisten untuk menyimpulkan.

Satu jawaban salah tidak otomatis mengonfirmasi blocker; jawaban benar atau hasil non-confirmed tidak berarti tidak ada kesulitan lain. Pemetaan pola semua-jawaban-benar tetap OI-03 Open.

4.3 Kemampuan Produk yang Termasuk MVP

Kemampuan	Tanggung jawab inti
C1 — Entri Belajar & Penemuan Topik	Katalog matematika, pemilihan SPL untuk diagnosis, dan penanda dukungan fitur yang tersedia.
C2 — Peta Belajar Hibrida	Explore (orientasi), Focus (berdasarkan bukti diagnosis), Progress (berdasarkan hasil validasi); tidak membuat diagnosis independen.
C3 — Diagnosis Terarah & Hasil	Q1–Q3, maksimal satu Q4, sintesis bukti, keluaran Confirmed/Rejected/Uncertain, dan kelanjutan yang sesuai.
C4 — Pembelajaran Terbimbing Terarah	Satu paket Distributive Property: penjelasan visual → contoh beralasan → latihan parsial aktif; bisa untuk recovery terkonfirmasi atau ulasan opsional.
C5 — Validasi Pengetahuan & Hasil	Independent Retry, lalu Transfer Problem hanya jika retry berhasil; menetapkan status blocker sesuai bukti.

Independent Retry	Transfer Problem	Status belajar
Berhasil	Berhasil	Understood
Berhasil	Gagal	Procedural Only
Gagal	Tidak diberikan	Needs Reinforcement

Jika retry gagal, pengguna diarahkan kembali ke paket belajar yang sama, tanpa diagnosis ulang otomatis atau remediasi adaptif. Kelayakan validasi sesudah Optional Concept Review tetap OI-02 Open.

4.4 Eksklusi, Landasan Masa Depan & Ketergantungan

Kelompok	Ketentuan
E1–E5 — Eksklusi	Tidak mencakup kurikulum luas, diagnosis/adaptivitas tingkat lanjut, orkestrasi instruksional dinamis, asesmen longitudinal komprehensif, atau fitur tutor chatbot/LMS generik.
F1 — Relasi konsep reusable	Hubungan topik–konsep dapat digunakan ulang.
F2 — Batas dukungan eksplisit	Representasi konsep dipisahkan dari kemampuan aktif.
F3 — Pemisahan definisi asesmen/pemulihan	Definisi dan paket untuk diagnosis, belajar, serta validasi tidak saling disamakan.
F4 — Status siswa dibatasi bukti	Informasi pembelajar hanya didasarkan pada bukti yang terkait.

Keputusan terbuka: OI-01 (metodologi/otoritas AI), OI-02 (validasi setelah ulasan opsional), OI-03 (kasus seluruh jawaban benar). Detail OPS-01–OPS-05 tetap spesifikasi operasional tertunda, bukan tambahan fitur MVP.

05. Ringkasan Persyaratan Produk

Dokumen persyaratan produk dipisahkan agar daftar kebutuhan fungsional yang rinci tetap mudah dicari, dikelola, dan diverifikasi tanpa membuat PRD utama terlalu panjang.

Kelompok	Jumlah requirement	Cakupan
C1 — Entri belajar dan penemuan topik	5	Katalog, topik yang didukung, pembukaan diagnosis
C2 — Peta Belajar Hibrida	5	Explore, Focus, Progress dan kejujuran status bukti
C3 — Asesmen & hasil diagnostik	10	Rancangan soal, evaluasi bukti, keluaran diagnosis, kelanjutan
C4 — Pembelajaran terbimbing	8	Urutan konten, partisipasi aktif, kelanjutan
C5 — Validasi & hasil belajar	8	Retry, transfer, dan status belajar sesuai bukti
Lintas kemampuan	3	Integritas konteks, transisi, dan representasi status
TOTAL	39	Approved functional requirements pada PRD v1.0

Ketentuan lintas kemampuan IP-01–IP-02 dan skenario verifikasi awal V-X-01–V-X-08 tetap berlaku. Persyaratan pada dokumen terpisah tetap menjadi referensi otoritatif setelah ditautkan dan diperiksa konsistensinya dengan baseline.

LINK DOKUMEN PRODUCT REQUIREMENTS LENGKAP: ReThink Documentation

07. Arsitektur Konseptual Produk & Skalabilitas

7.1 Batas Arsitektur Konseptual

Arsitektur ReThink membedakan secara logis lima domain informasi. Pemisahan ini bukan keputusan mengenai tabel database, layanan, atau teknologi tertentu.

Domain logis	Fungsi
Struktur Pengetahuan Konseptual	Topik, konsep, dan hubungan prasyarat yang dikurasi.
Definisi Asesmen	Butir diagnosis, bukti yang dinilai, dan kriteria evaluasi.
Konten Instruksional & Validasi	Penjelasan, contoh, latihan, serta item retry/transfer.
Ketersediaan Kemampuan	Penentuan dukungan diagnosis, belajar, atau validasi untuk setiap konsep.
Bukti Pembelajar	Hasil diagnosis individual, penyelesaian pembelajaran, dan status validasi.

AR-01 — Pemisahan Domain Informasi Logis: definisi konsep yang reusable, konfigurasi kemampuan, dan catatan bukti siswa diperlakukan sebagai jenis informasi berbeda.

7.2 Konten Modular & Aktivasi Kemampuan

Kondisi konsep	Interpretasi
Direpresentasikan	Konsep tercatat dalam struktur konseptual; tidak berarti tersedia asesmen.

Dikonfigurasi	Terdapat definisi atau paket yang terasosiasi; belum tentu tersedia bagi siswa.
Didukung Aktif	Kemampuan tertentu telah diaktifkan secara eksplisit dalam cakupan produk.

AR-02 — Konfigurasi Kemampuan Independen: cakupan diagnosis, pembelajaran, dan validasi dapat diperluas secara terpisah tanpa menganggap semuanya otomatis tersedia ketika satu materi ditambahkan.

7.3 Prinsip Skalabilitas & Ekstensi

AR-03 — Ekstensi Terkendali dan Berbasis Peninjauan Mutu: hubungan konsep, item asesmen, materi, dan kriteria penilaian yang baru harus ditinjau sebelum kemampuan diaktifkan. Prinsip skala di sini adalah keluwesan struktur dan keterpeliharaan, bukan bukti kapasitas produksi.

Prinsip tambahan: hubungan konsep reusable dan definisi kemampuan dapat diekstensikan secara terpisah. Ini bukan komitmen membangun diagnosis banyak blocker, tracing adaptif, generasi pelajaran dinamis, atau konfigurasi teknologi fisik dalam MVP.

08. Roadmap Produk

8.1 Prioritas Pengembangan Produk

Urutan berikut adalah prioritas perencanaan, bukan gerbang berurutan yang kaku. Pekerjaan dapat dilakukan paralel jika ketergantungannya memungkinkan.

Prioritas	Fokus	Luaran yang diharapkan
P1 — Fondasi MVP	Entri katalog, hubungan konsep terkurasi, dukungan kemampuan, konteks belajar.	Entri belajar dan struktur konseptual yang berfungsi.
P2 — Diagnosis	Q1–Q4 sesuai batas, evaluasi bukti, probing, hasil dan kelanjutan.	Pengalaman diagnosis SPL untuk Distributive Property.
P3 — Pembelajaran terbimbing	Paket penjelasan visual, contoh, latihan aktif, penyelesaian.	Paket pembelajaran Distributive Property yang dapat digunakan.
P4 — Validasi & integrasi	Retry, transfer kondisional, status belajar, Peta Belajar, transisi.	Jalur pemulihan terintegrasi pada tingkat blocker.
P5 — Verifikasi & penyempurnaan	Pengujian fungsi, konsistensi matematika, integritas keputusan, usability.	Bukti verifikasi, keterbatasan tercatat, kandidat MVP yang diperbaiki.
Pasca-MVP — ekstensi terkendali	Kajian penambahan konsep/paket terkurasi.	Usulan perluasan terpisah; bukan komitmen rilis MVP.

8.2 Ketergantungan & Batas Rilis

Ketergantungan	Dampak terhadap pekerjaan
OI-01	Menyelaraskan otoritas interpretasi dan keputusan diagnosis terutama P2.
OI-02	Menetapkan kelayakan validasi setelah ulasan opsional pada P3–P5.
OI-03	Menetapkan pemetaan respons semua benar pada P2 dan integrasi berikutnya.
OPS-01/02	Mekanisme umpan balik dan kriteria penyelesaian latihan terbimbing.
OPS-03	Penanganan respons validasi tidak lengkap, tak terinterpretasi, atau ekuivalen.
OPS-04	Butir matematika, solusi acuan, dan rubrik yang ditinjau.
OPS-05	Desain transisi dan konsistensi keadaan lintas kemampuan.

Kesiapan MVP bergantung pada implementasi dan verifikasi persyaratan yang berlaku, bukan hanya penyelesaian workstream atau demo. SC1–SC3 diverifikasi melalui skenario fungsional; SC4 melalui evaluasi pengguna terpisah. Tidak ada tanggal rilis, durasi sprint, atau janji efektivitas belajar yang ditetapkan dalam peta jalan ini.

09. Asumsi, Risiko & Ketergantungan

9.1 Risiko Kritis MVP & Mitigasi

Risiko berikut adalah potensi kegagalan, bukan cacat implementasi yang sudah terbukti. Mitigasi tidak menambah kemampuan MVP.

ID	Risiko / dampak	Mitigasi
R-01	Inferensi diagnosis tanpa dukungan bukti menimbulkan konfirmasi blocker salah.	Verifikasi bukti lintas soal, pengaman matematika dan penyelesaian OI-01.
R-02	Soal, contoh, atau rubrik tidak valid menghasilkan hasil yang tidak andal.	Tinjauan matematis konten, solusi acuan dan OPS-04.
R-03	Latihan menjadi pasif atau completion keliru.	Verifikasi respons siswa aktif, umpan balik dan OPS-01/02.
R-04	Penilaian validasi atau pemetaan status salah.	Uji evaluasi jawaban dan aturan tiga status secara terpisah; OPS-03.
R-05	Konteks atau keadaan antar kemampuan tidak konsisten.	Uji transisi dan integritas FR-X-01–03; OPS-05.
R-06	Implementasi dipakai untuk mengklaim usability/efektivitas tanpa bukti.	Pisahkan pengujian fungsional, usability, dan penelitian hasil belajar.

Tidak ada penilaian probabilitas atau tingkat keparahan numerik tanpa bukti. Penyelesaian implementasi tidak otomatis menutup risiko.

9.2 Tata Kelola Asumsi & Ketergantungan

A1–A6 mempertahankan status validasi di Bagian 02.6; keputusan OI dan spesifikasi OPS tetap dikelola melalui register dan ketergantungan Bagian 08.2. Bukti perlu dinilai sesuai batas metodenya, perubahan keputusan harus dapat ditelusuri, dan risiko yang masih menyisakan ketidakpastian tidak dinyatakan selesai tanpa bukti yang relevan.

Ketergantungan hanya membatasi persyaratan dan skenario penerimaan yang terkait; keputusan yang belum selesai tidak otomatis membatalkan persyaratan lain yang sudah disetujui.

10. Prinsip Penerimaan Produk & Tata Kelola PRD

10.1 Prinsip Penerimaan Produk

Penerimaan produk menilai apakah implementasi MVP memenuhi persyaratan yang disetujui berdasarkan bukti verifikasi yang terdokumentasi dan dapat ditinjau. Persetujuan dokumen, selesainya pengembangan, atau demonstrasi alur tunggal tidak sama dengan penerimaan implementasi.

ID	Prinsip penerimaan
AP-01 — Berbasis bukti	Suatu requirement atau skenario hanya dapat dianggap terverifikasi jika bukti memenuhi kondisi verifikasinya.
AP-02 — Hasil verifikasi eksplisit	Bedakan kondisi terpenuhi, gagal, dan belum terverifikasi/tidak dapat dinilai.
AP-03 — Penerimaan memperhitungkan ketergantungan	Perilaku yang membutuhkan keputusan atau kriteria tertunda belum dapat dinyatakan sepenuhnya terverifikasi; hasil untuk perilaku yang tak terkait tetap berlaku.
AP-04 — Ketertelusuran bukti	Kesimpulan penerimaan ditautkan pada requirements, skenario, dan bukti pendukung.
AP-05 — Klaim penerimaan dibatasi	Penerimaan fungsional bukan bukti efektivitas belajar, retensi, atau penguasaan matematika luas.

Bagian ini tidak menambah ambang penerimaan numerik, metode pengujian, atau FR baru. SC tetap merujuk Bagian 03, sedangkan verifikasi persyaratan merujuk dokumen Product Requirements terpisah.

10.2 Penutupan PRD & Tata Kelola Perubahan

PRD merupakan baseline definisi produk yang menjadi rujukan untuk desain produk, desain teknis, pengembangan konten matematika, dan verifikasi. Persetujuan PRD tidak membuktikan bahwa produk telah dibangun atau diterima.

ID	Prinsip tata kelola
PG-01 — Otoritas baseline	Bagian yang disetujui menjadi rujukan cakupan dan perilaku produk.
PG-02 — Item terbuka tetap terbuka	Keputusan dan spesifikasi yang belum selesai hanya ditutup melalui review terpisah.
PG-03 — Pengendalian perubahan perilaku	Perubahan material pada scope, diagnosis, status siswa, atau kondisi keberhasilan memerlukan analisis dampak dan approval.
PG-04 — Keselarasan spesifikasi turunan	Desain produk, teknis, dan konten harus sejalan dengan approved requirements.
PG-05 — Revisi dapat ditelusuri	Perubahan yang disetujui menyebut baseline, desain, dan kriteria verifikasi terdampak.

Penutupan baseline definisi produk tidak menutup OI-01–OI-03, OPS-01–OPS-05, tidak memastikan kelengkapan implementasi, dan tidak menetapkan efektivitas belajar. Revisi substantif sesudah persetujuan harus memperoleh otorisasi eksplisit.

Product Overview

1.1 Product Identity & Positioning

1.1.1 Product Identity

Attribute	Definition
Product Name	ReThink
Product Category	AI-Assisted Mathematics Conceptual Recovery Platform
Product Domain	Mathematics Education
Primary Target Users	High-school students experiencing difficulties with specific mathematics topics
Learning Mode	Self-directed learning
Core Product Focus	Evidence-based conceptual blocker identification, targeted guided learning, and understanding validation
AI Approach	Evidence-grounded AI assistance with bounded autonomy

The target-user profile represents the intended audience and remains subject to further user research validation.

1.1.2 Product Positioning

ReThink is an AI-assisted mathematics conceptual recovery platform that helps students identify evidence-supported conceptual blockers underlying difficulties with specific mathematics topics, engage in targeted guided learning, and validate their understanding through independent application and transfer.

Unlike general mathematics learning platforms that primarily organize instruction around broad topic coverage, ReThink focuses on identifying and recovering specific conceptual gaps that may interfere with a student's learning.

ReThink integrates evidence-based diagnosis, targeted guided learning, and knowledge validation into a closed-loop recovery process. The platform prioritizes the relevance and depth of supported conceptual recovery over comprehensive curriculum coverage.

ReThink does not assume that every mathematical difficulty originates from a conceptual blocker, nor does it claim to diagnose all possible prerequisite gaps. Its diagnostic and recovery capabilities

remain limited to concepts supported by appropriate assessment evidence, curated learning interventions, and validation mechanisms.

The platform is designed to support expansion across mathematics topics and reusable prerequisite concepts while maintaining controlled diagnostic and learning coverage.

1.2 Product Vision

- **Vision Statement**

To make mathematical learning difficulties easier to understand and overcome by helping students identify and address gaps in their understanding of foundational mathematical concepts.

- **Vision Context**

ReThink envisions a mathematics learning experience where students are better equipped to recognize the conceptual gaps that may contribute to their learning difficulties and address them through focused conceptual recovery. Rather than relying solely on repeated explanations or procedural assistance, students should have opportunities to rebuild relevant understanding and develop the ability to apply mathematical concepts independently.

This vision guides ReThink's long-term evolution toward broader conceptual coverage and more targeted, evidence-based recovery capabilities without shifting its focus toward becoming a general mathematics learning platform.

1.3 Product Description

ReThink provides a self-directed mathematics learning experience focused on identifying and recovering specific conceptual gaps that may contribute to difficulties with a selected mathematics topic.

Students begin by exploring a mathematics catalog, selecting a supported topic, and initiating a targeted diagnostic assessment. ReThink evaluates evidence from student responses to identify potential conceptual blockers associated with the selected topic. AI assists in interpreting and synthesizing diagnostic evidence within the boundaries of curated mathematical knowledge, mathematical verification, and controlled diagnostic decisions.

When a conceptual blocker is supported by sufficient diagnostic evidence, ReThink recommends a targeted recovery intervention. The intervention combines visual conceptual explanations, worked examples, and partially guided practice to help students rebuild relevant understanding while remaining actively engaged in mathematical reasoning.

Following guided learning, ReThink evaluates understanding through independent problem-solving and transfer tasks. The results provide evidence of the student's ability to apply the targeted

concept without instructional assistance and inform the corresponding concept-level learning status.

ReThink also provides a learning map that represents relationships between mathematical topics and prerequisite concepts, helping students understand relevant conceptual connections and view learning evidence for concepts that have been assessed.

Students without a confirmed blocker may access available concept reviews without such participation being treated as evidence of a diagnosed conceptual gap. When diagnostic evidence is insufficient or does not support a specific blocker, ReThink communicates the assessment limitations without assuming the student has a confirmed conceptual gap.

Through this integrated diagnosis, targeted recovery, and knowledge validation process, ReThink supports focused conceptual recovery rather than comprehensive curriculum-driven instruction. Its concept-based structure enables progressive expansion of supported diagnostic and recovery capabilities across mathematics topics.

1.4 Core Value Proposition

- **Core Value Proposition**

ReThink helps students better understand what may be holding them back in mathematics, focus on recovering relevant conceptual understanding, and verify what they can apply independently.

ReThink's core value lies in connecting conceptual blocker identification, targeted learning intervention, and independent understanding validation into one integrated recovery experience. Rather than treating diagnosis, learning assistance, and assessment as separate activities, ReThink uses evidence from students' mathematical responses to guide a more focused approach to addressing conceptual difficulties.

This value proposition is expressed through three complementary dimensions:

- **Clarity — Understanding What to Address:** Helping students recognize potential conceptual blockers supported by diagnostic evidence, rather than relying solely on whether their answers are correct or incorrect.
- **Focus — Recovering Relevant Understanding:** Directing learning efforts toward relevant conceptual gaps through structured guidance, rather than requiring students to review broad mathematics content without a specific recovery target.
- **Verification — Understanding What Can Be Applied Independently:** Providing evidence of students' ability to apply the targeted concept without instructional assistance and recognize its underlying structure across different problem representations.

Together, these dimensions support a more informed, focused, and evidence-based conceptual recovery experience. ReThink aims to help students move from uncertainty about their

mathematical difficulties toward clearer learning priorities and independently demonstrated understanding.

The value delivered remains dependent on the diagnostic and learning coverage supported by the platform. When evidence is insufficient to identify a conceptual blocker, ReThink prioritizes transparent assessment outcomes rather than making unsupported claims about students' understanding.

1.5 Product Differentiation

ReThink differentiates itself through its focused approach to conceptual recovery, connecting evidence-based diagnosis, targeted guided learning, and independent understanding validation within a unified learning experience. Rather than prioritizing comprehensive curriculum coverage or general-purpose problem-solving assistance, ReThink concentrates on identifying and addressing specific conceptual blockers that may contribute to students' mathematical difficulties.

A defining characteristic of ReThink is the relationship between its diagnostic, learning, and validation processes. Diagnostic evidence informs the selection of relevant conceptual recovery interventions, while subsequent independent application and transfer tasks provide evidence of students' understanding after instructional support is reduced. This creates a closed-loop recovery process in which assessment informs learning and learning outcomes are evaluated through independent application and transfer tasks within the supported assessment scope.

ReThink supports this approach through AI-assisted evidence interpretation combined with curated mathematical knowledge and controlled decision mechanisms. Its differentiation lies in the integration and focus of these capabilities rather than the novelty of any individual technique. This represents the intended product approach and does not imply demonstrated superiority over existing mathematics learning systems.

1.6 Core Product Principles

ReThink is guided by six core product principles that shape product decisions, preserve the integrity of its conceptual recovery approach, and provide a consistent foundation for future development.

1. Evidence Before Inference: Diagnostic conclusions and learning states must be grounded in observable student evidence rather than unsupported assumptions. ReThink recognizes uncertainty as a valid outcome and avoids inferring conceptual gaps or understanding beyond what the available evidence supports.

2. Recovery Over Content Coverage: ReThink prioritizes the relevance, depth, and completeness of supported conceptual recovery over the breadth of mathematics content offered. Expanding topic coverage should not compromise the quality of diagnosis, learning intervention, or understanding validation.

3. Active Reasoning Over Answer Delivery: ReThink's learning assistance should support students in developing mathematical reasoning rather than replacing their thinking with immediate solutions. Instructional guidance should encourage active engagement with mathematical concepts and support increasingly independent application.

4. Validation Before Understanding Claims: Completion of instructional activities does not automatically establish conceptual understanding. Learning outcomes must be supported by appropriate evidence of independent application and conceptual transfer. Claims about students' understanding must remain proportional to the scope and strength of the available assessment evidence.

5. Bounded AI Autonomy: AI plays an active role in interpreting and synthesizing student evidence and supporting diagnostic decisions. However, AI-assisted reasoning must operate within curated mathematical knowledge, supported diagnostic coverage, mathematical verification, and controlled decision safeguards. AI must not independently establish unsupported conceptual blockers or learning states.

6. Scalable Structure, Controlled Coverage: ReThink is designed to support progressive expansion across mathematics topics, prerequisite relationships, and conceptual recovery capabilities. The platform must distinguish between concepts represented in its knowledge structure and concepts with explicitly supported diagnostic assessments, curated learning interventions, and knowledge validation capabilities. Architectural extensibility must not be treated as evidence of operational capability.

Problem Statement & Target Users

2.1 Problem Background & Evidence

2.1.1 Mathematics Learning Difficulties & Prerequisite Knowledge

Mathematics learning involves interconnected concepts, where understanding more advanced topics can depend on knowledge established in earlier learning stages. Consequently, difficulties encountered in a particular mathematics topic may be associated with gaps in the understanding of relevant prerequisite concepts rather than the target topic alone.

Research discussed in the ReThink ICC 2026 paper supports the importance of prerequisite knowledge in mathematical learning progression. Preliminary self-reported survey findings also indicate that difficulties with conceptual understanding and previously learned material are relevant concerns among the surveyed students.

In an exploratory survey involving 12 high-school students, 10 respondents reported experiencing mathematics learning difficulties sometimes or frequently, while 6 respondents perceived that some of their difficulties were associated with previously learned material they had not fully understood.

However, these findings represent students' perceived experiences rather than objective evidence of prerequisite conceptual blockers. Prerequisite knowledge gaps remain one possible explanation for mathematical difficulties and should not be assumed to account for every instance of student struggle.

2.1.2 Difficulty Attribution & Conceptual Understanding

Identifying the source of a mathematical difficulty is not always straightforward. An incorrect answer may reflect different underlying causes, and answer correctness alone does not necessarily reveal the conceptual understanding or reasoning involved.

Similarly, students may be able to follow mathematical procedures or worked examples without consistently demonstrating an ability to apply the underlying concepts independently or across different problem representations.

Preliminary survey findings indicate that 8 of 12 respondents experienced difficulty understanding the meaning of mathematical formulas or concepts. Additionally, 7 respondents reported difficulties when learning examples differed from the problems they encountered, while 5 respondents reported being able to follow examples but struggling to solve problems independently.

These findings suggest that difficulties in conceptual understanding and independent problem application are relevant concerns within the surveyed group. However, reported difficulties with unfamiliar problem forms do not independently establish conceptual misunderstanding or failure of

mathematical transfer. Alternative explanations, including differences in problem complexity or familiarity, remain possible.

2.1.3 Evidence Boundaries

The educational literature provides a theoretical foundation for investigating prerequisite knowledge gaps, difficulties in attributing mathematical errors to underlying concepts, and differences between procedural performance and conceptual understanding.

Preliminary ReThink user research provides initial supporting evidence that some intended users experience difficulties understanding mathematical concepts, perceive gaps in previously learned material, and encounter challenges applying instructional examples to independent problem-solving.

The current evidence consists of an exploratory survey of 12 respondents, predominantly students in grades XI–XII. The survey uses self-reported experiences and a small, non-representative sample. Its findings therefore cannot establish the prevalence of these difficulties across high-school students, objectively confirm conceptual blockers, or demonstrate causal relationships between prerequisite gaps and learning difficulties.

The extent to which students can accurately identify the source of their mathematical difficulties also remains insufficiently established.

Further qualitative interviews and task-based mathematical observation are needed to examine the relationship between perceived difficulties, observable mathematical reasoning, and potential prerequisite knowledge gaps. Until stronger evidence is available, conclusions about the target population and underlying causes should remain provisional.

2.2 Core Problem Statement

Core Problem Statement

Students experiencing difficulties with specific mathematics topics may struggle to determine whether underlying conceptual gaps contribute to their difficulties, identify what understanding needs to be strengthened, and establish whether they can apply relevant mathematical concepts independently.

This problem is characterized by three interconnected dimensions:

1. Unclear Underlying Learning Difficulties

Students may recognize that they are struggling with a mathematics topic without being able to determine whether the difficulty is associated with gaps in prerequisite conceptual understanding or other contributing factors.

2. Challenges in Targeting Learning Recovery

When the source of difficulty is unclear, students may struggle to determine which mathematical concepts require further attention. Available learning assistance may explain procedures without necessarily resolving the specific understanding needed to address their difficulties.

3. Uncertainty About Independent Understanding

Following instructional explanations or worked examples does not necessarily demonstrate that students can apply the underlying mathematical concepts independently or across different problem representations. Students may therefore remain uncertain about the extent of their understanding.

These dimensions represent a literature-informed problem definition supported by preliminary exploratory survey findings. Their prevalence, underlying causes, and relationships within ReThink's intended target population remain subject to further user research and mathematical task observation.

2.3 Target Users & Segmentation

2.3.1 Primary Target User

ReThink's intended primary users are high-school students experiencing difficulties understanding or applying mathematics within specific topics, who may benefit from self-directed support in addressing relevant conceptual gaps.

These students are not required to have previously identified prerequisite knowledge gaps. Their difficulties may arise from different causes, and the relevance of conceptual recovery depends on the specific learning difficulty and the conceptual support available.

This target-user definition represents an intended audience rather than a validated user segment. Its suitability for ReThink requires further investigation through user research and observation of mathematical learning difficulties.

2.3.2 Initial Research Focus

ReThink's initial user research focuses on students in grades X–XII, with priority given to grades XI–XII who report difficulties understanding or applying mathematics within their current learning topics.

In the preliminary exploratory survey involving 12 respondents, 11 participants were students in grades XI–XII. This composition reflects the initial research focus and does not establish that students in these grades experience greater conceptual recovery needs than students in other grade levels.

The initial research context emphasizes algebra and related prerequisite concepts to maintain relevance to ReThink's current MVP focus. However, the selection of this mathematical domain is a

product development decision rather than a conclusion that algebra represents the most prevalent learning difficulty among the intended user population.

2.3.3 Provisional Segmentation Dimensions

ReThink uses two primary dimensions to explore variation among potential users:

Learning Difficulty

The frequency and characteristics of mathematics learning difficulties students report, including challenges understanding concepts or applying mathematical knowledge.

Help-Seeking Behavior

The ways students seek, use, or avoid learning assistance when encountering mathematical difficulties.

Problem Awareness is considered an additional analytical attribute, reflecting how specifically students can recognize and describe the difficulties they experience.

Preliminary survey findings indicate variation in students' reported learning difficulties and learning assistance practices. However, these findings do not yet establish distinct, validated, or mutually exclusive user segments.

Further user research is required to refine the target-user profile and determine which patterns of learning difficulty, help-seeking behavior, and problem awareness are most relevant to ReThink's intended users.

2.4 Context of Use & Current Behaviors

2.4.1 Learning Situations & Help-Seeking Triggers

ReThink's intended users may encounter mathematics learning difficulties in different situations, including challenges following classroom material, solving specific mathematics problems, or revisiting previously learned concepts.

Two initial contexts of use are considered relevant:

- **Falling Behind in Class:** Students experience difficulties understanding or keeping up with mathematics topics currently being taught and may seek additional learning support.
- **Struggling with a Specific Problem:** Students encounter mathematics problems they cannot solve confidently, even after receiving explanations or reviewing examples.

These contexts remain provisional. Preliminary survey findings indicate that students experience difficulties across different learning situations, but the available evidence does not establish which context most strongly motivates them to seek conceptual recovery support.

Potential help-seeking triggers include difficulties understanding explanations, completing mathematics tasks, preparing for assessments, or recognizing gaps in previously learned material. The relative importance of these triggers and their relationship to actual help-seeking decisions require further investigation.

2.4.2 Current Learning & Help-Seeking Behaviors

Preliminary exploratory survey findings indicate that students use multiple sources of assistance when encountering mathematics learning difficulties.

Among 12 surveyed high-school students, 10 reported seeking help from teachers, friends, or family members; 8 used written or video-based learning materials; 7 used AI assistants; and 6 attempted to solve problems independently. Tutoring services and skipping difficult problems were each reported by 2 respondents.

These findings indicate that learning support practices within the surveyed group involve a combination of interpersonal assistance, digital learning resources, and individual problem-solving efforts.

Reported AI usage suggests familiarity with AI-based assistance among some respondents. However, it does not establish trust in AI-generated diagnostic conclusions, preference for AI-assisted conceptual recovery, or willingness to use ReThink.

The survey also does not establish which learning resources students prioritize, how they combine different forms of assistance, or which approaches they consider most useful in specific situations.

2.4.3 Self-Directed Learning Conditions

Students' engagement in self-directed mathematics learning may be influenced by their available study time, motivation, and circumstances surrounding the learning difficulty.

In the preliminary survey, 6 of 12 respondents reported having approximately 15–30 minutes available for independent study, while 3 reported having no fixed study period. The remaining respondents reported longer available study durations.

These findings indicate variation in reported study-time availability but do not establish an appropriate duration for a ReThink learning session.

Self-directed learning in this context refers to students taking initiative in seeking and engaging with learning support. It does not require students to learn exclusively without teachers, peers, or other external resources.

Further user research is needed to clarify learning triggers, patterns of help-seeking, and the conditions influencing students' willingness and ability to participate in a structured conceptual recovery experience.

2.5 Unmet Needs & Opportunity

2.5.1 Potential Unmet Learning Needs

Preliminary user research indicates that some high-school students continue to encounter difficulties when using existing mathematics learning support, particularly in understanding explanations, connecting examples to the problems they face, and applying mathematical knowledge independently.

These findings suggest three potential unmet learning needs:

1. Clarity About Relevant Learning Gaps

Students may need greater clarity about which mathematical concepts or areas of understanding deserve attention when they struggle with a specific topic. However, current evidence is limited regarding students' ability to identify the underlying causes of their difficulties, and a prerequisite conceptual gap cannot be assumed from perceived difficulty alone.

2. Conceptually Relevant Learning Assistance

Students may need learning assistance that makes underlying mathematical ideas and reasoning easier to understand while supporting connections between explanations and the problems they encounter. Difficulties applying instructional examples may reflect conceptual gaps, differences in problem complexity, unfamiliar representations, or other contributing factors.

3. Evidence of Independent Application

Students may need clearer indications of whether they can apply relevant mathematical understanding independently rather than relying on their ability to follow instructional examples. This potential need does not establish that students prefer or are willing to complete additional formal assessments.

These needs are informed by educational literature and preliminary self-reported survey findings. Their relative importance, underlying causes, and the extent to which existing learning alternatives address them remain subject to further investigation.

2.5.2 Core Product Opportunity

The identified learning frictions suggest an opportunity for more focused support that helps students address potentially relevant conceptual difficulties and better understand the extent of their independent mathematical application.

This opportunity is particularly relevant where difficulties may involve prerequisite conceptual gaps, without assuming that such gaps explain every learning challenge.

The opportunity remains provisional. Available evidence supports the relevance of the problem area but does not establish unmet demand for ReThink, demonstrate the effectiveness of its proposed approach, or establish superiority over existing learning alternatives.

2.6 Critical Assumptions & Research Gaps

2.6.1 Critical Product Assumptions

ReThink's problem framing and intended user experience are informed by educational literature, preliminary user research, and established product decisions. However, several assumptions remain insufficiently validated within the intended target population.

Six critical assumptions are identified to guide further evaluation of ReThink's problem relevance, user needs, and potential product value.

ID	Critical Assumption	Current Evidence Status
A1 — Prerequisite Gap Relevance	Gaps in prerequisite conceptual understanding contribute to some mathematics learning difficulties experienced by intended users.	Literature-backed with preliminary self-reported indication; objective confirmation pending
A2 — Difficulty Attribution & Learning Direction	Some students struggle to recognize the source of their mathematical difficulties or determine which understanding requires attention.	Limited preliminary survey indication; objective evaluation pending
A3 — Learning Assistance Gaps	Some students experience difficulties that remain unresolved after using their available learning assistance.	Preliminary self-reported learning frictions; persistence of difficulties and comparative adequacy of alternatives not objectively established
A4 — Diagnostic Experience Acceptance	Students can recognize the purpose and potential value of targeted diagnosis, are willing to participate in the assessment process, and can meaningfully interpret its outcomes and limitations.	Not directly tested through the preliminary survey
A5 — Targeted Recovery Effectiveness	Targeted conceptual recovery can improve students' ability to independently apply the mathematical concepts being addressed.	Pedagogically informed hypothesis; ReThink learning effectiveness not established

A6 — Perceived Value of Understanding Validation	Students perceive evidence of independent conceptual application as useful for evaluating their understanding.	Indirect indication from reported learning goals; perceived value and acceptance of validation activities not directly tested

These assumptions do not represent confirmed characteristics of all intended users or evidence of demonstrated ReThink effectiveness.

Product development may proceed under explicitly documented assumptions. However, research findings, product claims, and subsequent decisions must remain proportional to the strength and scope of the available evidence.

2.6.2 Key Research Gaps

Four research areas remain relevant to evaluating the assumptions underlying ReThink's problem and user definition.

1. Observed Mathematical Difficulties

Further task-based mathematical observation is needed to investigate whether students' perceived learning difficulties correspond to observable prerequisite conceptual gaps, alternative explanations, or other mathematical learning challenges.

2. Learning Behaviors & Unmet Needs

Additional qualitative research is needed to understand how students recognize learning difficulties, seek and evaluate available assistance, and determine whether their learning needs have been addressed.

3. Product Experience Acceptance

Concept testing and usability evaluation are needed to examine whether students understand the purpose of targeted diagnosis, are willing to engage with the structured learning experience, and can meaningfully interpret diagnostic outcomes, including inconclusive results and assessment limitations.

Research should also investigate whether students perceive understanding validation as useful and acceptable within their learning experience.

4. Learning Outcome Evidence

Appropriate learning evaluation is needed to investigate whether targeted conceptual recovery helps students independently apply the mathematical concepts being addressed and recognize the same underlying structures across different problem representations.

Performance within ReThink's supported validation tasks should be distinguished from broader evidence of learning improvement, which requires an appropriate evaluation design.

Such evaluation must distinguish improvements within the assessed conceptual scope from broader claims of topic mastery, long-term retention, or general mathematics achievement.

2.6.3 Evidence & Decision Boundaries

The current evidence includes educational literature and a preliminary exploratory survey involving 12 high-school students, predominantly in grades XI–XII.

Survey findings provide initial evidence of students' reported learning difficulties, perceived conceptual understanding challenges, and learning assistance practices. However, the survey's self-reported nature and limited, non-representative sample do not establish population-level prevalence, objectively confirm prerequisite conceptual blockers, or demonstrate causal relationships between conceptual gaps and learning difficulties.

The effectiveness of ReThink's diagnostic and recovery approach, students' acceptance of the proposed learning experience, and the relative importance of the identified unmet needs remain insufficiently established.

Further findings should be used to evaluate and refine provisional assumptions while maintaining a distinction between observed user evidence, researcher interpretation, and established product decisions.

Unresolved methodological and operational decisions are tracked separately in the Product Decision Register and are not resolved through approval of this section.

03. Product Goals & Success Metrics

3.1 Product Goals & Intended Outcomes

ReThink aims to support evidence-grounded mathematics conceptual recovery by helping students understand relevant learning difficulties, strengthen targeted conceptual understanding, and evaluate their ability to apply mathematical concepts independently.

The product is guided by four outcome-oriented goals that define the results ReThink intends to enable for students within its supported learning coverage.

G1 — Provide Actionable, Evidence-Grounded Recovery Direction

ReThink aims to provide diagnostic outcomes that translate observable mathematical evidence into appropriately qualified learning directions. When evidence supports a conceptual blocker, the product should help students recognize the relevant concept requiring attention. When evidence is insufficient or does not support a blocker, ReThink should communicate the assessment outcome and its limitations without making unsupported conclusions.

Intended Outcome: Students receive understandable, evidence-grounded information about potential conceptual difficulties and appropriate available next steps.

G2 — Support Targeted Conceptual Recovery

ReThink aims to support students in strengthening their understanding of relevant mathematical concepts through focused learning interventions that preserve active reasoning and encourage independent application.

Intended Outcome: Students have opportunities to develop stronger conceptual understanding and improve their ability to apply the targeted mathematical concept without relying solely on instructional guidance.

G3 — Provide Evidence-Based Understanding Evaluation

ReThink aims to evaluate demonstrated conceptual understanding beyond the completion of instructional activities by examining students' ability to apply the targeted concept independently and recognize its underlying mathematical structure across different problem representations.

Intended Outcome: Students receive appropriately qualified evidence of their demonstrated understanding within the supported assessment scope, including indications of understanding that may require further reinforcement.

G4 — Enable an Understandable Self-Directed Recovery Experience

ReThink aims to make its supported conceptual recovery experience understandable and navigable for students learning through self-directed engagement. Students should be able to understand the purpose of diagnostic assessment, access relevant learning support, and interpret the meaning and limitations of their learning outcomes.

Intended Outcome: Students can meaningfully engage with the available recovery experience, understand their learning direction, and recognize what their assessment results indicate without requiring continuous external guidance.

Together, these goals define ReThink's intended product outcomes rather than demonstrated product effectiveness. Their achievement must be evaluated using appropriate evidence of functional reliability, user experience, and learning outcomes.

The goals remain bounded by ReThink's supported diagnostic, instructional, and validation coverage. They do not imply comprehensive mathematics mastery, guaranteed learning improvement, or demonstrated superiority over alternative learning approaches.

3.2 MVP Success Definition

ReThink's MVP success is evaluated through a tiered framework that distinguishes minimum functional product success from empirical validation of the user experience.

The MVP is designed to demonstrate one complete, evidence-grounded conceptual recovery pathway within its explicitly supported mathematical scope: Sistem Persamaan Linear as the selectable topic and Distributive Property as the primary supported conceptual blocker.

3.2.1 Minimum Functional MVP Success

Minimum functional MVP success requires the supported conceptual recovery system to operate correctly across its intended diagnostic, instructional, and validation processes.

Three success conditions define this requirement:

SC1 — Complete Supported Recovery Path

ReThink must support one complete conceptual recovery pathway, beginning with supported topic selection and diagnostic assessment, followed by targeted guided learning, independent understanding validation, and blocker-level learning results.

The pathway must function end-to-end for its supported learning scenario without implying comprehensive mathematics curriculum coverage or recovery of every prerequisite concept.

SC2 — Evidence-Grounded Decision Integrity

Diagnostic and validation decisions must remain consistent with observable mathematical evidence, supported conceptual knowledge, mathematical verification, and established product decision rules.

The system must distinguish supported blocker confirmation from rejected or inconclusive diagnostic outcomes and must not establish unsupported conceptual blockers or understanding claims beyond its assessment scope.

SC3 — Appropriate Outcome-Based Continuation

ReThink must communicate supported diagnostic and validation outcomes clearly and provide appropriate available next steps based on those outcomes.

Confirmed, Rejected, and Uncertain diagnostic outcomes must not be treated interchangeably. Similarly, completion of guided learning must not automatically establish demonstrated understanding, and blocker-level learning results must reflect the applicable validation evidence.

Together, SC1–SC3 define the minimum functional success standard for the ReThink MVP and require verification through supported functional test scenarios.

3.2.2 User Experience Validation

SC4 — Understandable Core Recovery Experience

ReThink should enable students to understand the purpose of diagnostic assessment, engage with the supported conceptual recovery experience, and meaningfully interpret the results and limitations of their diagnostic and learning assessments.

SC4 is evaluated separately through appropriate user testing. Its evaluation should examine whether intended users can understand and navigate the available experience without continuous external guidance.

Functional implementation alone does not establish that SC4 has been achieved. Likewise, the absence of completed usability testing must not be presented as evidence of successful user experience validation.

3.2.3 MVP Success Interpretation

Minimum functional MVP success is considered **verified only when all required functional scenarios within the approved MVP scope have been executed and satisfy their applicable acceptance criteria**.

Required scenarios that remain untested, fail, or lack finalized expected behavior prevent a claim of complete functional verification. The detailed scenario inventory and acceptance criteria are established separately in the product validation specification.

SC4 provides a separate assessment of user experience quality and must be reported according to the evidence available. Functional verification, user experience evaluation, demonstrated assessment performance, and evidence of learning improvement must not be treated as interchangeable outcomes.

Minimum functional MVP success must not be interpreted as demonstrated user experience quality or learning effectiveness. Applicable claim boundaries and non-goals are established in Section 3.4.

3.3 Success Dimensions & Metrics

ReThink evaluates product success through four distinct dimensions aligned with its product goals and tiered MVP success definition.

Each dimension uses primary metric groups supported by additional indicators. These measures provide evidence of functional performance, decision integrity, user experience, and learning outcomes without combining them into a single product success score.

3.3.1 Functional Completeness

Functional completeness evaluates whether ReThink correctly executes its supported conceptual recovery pathway and handles the relevant diagnostic and validation scenarios.

Primary Metric: Supported Scenario Pass Rate

The proportion of executed required functional scenarios that satisfy their defined expected outcomes.

Supporting Indicators:

- End-to-end recovery pathway completion in functional testing.
- Supported diagnostic and validation outcome routing coverage.
- Number of required scenarios executed, passed, failed, or not yet tested.

Functional success reporting must distinguish successful execution from incomplete test coverage. Passing all executed scenarios does not establish complete functional verification when required scenarios remain untested.

3.3.2 Decision Integrity

Decision integrity evaluates whether diagnostic conclusions and knowledge validation outcomes remain consistent with mathematical evidence, supported concept coverage, and established product decision rules.

Primary Metric: Decision Agreement Against Reviewed Cases

The degree of agreement between ReThink's diagnostic or validation decisions and the expected outcomes of appropriately reviewed reference cases.

Supporting Indicators:

- Unsupported blocker confirmation incidence.
- Uncertainty handling correctness.
- Validation state agreement with applicable assessment evidence.

Unsupported diagnostic confirmations and invalid understanding claims must remain explicitly visible in reporting rather than being obscured by an aggregate agreement rate.

Agreement against reviewed reference cases does not establish population-level diagnostic accuracy or the absence of errors in untested situations.

3.3.3 Usability & Interpretability

Usability and interpretability evaluate whether intended users can meaningfully engage with ReThink's supported recovery experience and correctly understand the outcomes communicated by the product.

Primary Metric Group: Core Task Success & Outcome Comprehension

Core Task Success evaluates whether students can complete representative tasks within the supported experience.

Outcome Comprehension evaluates whether students correctly interpret diagnostic results, learning outcomes, and relevant assessment limitations.

Supporting Indicators:

- Critical usability issues.
- Misinterpretation of diagnostic outcomes.
- Misinterpretation of learning status or assessment coverage.

Task completion and outcome comprehension must be reported separately because successful navigation does not necessarily indicate correct understanding of assessment results.

This dimension requires user testing and is evaluated independently from minimum functional MVP success.

3.3.4 Learning Outcome Evidence

Learning outcome evaluation examines students' demonstrated ability to apply targeted mathematical concepts and, where appropriate, investigates potential changes in mathematical performance following conceptual recovery.

Primary Metric Group: Independent Application & Transfer Performance

Independent Application Performance evaluates students' ability to apply the targeted concept without instructional scaffolding within the assessed mathematical structure.

Transfer Performance evaluates students' ability to recognize and apply the same underlying mathematical structure across different problem representations.

These primary measures describe demonstrated assessment outcomes within ReThink's supported validation scope. They do not, by themselves, establish that students' understanding improved because of the intervention.

Supporting Indicator:

- Preliminary pre/post concept performance change, where an appropriate evaluation design is available.

Pre/post performance change is considered separately as exploratory evidence of potential learning improvement. Its interpretation must account for assessment comparability and relevant alternative explanations. Observed improvement does not independently establish causal learning effectiveness.

Independent Retry and Transfer Problem results must be interpreted according to the applicable assessment sequence. Students who do not reach the transfer task must not automatically be classified as having failed transfer.

3.3.5 Measurement & Reporting Principles

ReThink's success measurements must follow these principles:

1. **Evidence-Specific Interpretation:** Each metric must be interpreted according to its evidence source, assessment coverage, and limitations.
2. **Separation of Evaluation Dimensions:** Functional correctness, decision integrity, user experience, and learning outcomes must be evaluated and reported distinctly.
3. **Transparent Coverage:** Functional and assessment results must indicate the scope of scenarios, concepts, or participants evaluated.
4. **Visibility of Critical Failures:** Unsupported diagnostic conclusions and invalid understanding claims must not be concealed by aggregate success rates.
5. **No Unsupported Numerical Targets:** Performance thresholds must be justified through functional requirements, methodological considerations, or appropriate evaluation evidence.

Reporting must distinguish between functionally verified and not fully verified product behavior, user experience findings and unevaluated usability, demonstrated assessment performance, and preliminary evidence of learning improvement. Each status must reflect the completeness and limitations of its supporting evidence.

Detailed measurement procedures, reference cases, test protocols, and acceptance thresholds are established separately in the relevant validation documentation.

The framework remains limited to ReThink's supported MVP capabilities and does not imply comprehensive mathematics mastery or demonstrated learning effectiveness across the intended student population.

3.4 Success Boundaries & Non-Goals

ReThink's success claims must remain proportional to its supported product capabilities, the mathematical concepts assessed, and the evidence available from product and learning evaluation.

The following boundaries distinguish what the MVP can demonstrate, what remains subject to empirical validation, and which broader outcomes are not required for MVP success.

3.4.1 Evidence & Claim Boundaries

Diagnostic outcomes represent evidence-grounded conclusions within ReThink's supported assessment scope. A Confirmed blocker does not establish the definitive cause of every mathematical difficulty experienced by a student, while Rejected or Uncertain outcomes do not establish the absence of learning difficulties.

Knowledge validation outcomes represent demonstrated performance within the concepts and assessment tasks evaluated. An Understood status indicates that the applicable blocker-level validation conditions have been met; it does not establish comprehensive topic mastery, long-term retention, or understanding across all mathematical contexts.

Functional verification, user experience validation, and learning effectiveness evaluation provide different forms of evidence. Success in one dimension must not be presented as proof of success in another.

Improving targeted conceptual understanding remains an intended product outcome. However, claims that ReThink has produced learning improvement require an appropriate evaluation design and supporting evidence. Neither functional completion nor successful assessment performance alone establishes causal learning effectiveness.

3.4.2 MVP Outcome Boundaries

The ReThink MVP demonstrates a targeted conceptual recovery experience within a deliberately limited mathematical scope.

Sistem Persamaan Linear is the selectable topic, and Distributive Property is the primary supported conceptual blocker. The MVP focuses on one complete recovery pathway connecting evidence-based diagnosis, guided conceptual learning, and blocker-level knowledge validation.

Diagnostic conclusions and learning outcomes are restricted to the concepts and evidence supported by this pathway.

Concepts that have not been assessed must not be classified as understood or misunderstood solely because they appear in the concept relationship structure.

Similarly, successful validation of Distributive Property does not establish mastery of the entire Sistem Persamaan Linear topic or guarantee successful application across every related mathematical problem.

The scope of demonstrated recovery is therefore limited to the supported conceptual blocker and the mathematical applications evaluated by the available assessment.

3.4.3 MVP Non-Goals & Deferred Outcomes

The following broader outcomes are not required to establish ReThink's minimum functional MVP success:

- **Comprehensive Concept Coverage:** Diagnostic and recovery support across the full mathematics curriculum.
- **Topic-Level Mastery:** Demonstrated mastery of all concepts and problem types within the selected mathematical topic.
- **Adaptive Multi-Level Recovery:** Dynamic recovery across multiple prerequisite layers or individually diagnosed competencies.
- **Longitudinal Understanding Evidence:** Demonstrated long-term retention or sustained recovery through repeated assessment over time.
- **Generalized Mathematics Improvement:** Demonstrated improvement in students' overall mathematics achievement beyond the concepts assessed.

These exclusions define the current MVP success boundary rather than permanent limitations on ReThink's product vision.

Learning effectiveness within the supported conceptual scope is not a non-goal. It remains an intended outcome requiring empirical evaluation, separate from minimum functional MVP verification.

Future product development may expand concept coverage, diagnostic granularity, adaptive recovery capabilities, and longitudinal learning evaluation. Such expansions must be supported by corresponding product capabilities and appropriate evidence before their outcomes can be claimed as demonstrated.

Together, these boundaries ensure that ReThink's MVP can be evaluated against its supported capabilities without overstating diagnostic certainty, conceptual mastery, or demonstrated educational impact.

Product Scope & MVP Definition

4.1 MVP Scope Statement & Product Boundary

4.1.1 MVP Scope Statement

ReThink's Minimum Viable Product (MVP) delivers one complete, evidence-grounded mathematics conceptual recovery experience within a deliberately controlled mathematical scope.

The MVP uses Sistem Persamaan Linear (SPL) as the selectable learning topic and Distributive Property as the primary supported conceptual blocker.

Its core implementation integrates targeted diagnostic assessment, curated guided learning, and blocker-level knowledge validation into a connected recovery experience.

The MVP is intended to demonstrate the functional feasibility and decision integrity of this conceptual recovery approach rather than provide comprehensive mathematics instruction, broad diagnostic coverage, or demonstrated topic-level mastery.

4.1.2 Supported Product Experience

The MVP provides a catalog-first, topic-based learning entry through which students can explore mathematics topics and initiate diagnostic assessment for supported content.

Within its supported scope, ReThink evaluates mathematical evidence to determine whether a relevant conceptual blocker can be confirmed, rejected, or remains uncertain.

When a supported blocker is confirmed, the student can proceed through the corresponding targeted guided learning intervention and blocker-level knowledge validation.

For non-confirmed diagnostic outcomes, ReThink communicates the result and its limitations without establishing unsupported conceptual weaknesses. The experience includes available continuation options, such as optional review of the supported concept or returning to topic exploration, without treating optional review as evidence of a confirmed blocker.

The supported experience also includes a bounded Hybrid Learning Map serving three complementary purposes:

- **Explore:** Providing orientation to available mathematics topics and conceptual relationships before diagnosis.
- **Focus:** Communicating the relevant conceptual focus after diagnostic assessment, where supported by evidence.
- **Progress:** Reflecting available learner evidence and applicable blocker-level learning outcomes after knowledge validation.

The Learning Map does not imply that all represented concepts have active diagnostic, instructional, or validation capabilities. Concepts without appropriate assessment evidence must not be classified as understood or misunderstood.

The MVP must communicate supported diagnostic and learning outcomes appropriately rather than relying exclusively on a successful confirmed-blocker recovery scenario.

Detailed outcome routing, optional review behavior, and learner-state implications are governed by the applicable product requirements and outstanding product decisions.

4.1.3 Product Boundary Principles

ReThink's MVP follows three product boundary principles:

1. Controlled Capability Coverage

The MVP supports one complete conceptual recovery pathway. Representing additional mathematics topics or prerequisite relationships does not imply that those concepts can be diagnosed, taught, or validated through the active MVP.

2. Evidence-Gated Diagnostic and Learning Outcomes

Diagnostic conclusions and learning states must be grounded in mathematical evidence available within the supported assessment scope. The system must distinguish supported conclusions from uncertain or unsupported interpretations and must not assign conceptual understanding states to unassessed concepts.

3. Future-Ready Structure Without Expanded Capability Claims

The MVP preserves a reusable concept-oriented product structure for future expansion. Extensibility does not imply that future diagnostic, instructional, or validation capabilities are already implemented.

These principles define the overall MVP boundary. The specific mathematical coverage, supported capabilities, exclusions, and extensibility constraints are established in Sections 4.2–4.4.

4.2 Mathematical & Diagnostic Coverage

4.2.1 Mathematical Content Coverage

ReThink's MVP operates within a deliberately controlled mathematical scope centered on Algebra, with Sistem Persamaan Linear (SPL) as the selectable learning topic and Distributive Property as the primary supported conceptual blocker.

The MVP examines Distributive Property through two mathematical application contexts:

1. Direct Concept Application

Application of Distributive Property in algebraic expressions and mathematical transformations, allowing the targeted concept to be examined relatively independently of the selected learning topic.

2. Application Within Sistem Persamaan Linear

Application of Distributive Property within mathematical transformations associated with SPL, allowing the targeted concept to be examined in the context of the selected learning topic.

These contexts provide complementary mathematical evidence about the same supported concept. Differences in student performance between them do not, by themselves, establish a separate conceptual blocker, prove a specific transfer difficulty, or justify additional diagnostic conclusions.

The MVP may represent additional curated mathematics concepts and prerequisite relationships through its underlying knowledge structure and bounded Learning Map.

However, concept representation is distinct from diagnostic, recovery, and validation coverage. A represented concept does not become an active diagnostic target merely because its relationship to another concept is known.

Distributive Property is the only conceptual blocker with a complete supported diagnostic, guided recovery, and blocker-level knowledge validation pathway in the initial MVP.

The MVP does not establish separate competency-level diagnoses or learner states for individual dimensions of Distributive Property.

4.2.2 Diagnostic Assessment Coverage

The MVP diagnostic assessment uses a defined blueprint consisting of three core questions and a maximum of one conditional targeted probe.

The three core questions serve complementary diagnostic roles:

Q1 — Target Performance

A multiple-choice question assessing student performance in a mathematical problem associated with the selected SPL topic.

Its role is to observe target-topic performance and provide initial evidence relevant to the diagnostic investigation. Q1 correctness alone does not establish the presence or absence of a conceptual blocker.

Q2 — Prerequisite Application

A short mathematical response assessing the student's application of Distributive Property in a more isolated algebraic expression or transformation.

Its role is to provide observable evidence of how the student applies the supported prerequisite concept.

Q3 — Conceptual Reasoning

A short explanation or reasoning-focused response examining the student's understanding of a relevant mathematical transformation.

Its role is to provide explanatory evidence that can inform interpretation of mathematical reasoning beyond final-answer correctness.

Q4 — Conditional Targeted Diagnostic Probe

When the available core-question evidence leaves a relevant information gap unresolved, the system may administer a maximum of one additional targeted probe.

The supported probe forms are:

- **Procedural Trace:** Examining where a mathematical procedure or transformation becomes incorrect.
- **Conceptual Probe:** Examining conceptual reasoning or representation relevant to the suspected blocker.

Q4 is conditional rather than mandatory. The maximum diagnostic assessment length for the MVP is four questions.

The supported assessment may consider three complementary evidence categories:

- **Outcome Evidence:** The mathematical correctness of submitted answers or selected responses.
- **Transformation Evidence:** Observable mathematical operations, expressions, or transformations performed by the student.
- **Explanatory Evidence:** Student-provided reasoning about mathematical relationships or procedures.

Not every evidence category must be present in every question. The relevance and sufficiency of available evidence determine which diagnostic conclusions can be justified.

The diagnostic assessment is restricted to evaluating the supported Distributive Property blocker. It does not provide comprehensive mathematics proficiency assessment, arbitrary misconception detection, or diagnosis across unsupported prerequisite concepts.

The three-core-question blueprint and maximum-one-probe configuration represent MVP-specific assessment coverage, not permanent limitations on future diagnostic designs.

4.2.3 Diagnostic Decision Boundaries

ReThink's MVP supports three diagnostic outcomes within its explicitly supported conceptual scope:

Confirmed

Available mathematical evidence sufficiently supports the examined Distributive Property conceptual blocker under the applicable diagnostic decision rules.

Rejected

The examined conceptual blocker is not supported or is contradicted by sufficient evidence under the applicable diagnostic decision rules.

Uncertain

Available evidence is insufficient, ambiguous, or inconsistent, preventing a justified conclusion about the examined conceptual blocker.

Diagnostic conclusions must remain bounded by the supported mathematical concept, available assessment evidence, and applicable decision constraints.

A mathematically incorrect response does not automatically establish a conceptual blocker. Similarly, a correct response or non-confirmed diagnostic outcome does not establish comprehensive conceptual mastery or the absence of other mathematical learning difficulties.

Mathematical errors outside supported diagnostic coverage must not automatically generate additional confirmed blockers.

Concepts that have not been diagnostically assessed must not receive unsupported conclusions about a student's understanding.

The MVP must preserve uncertainty when available evidence does not justify confirmation or rejection of the examined blocker.

The operational distribution of responsibilities between AI-led evidence interpretation and mathematical or system-level decision constraints remains subject to OI-01 — AI Methodology Alignment.

Likewise, formal diagnostic outcome mapping for all-correct response patterns remains unresolved under OI-03 and is not established by this coverage definition.

Detailed diagnostic reasoning procedures, evidence evaluation rules, response handling, and outcome routing are specified separately in the applicable product and technical documentation.

4.3 In-Scope Product Capabilities

The ReThink MVP includes five capability groups required to deliver its bounded, evidence-grounded conceptual recovery experience.

These capabilities establish supported product functions within the approved mathematical scope. Relevant result communication and continuation responsibilities are assigned to the capabilities that generate or use those outcomes.

Diagnostic outcomes are owned by C3, guided learning participation by C4, and validation-based learner outcomes by C5. C2 reflects available evidence and results without independently generating diagnostic or learning-state conclusions.

The capability groups do not prescribe detailed interface structures or technical implementation.

C1 — Learning Entry & Topic Discovery

ReThink provides a catalog-first learning entry through which students can explore mathematics subject areas and topics.

The MVP allows students to select Sistem Persamaan Linear as the supported learning topic and initiate the corresponding diagnostic assessment.

The product must distinguish supported topics from content represented for orientation but not yet providing active diagnosis or recovery capabilities.

C2 — Hybrid Learning Map

ReThink provides a bounded Hybrid Learning Map as a cross-cutting capability supporting three complementary purposes:

- **Explore:** Orienting students to mathematics topics and curated conceptual relationships before diagnosis.
- **Focus:** Communicating the relevant conceptual focus when supported by diagnostic evidence.
- **Progress:** Reflecting available blocker-level learner evidence and applicable learning outcomes following validation.

The Learning Map must distinguish represented concepts from actively assessed concepts and must not assign unsupported understanding states.

Concept representation and relationship visualization do not imply that broader prerequisite tracing, diagnosis, or recovery capabilities are available.

The Learning Map reflects diagnostic and validation results generated by their respective capabilities rather than independently making those decisions.

C3 — Targeted Diagnostic Assessment & Outcomes

ReThink provides targeted diagnostic assessment within the mathematical and evidence coverage established in Section 4.2.

The MVP supports three core diagnostic questions and a maximum of one conditional targeted probe, with assessment evidence directed toward the supported Distributive Property conceptual blocker.

Diagnostic evidence interpretation is AI-assisted and operates within bounded conceptual coverage and applicable mathematical and system-level constraints. The final allocation of interpretive and decision authority remains subject to OI-01 — AI Methodology Alignment.

The product must generate and communicate the applicable Confirmed, Rejected, or Uncertain diagnostic outcome, including its meaning and relevant limitations.

Confirmed outcomes support recommendation of the corresponding targeted conceptual recovery pathway.

Non-confirmed outcomes must not be presented as evidence of an established blocker. The product provides appropriate available continuation options, including optional concept review or return to topic exploration, consistent with applicable product decisions.

Formal outcome mapping for all-correct diagnostic responses remains unresolved under OI-03.

C4 — Targeted Guided Learning

ReThink provides one curated conceptual recovery package for Distributive Property.

The supported intervention follows a structured sequence:

Visual Explanation → Worked Example → Partial Guided Practice.

The sequence is designed to develop conceptual understanding and active mathematical reasoning while gradually reducing instructional assistance.

The same supported conceptual learning content may also be accessed through Optional Concept Review for non-confirmed diagnostic outcomes.

Participation in optional review does not retroactively establish a confirmed conceptual blocker.

Completion of guided learning does not automatically establish demonstrated conceptual understanding.

The continuation and learner-state implications of optional concept review remain subject to OI-02 — Optional Review Learner-State Behavior.

C5 — Knowledge Validation & Learning Outcomes

Knowledge validation is part of the supported confirmed-blocker recovery pathway.

It consists of Independent Retry and, when independent application is successful, a Transfer Problem examining the same underlying mathematical structure through a different representation.

The supported learning outcomes are:

- **Understood:** Applicable independent application and transfer conditions are satisfied.
- **Procedural Only:** Independent application is demonstrated, but the transfer requirement is not satisfied.
- **Needs Reinforcement:** Independent application has not been demonstrated under the applicable validation conditions.

The product must communicate the relevant learning result and its meaning within the supported conceptual and assessment scope.

Where additional reinforcement is indicated, continuation must remain consistent with the supported recovery process without requiring a new full diagnosis or adaptive prerequisite tracing.

Knowledge validation outcomes must not establish comprehensive mastery of Sistem Persamaan Linear or understanding of unassessed concepts.

The applicability of Knowledge Validation following Optional Concept Review and any resulting learner-state updates remain subject to OI-02.

Together, C1–C5 define the active functional scope of ReThink's initial MVP. Their responsibilities include appropriate result communication and continuation without introducing a separate results capability or expanding mathematical coverage.

4.4 Exclusions, Future-Readiness & Scope Dependencies

ReThink's initial MVP maintains a deliberately bounded product scope while preserving design constraints that support future extension of its conceptual recovery model.

The following exclusions, product-level foundations, and outstanding dependencies distinguish capabilities not included in the MVP, extensibility considerations within the approved scope, and decisions necessary to finalize specific product behaviors.

4.4.1 Explicit MVP Exclusions

The following capability groups are excluded from ReThink's initial MVP.

E1 — Expanded Mathematical Coverage

The MVP does not provide comprehensive mathematics curriculum instruction, a complete Sistem Persamaan Linear learning course, or active diagnostic and recovery pathways for multiple conceptual blockers.

Its complete supported conceptual recovery coverage remains limited to Distributive Property within the selected mathematical context.

E2 — Advanced Diagnostic Intelligence

The MVP does not include full computerized adaptive testing, probabilistic mastery estimation, unrestricted misconception diagnosis, or automatic confirmation of conceptual blockers outside its supported diagnostic knowledge.

AI-assisted evidence interpretation remains bounded by supported concept coverage and applicable mathematical and system-level constraints.

E3 — Adaptive Learning Orchestration

The MVP does not include dynamic scaffold selection, competency-specific recovery branching, or multi-level prerequisite tracing across the concept relationship structure.

Guided Learning follows the approved curated sequence rather than dynamically generating or selecting instructional pathways.

E4 — Expanded Assessment & Monitoring

The MVP does not include comprehensive topic-level mastery assessment, repeated adaptive knowledge validation, or longitudinal retention monitoring.

Knowledge Validation remains focused on independent application and transfer within the supported conceptual blocker.

E5 — General-Purpose Learning Features

The MVP does not include unrestricted conversational mathematics tutoring, unconstrained automatic lesson generation, or comprehensive learning management and progress tracking capabilities.

These exclusions do not remove the approved bounded Hybrid Learning Map, targeted guided learning, Optional Concept Review, or blocker-level knowledge validation from the MVP.

They also do not exclude empirical evaluation of learning outcomes within the supported mathematical scope. Supporting targeted conceptual understanding remains an intended product outcome, while claims of learning effectiveness require appropriate evidence.

4.4.2 Future-Ready Product Foundations

ReThink's MVP preserves four product-level design constraints intended to support future expansion without extending the active capabilities committed to the initial release.

These foundations apply within the approved C1–C5 capability groups. They do not constitute additional standalone features, independent product modules, or requirements to implement future adaptive functionality.

F1 — Reusable Concept Relationships

The product must preserve the distinction between mathematical topics, concepts, and their curated prerequisite relationships.

Its supported knowledge representation should allow relevant concepts and relationships to exist beyond the single actively supported blocker without automatically treating those concepts as diagnostically assessable.

This constraint does not require dynamic prerequisite traversal or a particular database or graph technology.

F2 — Explicit Concept Support Boundaries

The product must distinguish concept representation from active diagnostic, guided recovery, and knowledge validation support.

A concept may be available for orientation or relationship mapping without possessing a complete recovery pathway.

Expanding represented concept structures must not automatically activate additional diagnostic conclusions, learning interventions, or understanding states.

F3 — Separable Assessment & Recovery Definitions

The product must maintain a conceptual separation between supported mathematics concepts, their diagnostic assessment definitions, and their curated recovery and validation content.

This separation should allow future support packages to be introduced without redefining ReThink's core diagnosis–recovery–validation model.

The MVP does not require a generalized assessment authoring tool, automated curriculum generation system, or configurable multi-concept assessment platform.

F4 — Evidence-Bounded Learner State

The product must associate demonstrated learning outcomes with the applicable assessed concept and supporting assessment evidence.

A student's learning state must not be inferred solely from curriculum relationships, participation in learning activities, or the existence of a concept within the Learning Map.

This constraint does not require probabilistic mastery estimation, competency-specific learner states, or longitudinal learning intelligence.

Compliance with F1–F4 is assessed through the design and behavior of the applicable C1–C5 capabilities, rather than through independent feature deliverables or additional recovery pathways.

These foundations establish product-level extensibility and evidence-integrity constraints within the approved MVP scope. They must not be interpreted as evidence that future adaptive or multi-concept functionality has already been implemented.

4.4.3 Scope Dependencies & Outstanding Decisions

Several operational product decisions remain unresolved and may affect detailed functional requirements, supported continuation behavior, and acceptance criteria.

These decisions do not alter approved mathematical coverage or introduce additional MVP capabilities.

OI-01 — AI Methodology Alignment

The final allocation of responsibilities between AI-led evidence interpretation, mathematical verification, and system-level diagnostic decision constraints remains unresolved.

This dependency primarily affects C3 — Targeted Diagnostic Assessment & Outcomes.

The decision must preserve evidence-grounded interpretation, bounded diagnostic coverage, and safeguards against unsupported blocker conclusions.

OI-02 — Optional Review Learner-State Behavior

The availability of subsequent knowledge validation following Optional Concept Review, and the resulting learner-state implications, remain unresolved.

This dependency affects the relationship between C3 — Targeted Diagnostic Assessment & Outcomes, C4 — Targeted Guided Learning, and C5 — Knowledge Validation & Learning Outcomes.

Optional review remains an in-scope continuation using supported conceptual learning content without implying a previously confirmed blocker.

OI-03 — All-Correct Diagnostic Outcome Mapping

The formal diagnostic outcome and associated communication for all-correct response patterns remain unresolved.

This dependency affects C3 — Targeted Diagnostic Assessment & Outcomes.

Correct diagnostic performance must not automatically establish a comprehensive understanding of the selected mathematical topic.

Outstanding decisions affect only the specific behaviors and acceptance criteria dependent on them. They do not suspend or invalidate unrelated approved MVP requirements.

These decisions must be resolved before their affected expected behaviors and acceptance criteria can be finalized. Complete functional verification remains contingent on resolving applicable required scenarios.

Research assumptions A1–A6 remain subject to further empirical evaluation. Preliminary user research provides exploratory product context but does not independently establish diagnostic effectiveness, user acceptance, or learning improvement.

Any proposed expansion of supported mathematical coverage, product capabilities, or excluded functionality requires an explicit product scope decision rather than being treated as an automatic consequence of future-ready foundations.

Together, these boundaries preserve the integrity of ReThink's initial MVP while distinguishing active capabilities, controlled extensibility, unresolved operational requirements, and future development opportunities.

ReThink Summary

PRODUCT REQUIREMENTS DOCUMENT

Inventaris Persyaratan

Kelompok	Rentang ID	Jumlah
C1 — Akses & Penemuan Topik	FR-C1-01–05	5
C2 — Hybrid Learning Map	FR-C2-01–05	5
C3 — Diagnosis & Hasil	FR-C3-01–10	10
C4 — Pembelajaran Terarah	FR-C4-01–08	8
C5 — Validasi Pengetahuan	FR-C5-01–08	8
Lintas Kapabilitas	FR-X-01–03	3
TOTAL		39

5.1 Kerangka & Konvensi Persyaratan

Setiap FR memiliki ID stabil, kewajiban yang dapat diamati, prioritas, status keputusan, sumber/dependency, dan dasar verifikasi. Seluruh 39 FR aktif berprioritas Must Have dan telah disetujui pada baseline v1.0. Status Approved tidak berarti implementasi atau pengujian telah selesai; dependency yang tercatat tetap terbuka.

Prioritas lain (Should Have, Could Have, Won't Have — MVP) tidak mengesahkan perubahan scope. Product Design mengatur interaksi/UX, Technical Documentation mengatur implementasi, dan spesifikasi konten matematika mengatur butir soal, solusi acuan, serta rubrik. Perubahan kewajiban produk memerlukan impact review.

5.2 Akses Pembelajaran & Penemuan Topik

C1 mengelola akses, ketersediaan konten, pemilihan topik, dan kelayakan memasuki asesmen; tidak menetapkan diagnosis atau learner state.

ID & fokus	Kewajiban fungsional (ringkas)	Dasar verifikasi	Dependency
FR-C1-01 Akses katalog matematika	Menyediakan titik masuk berbasis katalog untuk menjelajahi bidang matematika.	Katalog dapat diakses; Aljabar teridentifikasi sebagai bidang.	—
FR-C1-02 Penemuan bidang dan topik	Mengorganisasi konten berdasarkan bidang dan topik tanpa mewajibkan layar terpisah per tingkat.	SPL ditemukan di bawah Aljabar.	—

FR-C1-03 Ketersediaan dukungan	Membedakan topik dengan dukungan aktif dari topik yang hanya direpresentasikan; topik tanpa dukungan dinyatakan belum tersedia.	Status ketersediaan jelas; tidak ada akses palsu ke fitur yang belum aktif.	—
FR-C1-04 Pemilihan topik didukung	Memungkinkan memilih SPL sebagai topik MVP dan mempertahankannya sebagai konteks diagnosis.	Pemilihan SPL membuka konteks diagnosis SPL yang benar.	—
FR-C1-05 Kelayakan memulai diagnosis	Hanya topik dengan dukungan diagnosis aktif yang dapat memulai asesmen.	SPL dapat memulai diagnosis; topik tanpa dukungan tidak dapat.	—

5.3 Persyaratan Hybrid Learning Map

C2 hanya merepresentasikan hubungan terkurasi dan evidence yang dihasilkan C3/C5; tidak mengambil keputusan diagnosis maupun validasi.

ID & fokus	Kewajiban fungsional (ringkas)	Dasar verifikasi	Dependency
FR-C2-01 Relasi konsep terkurasi	Menampilkan relasi matematika terkurasi antara topik terpilih dan konsep relevan.	Relasi SPL–Distributive Property benar; tanpa relasi yang belum diverifikasi.	—
FR-C2-02 Explore	Menampilkan orientasi topik dan hubungan konsep sebelum evidence diagnosis individual tersedia.	Peta pra-diagnosis tidak memberikan label kelemahan atau pemahaman personal.	—
FR-C2-03 Focus diagnosis	Merefleksikan hasil diagnosis C3 dan membedakan blocker Confirmed dari Rejected/Uncertain.	Confirmed berfokus pada Distributive Property; hasil non-confirmed tidak dilabelkan sebagai kelemahan terbukti.	OI-03
FR-C2-04 Progress tervalidasi	Menampilkan status tingkat blocker yang telah ditentukan C5 setelah Knowledge Validation yang berlaku.	Understood / Procedural Only / Needs Reinforcement sesuai keluaran C5.	OI-02
FR-C2-05 Integritas cakupan dan state	Tidak menetapkan status paham, tidak paham, atau blocker Confirmed pada konsep tanpa evidence asesmen.	Konsep tak diases tetap tanpa learner state walau tampil di peta.	—

5.4 Persyaratan Diagnosis & Hasil

C3 bertanggung jawab atas evidence asesmen, interpretasi, outcome diagnosis dan kelanjutan sesuai outcome. Batas MVP: tiga soal inti dan maksimal satu probe.

ID & fokus	Kewajiban fungsional (ringkas)	Dasar verifikasi	Dependency
FR-C3-01 Konteks diagnosis didukung	Melakukan asesmen dalam konteks SPL, hanya mengevaluasi Distributive Property sebagai blocker aktif.	Tidak ada diagnosis blocker tambahan yang belum didukung.	—
FR-C3-02 Blueprint Q1–Q3	Memberikan Q1 pilihan ganda performa SPL; Q2 respons matematika singkat Distributive Property; Q3 penjelasan konseptual.	Seluruh tiga soal inti tersedia sesuai fungsi dan formatnya.	—

FR-C3-03 Pengumpulan respons	Mengumpulkan respons menurut format soal dan mempertahankan asosiasi dengan peran soal dan konteks matematika.	Respons Q1–Q3 dan Q4 jika ada dapat ditelusuri ke soal yang tepat.	—
FR-C3-04 Evaluasi evidence matematika	Menilai outcome, transformasi, dan penjelasan yang relevan dari tiap respons; satu kesalahan tidak cukup untuk mengonfirmasi blocker.	Kategori evidence yang berlaku diperiksa tanpa inferensi dari satu error.	—
FR-C3-05 Sintesis lintas soal	Mempertimbangkan evidence mendukung, bertentangan, dan tidak cukup dari semua soal yang diberikan sebelum menarik kesimpulan.	Kasus acuan yang kontradiktif/ambigu tidak dipaksa menjadi Confirmed.	OI-01
FR-C3-06 Probe Q4 kondisional	Memberikan paling banyak satu probe ketika ada kekosongan evidence yang relevan; bentuk Procedural Trace atau Conceptual Probe.	Total maksimal empat soal; Q4 tidak wajib dan tidak berulang.	OI-01
FR-C3-07 Hasil diagnosis berbasis evidence	Menghasilkan Confirmed, Rejected, atau Uncertain untuk blocker Distributive Property sesuai evidence dan aturan yang berlaku.	Kasus acuan memetakan outcome secara sah; evidence tidak cukup dapat menghasilkan Uncertain.	OI-01; OI-03
FR-C3-08 Interpretasi AI berbatas	Membatasi interpretasi AI pada konsep didukung, respons siswa, verifikasi matematika dan safeguard sistem.	AI tidak menetapkan blocker unsupported atau mengabaikan guardrail.	OI-01
FR-C3-09 Komunikasi hasil diagnosis	Menjelaskan outcome, cakupan konsep yang diperiksa, dan keterbatasan; tidak mengklaim mastery SPL.	Confirmed, Rejected, Uncertain dibedakan secara jelas dan tidak overclaim.	OI-03
FR-C3-10 Kelanjutan menurut outcome	Confirmed memberi jalur recovery; Rejected/Uncertain menyediakan eksplorasi topik atau Optional Concept Review tanpa klaim blocker Confirmed.	Opsi setelah diagnosis cocok dengan hasil dan konteksnya.	OI-02; OI-03

5.5 Persyaratan Guided Learning Terarah

C4 memiliki paket pembelajaran dan completion, bukan diagnosis atau penetapan learner state. Paket yang sama dapat diakses melalui dua konteks masuk.

ID & fokus	Kewajiban fungsional (ringkas)	Dasar verifikasi	Dependency
FR-C4-01 Paket pembelajaran terkurasi	Menyediakan satu paket Distributive Property: Visual Explanation, Worked Example, Partial Guided Practice.	Ketiga komponen tersedia dalam paket didukung.	OPS-04
FR-C4-02 Konteks masuk pembelajaran	Membedakan Confirmed Recovery dari Optional Concept Review; review tidak mengubah diagnosis awal menjadi Confirmed.	Asal jalur dan outcome diagnostik tetap terlacak.	OI-02
FR-C4-03 Visual Explanation	Menyajikan ilustrasi konseptual yang sah untuk hubungan Distributive Property, bukan hanya rumus.	Representasi visual menjelaskan hubungan matematika dengan benar.	OPS-04

FR-C4-04 Worked Example	Menyajikan contoh terkurasi dengan transformasi benar dan alasan atas langkah penting.	Solusi sah dan alasan transformasi tersedia.	OPS-04
FR-C4-05 Partial Guided Practice aktif	Meminta siswa mengisi langkah matematika atau reasoning dalam solusi yang sebagian dibimbing.	Minimal satu respons matematika aktif diperlukan; melihat contoh saja tidak cukup.	OPS-01; OPS-04
FR-C4-06 Urutan tetap	Mengikuti Visual Explanation → Worked Example → Partial Guided Practice tanpa pemilihan scaffold adaptif.	Urutan selalu sesuai rancangan pada jalur didukung.	—
FR-C4-07 Completion & handoff	Membedakan pembelajaran selesai vs belum, mempertahankan konsep dan konteks asal untuk handoff; recovery Confirmed yang selesai dapat lanjut ke validasi.	Handoff mengikuti kriteria completion dan konteks asal.	OPS-01; OPS-02; OI-02
FR-C4-08 Batas klaim hasil belajar	Partisipasi atau completion pembelajaran tidak otomatis membuktikan pemahaman tervalidasi.	Tanpa validasi berlaku, tidak muncul tiga learner state C5.	—

5.6 Persyaratan Validasi Pengetahuan & Hasil Belajar

C5 mengevaluasi Independent Retry dan Transfer Problem, lalu menetapkan learner state tingkat blocker. Tidak melakukan re-diagnosis.

ID & fokus	Kewajiban fungsional (ringkas)	Dasar verifikasi	Dependency
FR-C5-01 Kelayakan validasi	Knowledge Validation tersedia setelah Confirmed Recovery selesai dan terkait Distributive Property.	Tidak terbuka hanya karena memilih topik atau belum selesai belajar.	OI-02
FR-C5-02 Independent Retry	Memberikan soal Distributive Property tanpa scaffolding; struktur matematika sejenis namun tidak menyalin contoh.	Siswa bekerja tanpa petunjuk/solusi yang disiapkan.	OPS-04
FR-C5-03 Transfer Problem kondisional	Hanya setelah retry berhasil; konteks atau representasi harus berbeda bermakna, bukan sekadar angka pengganti.	Retry gagal tidak membuka transfer; retry berhasil membuka soal transfer.	OPS-04
FR-C5-04 Evaluasi respons validasi	Menilai setiap respons retry/transfer terhadap kriteria matematika terkurasi dan menghasilkan hasil item-level.	Hasil item valid tersedia untuk keputusan state; respons ambigu tidak otomatis dianggap salah konsep.	OPS-03; OPS-04
FR-C5-05 Penetapan learner state	Retry gagal → Needs Reinforcement (tanpa transfer); retry lulus + transfer gagal → Procedural Only; keduanya lulus → Understood.	Ketiga kombinasi sah menghasilkan state tepat; transfer tidak diberikan saat retry gagal.	OI-02; OPS-03
FR-C5-06 Komunikasi learner state	Menyampaikan state validasi dan artinya hanya untuk Distributive Property yang dinilai.	Ketiga hasil dapat dibedakan dan tidak menyatakan mastery SPL.	—
FR-C5-07 Kelanjutan menurut learner state	Needs Reinforcement kembali ke paket Guided Learning yang sama; tanpa re-diagnosis atau paket adaptif baru.	Arah kelanjutan sesuai state tanpa memulai proses tambahan yang excluded.	—

FR-C5-08 Batas evidence validasi	Tidak membuat diagnosis blocker baru atau menelusuri prasyarat lain dari kesalahan validasi.	Validasi hanya memberi hasil dalam cakupan blocker yang diuji.	—

Matriks keputusan C5 (berlaku setelah item-level evaluation yang sah):

Independent Retry	Transfer Problem	Learner state
Gagal	Tidak diberikan	Needs Reinforcement
Lulus	Gagal	Procedural Only
Lulus	Lulus	Understood

5.7 Persyaratan Lintas Kapabilitas

FR-X mengikat konteks, transisi, dan komunikasi hasil antarkapabilitas, tanpa menjadi capability keenam.

ID & fokus	Kewajiban fungsional (ringkas)	Dasar verifikasi	Dependency
FR-X-01 Kontinuitas konteks	Mempertahankan topik SPL, konsep Distributive Property, serta konteks masuk saat berpindah C3–C5.	Konteks tidak tertukar; Confirmed Recovery tetap berbeda dari Optional Review.	OPS-05
FR-X-02 Transisi berbasis outcome	Memastikan transisi C3–C5 mematuhi outcome diagnosis, konteks masuk, dan syarat completion pembelajaran.	Rejected/Uncertain tidak otomatis boleh validasi atau dianggap recovery Confirmed.	OI-02; OI-03; OPS-05
FR-X-03 Konsistensi peta & hasil	Menjaga makna hasil diagnosis, learner state tervalidasi, dan Hybrid Learning Map tetap sama untuk evidence siswa yang sama.	Peta tidak menampilkan Understood hanya karena diagnosis Confirmed atau belajar selesai.	OI-02; OI-03; OPS-05

Prinsip integritas & cakupan verifikasi

IP-01 — Kepemilikan evidence dan learner state: C3 menentukan hasil diagnosis; C4 memiliki partisipasi/completion; C5 menetapkan learner state berdasarkan validasi; C2 merepresentasikan evidence tanpa keputusan independen. Diagnosis Confirmed ataupun completion tidak dengan sendirinya memberikan state C5.

IP-02 — Batas dukungan eksplisit: kehadiran topik pada katalog atau peta relasi tidak membuktikan aktifnya kemampuan diagnosis, recovery, atau validasi. Konsep tanpa asesmen tidak menerima label learner state.

V-X-01–V-X-08 merupakan kategori verifikasi integrasi awal, bukan FR baru atau test script lengkap. Nama kategori sumber dipertahankan sebagai label referensi.

ID	Skenario referensi	Fokus verifikasi sumber
V-X-01	Penyelesaian recovery Confirmed	Masuk dari SPL, konfirmasi blocker, pembelajaran terkurasi, validasi yang berlaku, hasil tingkat blocker yang benar

V-X-02	Kelanjutan diagnosis non-confirmed	Rejected dan Uncertain tidak menjadi Confirmed; hanya kelanjutan yang sesuai
V-X-03	Independent Retry gagal	Needs Reinforcement, tanpa Transfer Problem, kembali ke pembelajaran yang didukung
V-X-04	Transfer Problem gagal	Procedural Only jika retry berhasil dan transfer gagal
V-X-05	Validasi berhasil	Understood hanya jika retry dan transfer berhasil
V-X-06	Batas konsep tanpa dukungan	Tidak mengaktifkan diagnosis, recovery, atau validasi untuk konsep unsupported
V-X-07	Pola semua jawaban diagnosis benar	Outcome dan kelanjutannya masih menunggu OI-03
V-X-08	Kelanjutan setelah Optional Review	Kelayakan validasi dan dampak learner state masih menunggu OI-02

Register keputusan dan spesifikasi terbuka

ID	Status	Pokok keputusan / spesifikasi
OI-01	Open	Pembagian wewenang interpretasi AI, verifikasi matematika, dan batas sistem.
OI-02	Open	Kelayakan validasi dan learner state setelah Optional Concept Review.
OI-03	Open	Pemetaan outcome ketika seluruh jawaban diagnosis benar.
OPS-01	Pending	Mekanisme umpan balik Guided Practice.
OPS-02	Pending	Kriteria completion Guided Learning.
OPS-03	Pending	Penanganan respons validasi kosong, tidak lengkap, tak terbaca, atau ekuivalen secara matematika.
OPS-04	Pending	Butir matematika terkurasi, solusi acuan, dan rubrik.
OPS-05	Pending	Desain operasional transisi dan state lintas kapabilitas.

5.1 Requirements Framework & Conventions

This section establishes the conventions used to define, classify, trace, and verify ReThink's product requirements.

Product requirements translate the approved MVP capabilities and product decisions into observable product behaviors without prescribing detailed user interface design or technical implementation.

Requirements must remain consistent with the approved product scope, success boundaries, and outstanding decisions established in Sections 01–04.

5.1.1 Requirement Structure & Identification

Each functional requirement is documented using six attributes:

- **Requirement ID:** A stable identifier for reference, traceability, and revision tracking.
- **Requirement Statement:** A specific and observable product behavior or obligation.
- **Priority:** The importance of the requirement for delivery within the approved MVP scope.
- **Decision Status:** The approval state of the specific requirement, independently of the approval state of its source.
- **Source / Dependency:** The applicable approved product capability, decision, source reference, and any unresolved dependency affecting the requirement.
- **Verification Basis:** The observable condition or supporting evidence used to determine whether the requirement has been satisfied.

Functional requirements use capability-based identifiers:

Identifier	Requirement Ownership
FR-C1-XX	Learning Entry & Topic Discovery
FR-C2-XX	Hybrid Learning Map
FR-C3-XX	Targeted Diagnostic Assessment & Outcomes
FR-C4-XX	Targeted Guided Learning
FR-C5-XX	Knowledge Validation & Learning Outcomes
FR-X-XX	Cross-Capability Requirements

FR-X requirements define shared behaviors and integrity constraints. They do not introduce a sixth MVP capability group.

Requirement identifiers remain stable after approval. Substantive modifications to approved requirements require explicit review and revision tracking.

5.1.2 Requirement Priority & Decision Status

Requirements use the following delivery priority classifications:

Must Have: Necessary to deliver an approved MVP capability, satisfy a defined functional success condition, or preserve the integrity of the supported conceptual recovery experience.

Should Have: An enhancement considered important for product quality but not necessary for minimum functional MVP success, provided it receives the applicable scope approval.

Could Have: An optional enhancement subject to explicit scope approval and available delivery capacity.

Won't Have — MVP: A capability or requirement excluded from the active MVP under the approved scope boundaries.

Priority does not authorize the omission of mandatory C1–C5 behavior or the introduction of additional capabilities without a scope decision.

Each active requirement also has a separate Decision Status:

- **Proposed:** The specific requirement awaits review and approval.
- **Approved:** The requirement statement and its defined behavioral obligations have been approved.
- **Pending Decision:** Applicable behavior cannot be finalized until an outstanding decision is resolved.

The approval status of a requirement is distinct from the maturity of its source. A requirement derived from an approved capability may remain Proposed until its wording and obligations receive explicit approval.

Likewise, a Must Have requirement may have unresolved operational details requiring additional decisions.

Within this consolidated draft, the 39 requirements carry **Approved** decision status because their individually reviewed behavioral obligations have been approved. This approval does not resolve their explicitly documented dependencies or establish that they have been implemented or verified.

Future and excluded items are controlled through scope and priority classification, not counted as active functional commitments.

5.1.3 Requirement Quality & Verification Principles

Each requirement should be:

Atomic: Express a distinct product obligation wherever practical.

Observable: Describe behavior or conditions that can be evaluated through product execution, assessment evidence, or another appropriate verification method.

Traceable: Connect to the relevant capability, approved product decision, source reference, and verification basis.

Verification Basis identifies the conditions or evidence needed to assess requirement fulfillment. It does not replace detailed test procedures, datasets, or execution scripts.

Vague qualities such as intelligent, appropriate, or adaptive must not replace observable obligations unless the applicable interpretation criteria are established.

Requirements dependent on outstanding decisions must preserve the unresolved behavior explicitly. Dependent acceptance criteria cannot be treated as final before the relevant decision is resolved.

5.1.4 Requirement Governance & Documentation Boundaries

Section 05 specifies product-level functional requirements within the approved MVP scope established in Section 04.

Product Design Documentation addresses detailed user journeys, interface interactions, screen behavior, navigation, and user-facing presentation states.

Technical Documentation addresses architecture, data structures, AI implementation, mathematical evaluation mechanisms, APIs, and other implementation details.

Mathematical content specifications establish the curated questions, examples, accepted responses, reference solutions, and assessment rubrics required for implementation and testing.

Changes to approved requirement wording that materially alter expected product behavior require explicit review. Changes that expand supported mathematical coverage, introduce capabilities, or contradict exclusions require a separate scope decision.

Outstanding decisions OI-01–OI-03 remain open. Their effects are limited to the specific requirements and verification scenarios that depend on them. Unaffected requirements may proceed independently through implementation and verification.

The conventions defined in this section apply to Sections 5.2–5.7.

5.2 Learning Entry & Topic Discovery Requirements

Learning Entry & Topic Discovery enables students to access ReThink through a catalog-first experience, explore represented mathematics subjects and topics, identify available learning support, and initiate diagnosis for the supported MVP topic.

A represented topic is one included in the catalog or conceptual structure. A supported topic has the relevant active product capabilities. An unavailable topic may be represented for orientation without having active learning or diagnostic support.

The presence of a topic in the catalog does not independently establish diagnostic, recovery, or validation coverage.

FR-C1-01 — Mathematics Catalog Access

Requirement Statement

The system must provide a catalog-first entry that enables students to access and explore represented mathematics subject areas.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved Section 4.1.2 — Supported Product Experience; C1 — Learning Entry & Topic Discovery; Catalog-First Entry Product Decision.
- **Verification Basis:** The mathematics catalog is accessible, and students can identify Algebra as a represented subject area.

FR-C1-02 — Subject & Topic Discovery

Requirement Statement

The system must organize represented mathematics content by subject and topic, allowing students to identify represented topics within the selected subject area.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved Sections 4.1.2 and 4.2.1; C1 — Learning Entry & Topic Discovery.
- **Verification Basis:** Students can identify Sistem Persamaan Linear within the Algebra subject area through the supported catalog structure.

The subject–topic hierarchy defines logical content organization and does not require separate screens for every level.

FR-C1-03 — Supported Content Availability

Requirement Statement

The system must distinguish topics with active MVP support from represented topics without active support and communicate unsupported topics as unavailable.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved C1; F2 — Explicit Concept Support Boundaries; Catalog Availability Product Decision.
- **Verification Basis:** Supported and unsupported topics have distinguishable availability states. Unsupported topics are communicated as unavailable rather than presented as accessible learning or diagnostic experiences.

Unavailable content must not be represented as locked because of the student's ability, progress, or achievement.

FR-C1-04 — Supported Topic Selection

Requirement Statement

The system must allow students to select Sistem Persamaan Linear as the supported MVP learning topic and establish the selected topic as the mathematical context for the subsequent diagnostic experience.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved Section 4.2.1; C1 — Learning Entry & Topic Discovery; C3 — Targeted Diagnostic Assessment & Outcomes.
- **Verification Basis:** Selecting Sistem Persamaan Linear establishes the intended topic context, and the subsequent supported diagnostic experience corresponds to that topic.

Topic selection does not independently establish Distributive Property as a confirmed conceptual weakness or assign a learner understanding state.

FR-C1-05 — Diagnostic Entry Eligibility

Requirement Statement

The system must permit diagnostic assessment initiation only for topics with active diagnostic support and prevent represented but unsupported topics from initiating an active diagnostic session.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved C1 and C3; F2 — Explicit Concept Support Boundaries; E1 — Expanded Mathematical Coverage.

- **Verification Basis:** Sistem Persamaan Linear permits initiation of its supported MVP diagnostic assessment. Represented topics without active diagnostic support cannot initiate unsupported assessment.

Diagnostic entry eligibility depends on available diagnostic coverage, not catalog presence alone.

C1 Capability Boundaries

C1 establishes access, content availability, topic selection, and eligible diagnostic entry. It does not evaluate mathematical responses, confirm conceptual blockers, or assign learner states.

The MVP does not require personalized topic ranking, comprehensive curriculum coverage, or separate screens for every content hierarchy level.

C3 owns diagnostic evidence interpretation and outcomes.

5.3 Hybrid Learning Map Requirements

The Hybrid Learning Map is a bounded product capability that communicates curated mathematical relationships, relevant diagnostic focus, and applicable learner evidence throughout the supported conceptual recovery experience.

Its three contexts are **Explore, Focus, and Progress**. These are contextual purposes rather than mandatory standalone screens.

C2 represents results produced by the responsible capabilities without generating independent diagnostic or validation conclusions.

FR-C2-01 — Curated Concept Relationship Representation

Requirement Statement

The system must represent curated mathematical relationships between the selected learning topic and relevant concepts within the supported knowledge structure.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved C2 — Hybrid Learning Map; Section 4.2.1; F1 — Reusable Concept Relationships.
- **Verification Basis:** The Learning Map represents the curated relationship between Sistem Persamaan Linear and Distributive Property without introducing unsupported or unverified prerequisite relationships.

Representing a concept does not establish active diagnostic coverage or prove that the student has difficulty understanding it.

FR-C2-02 — Explore Context

Requirement Statement

The system must provide an Explore context that enables students to understand the selected mathematics topic's position within the represented conceptual structure before individual diagnostic evidence is available.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved C2; Section 4.1.2; Hybrid Learning Map Product Decision.
- **Verification Basis:** Before diagnostic assessment, the map provides topic orientation and relevant curated conceptual relationships without assigning individual conceptual weaknesses or understanding states.

Explore complements rather than replaces the mathematics catalog.

FR-C2-03 — Diagnostic Focus Representation

Requirement Statement

The system must communicate the conceptual focus supported by the applicable diagnostic outcome produced by C3, while distinguishing confirmed conceptual blockers from non-confirmed outcomes.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved C2 and C3; Section 4.1.2; OI-03 — All-Correct Diagnostic Outcome Mapping.
- **Verification Basis:** A Confirmed diagnostic outcome identifies Distributive Property as the supported recovery focus. Rejected and Uncertain outcomes do not represent it as a confirmed conceptual weakness.

The map may identify the concept examined or available for Optional Concept Review without implying confirmation.

Pending Dependency — OI-03: Formal mapping and map representation of all-correct diagnostic response patterns remain unresolved. Such responses must not independently establish topic-level mastery.

FR-C2-04 — Validation-Based Progress Representation

Requirement Statement

The system must reflect applicable blocker-level learning outcomes produced by C5 following supported Knowledge Validation.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved C2 and C5; Section 4.3; F4 — Evidence-Bounded Learner State; OI-02 — Optional Review Learner-State Behavior.
- **Verification Basis:** Following applicable Knowledge Validation, the map reflects the resulting Understood, Procedural Only, or Needs Reinforcement outcome for the assessed Distributive Property concept.

The map must not assign understanding solely because Guided Learning has been completed.

Pending Dependency — OI-02: Applicability of validation and learner-state updates after Optional Concept Review remains unresolved.

FR-C2-05 — Assessment Coverage & State Integrity

Requirement Statement

The system must prevent mathematical concepts without applicable assessment evidence from being represented as understood, misunderstood, or associated with a confirmed conceptual blocker.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved C2; F2 — Explicit Concept Support Boundaries; F4 — Evidence-Bounded Learner State; Section 4.1.3.
- **Verification Basis:** Unassessed concepts do not receive diagnostic or understanding states, regardless of their map presence or mathematical relationship to supported concepts.

A curated concept relationship must not independently count as evidence of student understanding or difficulty.

C2 Capability Boundaries

C2 communicates supported concept relationships and evidence generated by other applicable capabilities.

It does not evaluate mathematical responses, confirm blockers, determine learner states, infer SPL mastery, or perform adaptive prerequisite tracing.

Cross-capability alignment between Learning Map representations and source outcomes is specified in FR-X-03.

5.4 Diagnostic Assessment & Outcomes Requirements

Targeted Diagnostic Assessment & Outcomes enables ReThink to collect and interpret mathematical evidence, investigate the supported conceptual blocker when additional evidence is required, communicate an evidence-grounded diagnostic conclusion, and provide appropriate continuation.

The supported diagnostic context is Sistem Persamaan Linear, with Distributive Property as the only active conceptual blocker.

The MVP supports three core questions and a maximum of one conditional targeted probe. AI-assisted interpretation operates within curated mathematical coverage and applicable safeguards.

FR-C3-01 — Supported Diagnostic Context

Requirement Statement

The system must conduct diagnostic assessment in the context of the selected Sistem Persamaan Linear topic and restrict active conceptual blocker evaluation to Distributive Property.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved Section 4.2.1; C3 — Targeted Diagnostic Assessment & Outcomes; F2 — Explicit Concept Support Boundaries; E1 — Expanded Mathematical Coverage.
- **Verification Basis:** A supported diagnostic session evaluates the Distributive Property blocker within the selected SPL context without generating unsupported additional blocker diagnoses.

Diagnostic coverage does not establish comprehensive SPL proficiency or assess every prerequisite of SPL.

FR-C3-02 — Core Diagnostic Question Blueprint

Requirement Statement

The system must administer the three approved core diagnostic questions according to their assessment roles and response formats:

- **Q1 — Target Performance:** Multiple-choice question within SPL.
- **Q2 — Prerequisite Application:** Short mathematical response examining Distributive Property.
- **Q3 — Conceptual Reasoning:** Short explanation examining relevant mathematical reasoning.
- **Priority:** Must Have

- **Decision Status:** Approved
- **Source / Dependency:** Approved Section 4.2.2; C3 — Targeted Diagnostic Assessment & Outcomes.
- **Verification Basis:** Every supported core diagnostic assessment includes Q1, Q2, and Q3 with their corresponding roles and response formats.

The three question roles provide complementary evidence. No single question is designated as independently sufficient to confirm the supported blocker.

FR-C3-03 — Student Response Collection

Requirement Statement

The system must collect responses to administered diagnostic questions using their supported formats and preserve their association with the corresponding question role and mathematical context for diagnostic evaluation.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved C3; Section 4.2.2; F3 — Separable Assessment & Recovery Definitions.
- **Verification Basis:** Submitted responses to Q1, Q2, Q3, and any administered Q4 remain attributable to their correct question roles and relevant mathematical context.

This requirement does not prescribe database structure, mathematical normalization technology, or assessment submission interface design.

FR-C3-04 — Mathematical Evidence Evaluation

Requirement Statement

The system must evaluate individual diagnostic responses for applicable outcome, transformation, and explanatory evidence relevant to the supported Distributive Property blocker.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved Section 4.2.2; C3; Evidence Before Inference product principle.
- **Verification Basis:** Individual responses are evaluated using their applicable evidence categories. An incorrect answer or isolated error observation is not independently treated as sufficient confirmation of a conceptual blocker.

Outcome evidence concerns answer correctness. Transformation evidence concerns observable mathematical operations. Explanatory evidence concerns student-provided mathematical reasoning.

Not every evidence category must be available for every question.

This requirement concerns the evaluation of observed responses, not cross-question synthesis or final diagnostic authority.

FR-C3-05 — Cross-Question Evidence Synthesis

Requirement Statement

The system must consider relevant supporting, contradictory, and insufficient evidence across administered diagnostic questions when evaluating whether the supported blocker can be confirmed, rejected, or remains uncertain.

- **Priority:** Must Have
- **Decision Status:** Approved — defined behavioral obligation only.
- **Source / Dependency:** Approved Section 4.2.3; C3; OI-01 — AI Methodology Alignment.
- **Verification Basis:** Reviewed diagnostic cases demonstrate that conflicting or insufficient observations are considered rather than automatically converting an isolated error into blocker confirmation.

The process distinguishes observations from diagnostic inference and does not automatically treat multiple observations from one underlying response as independent evidence.

Pending Dependency — OI-01: Final allocation of interpretive and diagnostic decision responsibilities among AI, mathematical verification, and system-level constraints remains unresolved.

FR-C3-06 — Conditional Targeted Diagnostic Probe

Requirement Statement

The system must support a maximum of one conditional targeted diagnostic probe (Q4) when an identified information gap requires additional relevant evidence.

The probe must use one approved form:

- **Procedural Trace:** Examines procedural steps or mathematical transformations.
- **Conceptual Probe:** Examines reasoning or conceptual understanding relevant to the suspected blocker.
- **Priority:** Must Have
- **Decision Status:** Approved — defined behavioral obligation only.
- **Source / Dependency:** Approved Section 4.2.2; C3; OI-01 — AI Methodology Alignment.
- **Verification Basis:** Supported sessions administer Q1–Q3 and no more than one Q4. Any administered Q4 uses one of the approved forms and addresses a relevant information gap.

Q4 is conditional, not mandatory. The maximum assessment length is four questions, with no repeated probing cycles.

Pending Dependency — OI-01: Operational responsibilities for information-gap interpretation and probe selection remain to be finalized.

FR-C3-07 — Evidence-Grounded Diagnostic Outcome

Requirement Statement

The system must produce an applicable diagnostic conclusion about the supported Distributive Property blocker using the approved categories:

- **Confirmed:** Evidence sufficiently supports the examined blocker under the applicable decision rules.
- **Rejected:** Evidence sufficiently does not support or contradicts the examined blocker under the applicable decision rules.
- **Uncertain:** Evidence does not justify reliable confirmation or rejection.
- **Priority:** Must Have
- **Decision Status:** Approved — defined behavioral obligation only.
- **Source / Dependency:** Approved Section 4.2.3; C3; OI-01 — AI Methodology Alignment; OI-03 — All-Correct Diagnostic Outcome Mapping.
- **Verification Basis:** Reviewed supported diagnostic cases produce conclusions consistent with available evidence and applicable decision rules. Insufficient evidence can result in Uncertain instead of unsupported confirmation.

Incorrect performance must not automatically establish a blocker. Correct performance or a non-confirmed outcome must not establish comprehensive mathematical mastery or prove the absence of other learning difficulties.

Pending Dependency — OI-01: Final operational decision authority and supporting mechanisms.

Pending Dependency — OI-03: Formal diagnostic outcome mapping for all-correct response patterns. This requirement does not assign such patterns to Confirmed, Rejected, or Uncertain.

FR-C3-08 — Bounded AI-Assisted Interpretation

Requirement Statement

The system must constrain AI-assisted diagnostic interpretation to the supported mathematical concept, available student response evidence, and applicable mathematical and system-level safeguards.

- **Priority:** Must Have
- **Decision Status:** Approved — defined behavioral obligation only.

- **Source / Dependency:** Approved C3; Bounded AI Autonomy product principle; E2 — Advanced Diagnostic Intelligence; OI-01 — AI Methodology Alignment.
- **Verification Basis:** Diagnostic interpretation does not independently activate unsupported blockers, disregard applicable mathematical constraints, or present unsupported inferences as established diagnoses.

AI-assisted interpretation may contribute to analyzing student reasoning and synthesizing evidence within the approved boundaries.

The requirement does not authorize unrestricted diagnostic autonomy, arbitrary misconception detection, or probabilistic mastery estimation.

Pending Dependency — OI-01: Detailed decision authority and operational acceptance criteria for AI, mathematical verification, and system constraints remain unresolved.

FR-C3-09 — Diagnostic Result Communication

Requirement Statement

The system must communicate the applicable diagnostic outcome, examined conceptual scope, and relevant limitations while distinguishing confirmed blockers from rejected or uncertain interpretations.

- **Priority:** Must Have
- **Decision Status:** Approved — defined behavioral obligation only.
- **Source / Dependency:** Approved C3; Sections 4.1.2 and 4.2.3; OI-03 — All-Correct Diagnostic Outcome Mapping.
- **Verification Basis:** Supported Confirmed, Rejected, and Uncertain results are distinguishable, identify the examined scope, and do not make unsupported claims about SPL mastery, comprehensive diagnosis, or confirmed weakness.

Communication must remain consistent with the evidence-bounded diagnostic conclusion.

Pending Dependency — OI-03: The outcome-specific message and continuation meaning for all-correct responses remain unresolved. No topic-level mastery conclusion may be inferred from the all-correct pattern alone.

FR-C3-10 — Outcome-Based Continuation

Requirement Statement

The system must provide an appropriate supported continuation following the applicable diagnostic outcome without presenting a non-confirmed blocker as confirmed.

- **Priority:** Must Have
- **Decision Status:** Approved — defined behavioral obligation only.

- **Source / Dependency:** Approved Section 4.1.2; C3 and C4; OI-02 — Optional Review Learner-State Behavior; OI-03 — All-Correct Diagnostic Outcome Mapping where applicable.
- **Verification Basis:** A Confirmed outcome provides access to the supported Distributive Property recovery pathway. Rejected and Uncertain outcomes provide available topic-exploration or Optional Concept Review continuation without implying confirmed-blocker status.

Optional Concept Review may use the supported curated learning package but does not retroactively confirm the blocker.

Pending Dependency — OI-02: Knowledge Validation eligibility and learner-state implications after Optional Concept Review.

Pending Dependency — OI-03: Formal outcome-based continuation for all-correct response patterns.

C3 Capability Boundaries

C3 owns diagnostic evidence collection, interpretation, supported blocker-level outcomes, result communication, and diagnostic continuation.

C3 does not assign post-intervention learner states, provide instructional content, perform unrestricted prerequisite diagnosis, or operate as a general-purpose mathematics tutor.

C4 owns guided learning and optional review content. C5 owns validation-based learner outcomes. C2 reflects applicable diagnostic evidence without generating independent conclusions.

OI-01 and OI-03 remain unresolved where they affect diagnostic decision rules; OI-02 governs the downstream implications of optional review.

5.5 Targeted Guided Learning Requirements

Targeted Guided Learning provides one curated Distributive Property intervention through conceptual explanation, demonstrated mathematical reasoning, and structured student participation.

The approved sequence is:

Visual Explanation → Worked Example → Partial Guided Practice.

The same supported package is available for confirmed-blocker recovery and Optional Concept Review. The entry context must remain distinguishable.

Instructional completion does not independently establish validated conceptual understanding.

FR-C4-01 — Supported Learning Package

Requirement Statement

The system must provide one curated conceptual learning package for Distributive Property within the approved MVP mathematical scope.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved Section 4.2.1; C4 — Targeted Guided Learning; F3 — Separable Assessment & Recovery Definitions.
- **Verification Basis:** A complete curated Distributive Property package contains the supported Visual Explanation, Worked Example, and Partial Guided Practice components.

The requirement does not establish additional recovery packages, comprehensive SPL lessons, or unrestricted content generation.

FR-C4-02 — Recovery Entry Context

Requirement Statement

The system must distinguish access to the supported learning package through Confirmed blocker recovery from access through Optional Concept Review following a non-confirmed diagnostic outcome.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved Section 4.1.2; C3 and C4; OI-02 — Optional Review Learner-State Behavior.
- **Verification Basis:** Confirmed recovery retains the confirmed Distributive Property context. Optional Concept Review retains the non-confirmed diagnostic context without converting the originating diagnosis to Confirmed.

Both entry contexts may use the same curated package.

Pending Dependency — OI-02: Downstream Knowledge Validation eligibility and learner-state implications after Optional Concept Review remain unresolved.

FR-C4-03 — Visual Explanation

Requirement Statement

The system must provide a curated Visual Explanation that illustrates the conceptual meaning of Distributive Property and the relevant relationships between its mathematical components.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved C4; Section 4.3.
- **Verification Basis:** The explanation uses a mathematically valid conceptual representation to show how the distributive relationship applies, rather than presenting only a symbolic formula or memorized rule.

Specific presentation formats and example content belong in the supporting content and Product Design specifications.

FR-C4-04 — Worked Example

Requirement Statement

The system must provide a curated Worked Example demonstrating the application of Distributive Property through mathematically valid transformations accompanied by explanations of meaningful solution steps.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved C4; Section 4.3.
- **Verification Basis:** The example presents a mathematically valid procedure and explains relevant transformations, linking conceptual representation with procedural application.

The example must not be limited to displaying the final answer without the reasoning needed to understand the relevant steps.

FR-C4-05 — Active Partial Guided Practice

Requirement Statement

The system must provide curated Partial Guided Practice that requires the student to supply mathematical responses for selected incomplete steps or reasoning within a partially supported solution.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved C4; Section 4.3; Active Reasoning Over Answer Delivery product principle.
- **Verification Basis:** The student must contribute at least one mathematical response to a supported incomplete solution step or reasoning task. Merely viewing a completed worked solution does not satisfy this requirement.

The practice retains structured instructional assistance while requiring active participation.

Response input, mathematical feedback, and correction mechanics belong in the applicable operational and Product Design specifications.

FR-C4-06 — Fixed Learning Sequence

Requirement Statement

The system must deliver the supported Guided Learning package in the approved instructional order: Visual Explanation → Worked Example → Partial Guided Practice, without requiring adaptive scaffold selection or dynamic instructional branching.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved C4; Section 4.3; E3 — Adaptive Learning Orchestration.
- **Verification Basis:** The supported sequence presents the three instructional stages in the approved progression without dynamic recovery-path selection or learner-state-driven scaffold changes.

This does not prohibit ordinary navigation for reviewing previously presented content.

FR-C4-07 — Learning Completion & Applicable Handoff

Requirement Statement

The system must distinguish completion of the supported Guided Learning sequence from incomplete participation and make the applicable concept, learning-entry context, and completion information available for subsequent product behavior.

- **Priority:** Must Have
- **Decision Status:** Approved — defined behavioral obligation only.
- **Source / Dependency:** Approved C4 and C5; Section 4.3; FR-C4-05; OI-02 — Optional Review Learner-State Behavior.
- **Verification Basis:** Completion is established according to the applicable instructional completion requirements, remains associated with Distributive Property and the original entry context, and supports Knowledge Validation continuation following completed confirmed-blocker recovery.

Completion information is not independent evidence of validated understanding.

Pending Operational Specification: Detailed Guided Practice feedback and completion rules, including treatment of incorrect attempts, remain to be finalized.

Pending Dependency — OI-02: Optional Concept Review completion does not, by itself, establish validation eligibility. Whether it may qualify under additional conditions remains unresolved.

FR-C4-08 — Learning Outcome Claim Boundary

Requirement Statement

The system must not interpret participation in or completion of Guided Learning as independent evidence that the student has demonstrated validated conceptual understanding.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved C4 and C5; F4 — Evidence-Bounded Learner State; Validation Before Understanding Claims principle.
- **Verification Basis:** Completing the instructional sequence without applicable Knowledge Validation does not independently assign Understood, Procedural Only, or Needs Reinforcement.

This boundary applies to confirmed recovery and Optional Concept Review.

C4 Capability Boundaries

C4 owns the supported curated learning package, instructional sequence, participation, completion evidence, and distinction between confirmed recovery and non-confirmatory review.

C4 does not independently diagnose blockers, conduct additional diagnostic assessment, assign learner outcomes, or establish SPL topic mastery.

The MVP does not require adaptive scaffolding, additional recovery packages, dynamic lesson generation, or unrestricted mathematics tutoring.

Completion criteria and feedback mechanics remain pending operational specification. Optional Review validation eligibility remains OI-02.

5.6 Knowledge Validation & Learning Outcomes Requirements

Knowledge Validation assesses whether students can independently apply Distributive Property and recognize the same underlying mathematical structure through a different representation or context.

It uses a bounded sequence:

Independent Retry → Conditional Transfer Problem → Blocker-Level Learner State.

Knowledge Validation is included in the confirmed-blocker recovery pathway. Applicability after Optional Concept Review remains unresolved under OI-02.

FR-C5-01 — Validation Context & Eligibility

Requirement Statement

The system must make Knowledge Validation available following completion of the supported confirmed-blocker recovery pathway and associate validation with the applicable Distributive Property concept.

- **Priority:** Must Have
- **Decision Status:** Approved — confirmed-recovery behavior only.
- **Source / Dependency:** Approved Section 4.3; C4 and C5; FR-C4-07; OI-02 — Optional Review Learner-State Behavior.
- **Verification Basis:** Completed confirmed-blocker recovery permits the corresponding Distributive Property validation. Topic selection or incomplete instructional participation alone does not establish validation eligibility.

Pending Dependency — OI-02: Validation eligibility after Optional Concept Review and resulting learner-state applicability remain unresolved.

FR-C5-02 — Independent Retry

Requirement Statement

The system must provide a curated Independent Retry requiring the student to apply Distributive Property without instructional scaffolding, using a problem that preserves the learned mathematical structure without reproducing the previously completed example.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved C5; Section 4.3.
- **Verification Basis:** An eligible student attempts a relevant Distributive Property problem without visual instructional assistance, worked-solution guidance, or partially completed solution scaffolds. The item preserves the applicable mathematical structure while differing from the earlier example.

Successful retry provides evidence of independent application within the tested task, not independently demonstrated transfer.

FR-C5-03 — Conditional Transfer Problem

Requirement Statement

The system must administer a curated Transfer Problem only after successful Independent Retry, using a meaningfully different representation or context while preserving the mathematical concept being validated.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved C5; Section 4.3; E4 — Expanded Assessment & Monitoring.
- **Verification Basis:** Successful Independent Retry permits Transfer Problem; unsuccessful retry prevents its administration. The transfer task differs meaningfully from the retry and does not simply substitute numbers.

Repeated adaptive transfer-question loops are outside MVP scope.

FR-C5-04 — Validation Response Evaluation

Requirement Statement

The system must evaluate each administered Independent Retry and Transfer Problem response against applicable curated mathematical assessment criteria and produce the corresponding item-level evaluation result for learner-state determination.

- **Priority:** Must Have
- **Decision Status:** Approved — defined behavioral obligation only.
- **Source / Dependency:** Approved C5; Section 4.3; bounded Knowledge Validation approach.
- **Verification Basis:** Supported, interpretable responses are assessed consistently against their mathematical criteria, and resulting item-level success or failure is available for learner-state determination.

This requirement evaluates responses; it does not assign learner states.

Pending Operational Specification — Validation Response Handling: Accepted mathematically equivalent expressions, missing or incomplete responses, and uninterpretable submissions require explicit evaluation rules.

Responses that cannot be evaluated must not automatically be treated as evidence of conceptual misunderstanding without an applicable approved rule.

FR-C5-05 — Evidence-Based Learner State Assignment

Requirement Statement

The system must assign the applicable blocker-level learner state using the following approved Knowledge Validation rules:

Independent Retry	Transfer Problem	Learner State
Successful	Successful	Understood

Successful	Unsuccessful	Procedural Only
Unsuccessful	Not administered	Needs Reinforcement

- **Priority:** Must Have
- **Decision Status:** Approved — supported validation path.
- **Source / Dependency:** Approved C5; Section 4.3; F4 — Evidence-Bounded Learner State; FR-C5-04; OI-02 where applicable.
- **Verification Basis:** Each approved combination of validly evaluated item results produces its corresponding learner state. Transfer is not administered after unsuccessful Independent Retry.

The mapping applies after the relevant validation-item outcomes have been validly established. Missing or uninterpretable response handling remains subject to FR-C5-04's pending operational criteria.

Learner states describe supported blocker-level performance and do not establish mastery of SPL.

Pending Dependency — OI-02: Whether this mapping is applicable following Optional Concept Review remains unresolved.

FR-C5-06 — Learning Outcome Communication

Requirement Statement

The system must communicate the applicable validation-based learner state and its meaning in relation to demonstrated performance on Distributive Property without implying broader mastery than the evidence supports.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved C5; C2 — Hybrid Learning Map; Section 4.3; F4 — Evidence-Bounded Learner State.
- **Verification Basis:** Understood, Procedural Only, and Needs Reinforcement are distinguishable and explained consistently with applicable validation results.

The interpretations are limited as follows:

- **Understood:** Successful independent application and transfer within supported tasks.
- **Procedural Only:** Successful independent retry, without successful transfer in the administered task.
- **Needs Reinforcement:** Unsuccessful independent retry, with transfer not assessed.

Outcomes must not be communicated as comprehensive SPL mastery, permanent learning classifications, or proof of long-term retention.

FR-C5-07 — Outcome-Based Continuation

Requirement Statement

The system must provide continuation consistent with the applicable learner state, including access to the existing Distributive Property Guided Learning package when the outcome is Needs Reinforcement.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved C5 and C4; Section 4.3; E3 — Adaptive Learning Orchestration; E4 — Expanded Assessment & Monitoring.
- **Verification Basis:** Needs Reinforcement provides a route to revisit the supported Guided Learning package. Procedural Only and Understood can be communicated as completed validation outcomes without automatically initiating new diagnoses, adaptive transfer loops, or additional recovery packages.

The requirement does not introduce new remediation workflows. Detailed navigation choices for Procedural Only and Understood may be specified separately within approved product scope.

FR-C5-08 — Validation Scope & Evidence Integrity

Requirement Statement

The system must restrict Knowledge Validation conclusions to the assessed Distributive Property concept and must not use unsuccessful validation responses to initiate new conceptual blocker diagnoses, comprehensive re-diagnosis, or unsupported prerequisite tracing.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved C5; F2 — Explicit Concept Support Boundaries; F4 — Evidence-Bounded Learner State; E2 and E4.
- **Verification Basis:** Validation produces only applicable supported concept-level outcomes. Failure does not activate a new blocker hypothesis, launch C3 re-diagnosis, or establish an unsupported learner state.

Validation tests whether the supported concept can be applied within curated tasks; it does not independently determine the cause of every unsuccessful response.

C5 Capability Boundaries

C5 owns Independent Retry, conditional Transfer Problem, response evaluation, learner-state assignment, outcome communication, and applicable validation continuation.

It does not independently diagnose the original blocker, deliver new instructional content, or establish complete topic-level SPL mastery.

The MVP excludes adaptive validation cycles, probabilistic mastery estimates, comprehensive SPL assessment, competency-level learner states, and longitudinal retention monitoring.

Functional validation outcomes remain distinct from empirical assessment of learning effectiveness under D4.

5.7 Cross-Capability Requirements & System Behaviors

Cross-Capability Requirements preserve context continuity, valid transitions, and consistent learner-evidence communication across ReThink's five approved capability groups.

They complement capability-specific requirements without introducing a sixth group or expanding mathematical coverage.

FR-X-01 — Supported Context Continuity

Requirement Statement

The system must preserve the applicable selected learning topic, supported mathematical concept, and originating learning context when transferring information between diagnostic assessment, guided learning, and knowledge validation capabilities.

- **Priority:** Must Have
- **Decision Status:** Approved
- **Source / Dependency:** Approved C1–C5; F2 and F3; FR-C1-04; FR-C3-01; FR-C4-02; FR-C4-07; FR-C5-01.
- **Verification Basis:** Supported integration scenarios retain SPL as the selected topic and Distributive Property as the relevant concept, together with the originating diagnostic and learning-entry context required by subsequent behavior.

Context continuity distinguishes confirmed recovery from Optional Concept Review where relevant.

The requirement does not prescribe storage structures, session architecture, or API contracts.

FR-X-02 — Outcome-Gated Learning Transitions

Requirement Statement

The system must ensure that transitions between diagnostic assessment, guided learning, and knowledge validation respect the applicable diagnostic outcome, learning-entry context, and supported instructional completion conditions.

- **Priority:** Must Have
- **Decision Status:** Approved — defined transition obligations only.
- **Source / Dependency:** Approved C3–C5; FR-C3-10; FR-C4-02; FR-C4-07; FR-C5-01; OI-02 and OI-03.
- **Verification Basis:** Confirmed Distributive Property recovery permits the supported learning pathway and, after applicable completion, Knowledge Validation. Rejected and Uncertain outcomes do not automatically establish confirmed recovery or validation eligibility.

Optional Concept Review may reuse the learning package without changing the original diagnostic conclusion.

Pending Dependency — OI-02: Validation eligibility and learning-state implications following Optional Concept Review.

Pending Dependency — OI-03: Outcome and resulting continuation rules for all-correct diagnostic response patterns.

This requirement does not decide either unresolved behavior.

FR-X-03 — Contextual Map & Result Consistency

Requirement Statement

The system must maintain consistency between the selected mathematics topic, applicable diagnostic outcomes, and available validation-based learner states as communicated through diagnostic results, learning outcomes, and the Hybrid Learning Map.

- **Priority:** Must Have
- **Decision Status:** Approved — defined consistency obligations only.
- **Source / Dependency:** Approved C1, C2, C3, and C5; FR-C2-03–05; FR-C3-09; FR-C5-06; F4; OI-02 and OI-03 where applicable.
- **Verification Basis:** For the same applicable learner evidence, diagnostic result communication and the Learning Map convey compatible diagnostic meanings; validation results and the map convey compatible blocker-level learner states without independent or conflicting conclusions.

Confirmed diagnostic evidence must not appear as a validated Understood outcome before applicable Knowledge Validation.

Rejected or Uncertain results must not appear as confirmed weaknesses.

An applicable Procedural Only, Needs Reinforcement, or Understood result must not conflict with the Learning Map's representation of the same assessed concept.

Pending Dependencies: Representation of all-correct diagnostic outcomes and Optional Review follow-on states remains subject to OI-03 and OI-02 respectively.

Cross-Capability Integrity Principles

The following principles are implemented through approved capability-specific requirements and applicable cross-capability behavior. They are not additional independent FR-X requirements.

IP-01 — Evidence & Learner-State Ownership Integrity

C3 owns diagnostic conclusions; C4 owns guided learning participation and completion; C5 owns validation-based learner states. C2 communicates applicable evidence without making independent assessment decisions.

Neither confirmed diagnosis nor instructional completion is sufficient to assign Understood, Procedural Only, or Needs Reinforcement.

Implementation is governed by FR-C2-05, FR-C3-07, FR-C4-08, FR-C5-05, and applicable integration requirements.

IP-02 — Explicit Support-Boundary Enforcement

Catalog presence and curated concept relationships do not independently establish diagnostic, recovery, or validation support.

Unassessed concepts must not receive unsupported diagnostic conclusions or learner states.

Implementation is governed by FR-C1-03, FR-C1-05, FR-C2-05, FR-C3-01, FR-C5-08, and applicable integration requirements.

End-to-End Verification Coverage

The following eight categories define preliminary integration verification coverage. They are not additional functional requirements or complete test scripts.

Scenario ID	Scenario		Verification Focus
V-X-01	Confirmed Completion	Recovery	SPL entry, supported blocker confirmation, curated learning, applicable validation, correct blocker-level outcome

V-X-02	Non-Confirmed Diagnostic Continuation	Rejected and Uncertain retain non-confirmed meaning and allow only applicable continuation
V-X-03	Failed Independent Retry	Needs Reinforcement, no Transfer Problem, supported reinforcement route
V-X-04	Failed Transfer Problem	Procedural Only after successful retry and unsuccessful transfer
V-X-05	Successful Validation	Understood only after successful retry and transfer
V-X-06	Unsupported Concept Boundary	No unsupported diagnostic, recovery, or validation activation
V-X-07	All-Correct Diagnostic Pattern	Expected outcome and continuation remain pending OI-03
V-X-08	Optional Review Follow-On	Validation eligibility and learner-state implications remain pending OI-02

The detailed verification specification must additionally cover relevant negative and boundary conditions, including:

- Insufficient diagnostic evidence resulting in an appropriate non-confirmatory decision.
- Unsupported topic entry being rejected.
- Guided Learning completion not independently assigning Understood.
- Uninterpretable or incomplete validation submissions following their approved response-handling rules.
- Consistency of diagnosis, instructional participation, and learning-state representations across the core experience.

These are verification refinements, not new functional requirements. Relevant expected behaviors that remain dependent on OI-01–OI-03 or operational specifications cannot be finalized prematurely.

The integration verification suite supports Section 03's functional success conditions:

- **SC1 — Complete Supported Recovery Path**
- **SC2 — Evidence-Grounded Decision Integrity**
- **SC3 — Appropriate Outcome-Based Continuation**

SC1–SC3 require successful verification of all applicable required scenarios; completion of one happy-path demonstration is insufficient.

Functional verification, user-experience usability, and empirical learning effectiveness are assessed separately.

Cross-Capability Boundaries

FR-X-01–03 establish shared context, transition, and representation integrity without introducing independent assessment engines or additional product capabilities.

The MVP remains limited to SPL as the supported selectable topic and Distributive Property as the single fully supported blocker.

OI-01–OI-03 remain open. Their unresolved behavior affects only the requirements and verification scenarios that depend upon them.

Detailed interface transitions, data contracts, failure handling, and implementation architecture remain responsibilities of the appropriate Product Design and Technical Documentation.

7.1 Conceptual Architecture Boundaries

ReThink's conceptual architecture separates reusable mathematical knowledge definitions, capability configurations, curated assessment and instructional content, and learner-specific evidence.

This separation establishes the logical responsibilities of the information domains supporting the product without prescribing their physical implementation.

The conceptual architecture comprises five distinguishable domains:

Logical Domain	Architectural Responsibility
Conceptual Knowledge Structure	Defines mathematical subjects, topics, concepts, and curated relationships independently of individual learner performance.
Assessment Definitions	Organizes assessment tasks, intended evidence, and evaluation criteria associated with supported concepts.
Instructional & Validation Content	Organizes curated conceptual explanations, worked examples, guided practice, and validation materials associated with supported concepts.
Capability Availability	Distinguishes which assessment, instructional, and validation capabilities are configured and available for a given concept.
Learner Evidence	Distinguishes learner-specific diagnostic conclusions, instructional completion information, and validation-based outcomes from reusable mathematical definitions.

These domains describe logical information responsibilities rather than separate applications, services, databases, or mandatory storage entities.

A mathematical concept may be represented independently of whether it has an active assessment or instructional package. Likewise, learner-specific evidence is associated with the relevant concept without becoming an intrinsic property of the reusable concept definition.

The architecture therefore preserves a distinction between:

- What a mathematical concept represents.
- Which product capabilities support that concept.
- What relevant evidence has been established for an individual learner.

This conceptual separation supports the product boundaries already defined in Sections 04–05 and does not introduce additional product behavior.

Architectural status: Proposed logical separation. Physical domain boundaries, data models, storage mechanisms, and integration contracts remain subject to technical design.

7.2 Modular Content & Capability Activation

ReThink's conceptual architecture should allow mathematical concepts to be represented independently from the assessment, instructional, and validation packages associated with them.

Capability support is treated as an explicit configuration relationship rather than an automatic consequence of concept representation or graph connectivity.

At the conceptual level, three distinctions are relevant:

Conceptual Condition	Meaning
Represented	The concept exists within the curated mathematical knowledge structure.
Configured	The concept has an associated definition or content configuration for a particular product capability.
Actively Supported	The associated capability has been explicitly enabled for use within the applicable approved product coverage.

These conditions describe architectural distinctions, not new user-facing learning states or additional MVP features.

A concept may have a curated relationship with a topic without having assessment, recovery, or validation support. The existence of content configuration alone does not require a capability to become available to learners.

The conceptual model should permit assessment definitions and instructional content to be associated with reusable concepts without duplicating their underlying mathematical definitions for every related topic.

The architecture should also distinguish capability availability from learner-specific diagnostic and validation evidence.

This approach supports independent management of concept relationships and content readiness while leaving the exact configuration mechanism to technical design.

Architectural status: Proposed modular association and activation principle. No requirement is established for a particular registry implementation, configuration schema, or feature-flag technology.

7.3 Scalability & Extension Principles

ReThink's conceptual architecture is intended to accommodate controlled extension of mathematical concept coverage without requiring the fundamental distinction between knowledge definitions, capability support, and learner evidence to be redesigned.

Scalability in this context refers to **conceptual extensibility and maintainability**, not a claim of demonstrated production capacity, automated curriculum expansion, or broad diagnostic intelligence.

Future extension of concept support should follow three principles.

1. Reusable Concept Relationships

New mathematical topics may reference existing curated concepts and relationships where mathematically appropriate. Reuse should not require duplicating equivalent concept definitions for each topic.

2. Separately Extensible Capability Definitions

Assessment configurations, instructional packages, and validation materials may be extended independently at the logical level.

The existence of one type of supporting material does not imply that all other capabilities have been implemented or verified for the same concept.

3. Quality-Gated Capability Activation

Additional capability coverage should be activated only after the relevant mathematical relationships, assessment definitions, content packages, and evaluation criteria have received appropriate review for their intended use.

Proposed concept relationships or generated content do not become authoritative solely because they are available in the system.

These principles describe the intended direction of future extension. They do not commit the MVP to implementing additional supported concepts, multi-level prerequisite tracing, dynamically generated learning interventions, or unrestricted AI-based concept mapping.

The architecture's extensibility should be evaluated through its ability to preserve conceptual distinctions and support controlled addition of approved content, rather than through unverified claims of broad scalability.

Architectural status: Proposed extension principles. Specific extension workflows, quality criteria, implementation mechanisms, and performance expectations remain outside this Overall PRD section.

Section 07 Architectural Boundaries

Section 07 establishes conceptual architecture principles only. It does not redefine the MVP scope, modify approved functional requirements, or introduce an additional product capability.

The following remain outside its decision scope:

- Physical architecture, database technology, storage schemas, and deployment topology.
- API contracts, service decomposition, and detailed persistence strategies.
- Final diagnostic AI decision authority and mathematical verification implementation.
- New adaptive instructional or diagnostic capabilities.
- Additional supported mathematical topics or conceptual blockers.
- Unverified production scalability and performance claims.

Product Roadmap

8.1 Product Development Roadmap

ReThink's development roadmap prioritizes delivery and verification of the approved MVP conceptual recovery experience before expanding mathematical content coverage or introducing additional learning capabilities.

The roadmap is organized into indicative development priorities rather than mandatory sequential delivery gates. Activities may overlap or proceed in parallel where their dependencies permit.

Development Priority	Focus	Key Activities	Expected Deliverable
P1 — MVP Foundations	Establish the supporting product structure	Implement catalog-based entry, curated mathematical relationships, supported capability availability, and the foundational learning context	Functional learning entry and supported conceptual structure
P2 — Diagnostic Capability	Deliver bounded, evidence-grounded diagnosis	Implement the approved diagnostic questions, evidence evaluation, cross-question synthesis, conditional probing, outcome communication, and supported continuation	Working SPL diagnostic experience for the supported Distributive Property blocker
P3 — Guided Learning	Deliver the supported conceptual intervention	Integrate curated Visual Explanation, Worked Example, Partial Guided Practice, participation requirements, and completion handling	Functional Distributive Property learning package
P4 — Knowledge Validation & Integration	Complete the supported recovery experience	Implement Independent Retry, conditional Transfer Problem, learner-state assignment, Learning Map consistency, and cross-capability transitions	Integrated conceptual recovery pathway with applicable blocker-level learning outcomes

P5 — MVP Verification & Refinement	Evaluate MVP readiness	Verify applicable functional scenarios, mathematical correctness, diagnostic decision integrity, transition consistency, and core experience usability; address identified defects	Verification evidence, documented limitations, and a refined MVP candidate
Post-MVP — Controlled Extension	Evaluate potential expansion	Assess additional curated mathematical concept coverage and supporting packages through separate product and content-quality review	Proposed extensions, subject to future approval

Roadmap Interpretation

The priorities describe the main workstreams required to realize the approved product baseline. They are not fixed calendar phases, mandatory sprint sequences, or statements of completed implementation.

Relevant workstreams may advance concurrently. In particular, curated mathematical content preparation, detailed operational specifications, integration design, and verification planning may begin before their corresponding capability implementations are complete.

The Post-MVP item represents a possible extension direction rather than a committed release or additional MVP deliverable.

8.2 Roadmap Dependencies & Release Boundaries

Execution of the MVP roadmap depends on resolving the outstanding decisions and operational specifications relevant to each workstream.

Dependency	Affected Workstreams	Required Treatment
OI-01 — AI Methodology Alignment	P2; applicable P4 integration and P5 verification	Resolve the division of diagnostic interpretation and decision authority among AI, mathematical verification, and system constraints before finalizing dependent behavior and acceptance criteria.
OI-02 — Optional Review Learner-State Behavior	P3–P5	Resolve whether Optional Concept Review permits subsequent Knowledge Validation and how any resulting learner states are handled.

OI-03 — All-Correct Diagnostic Outcome Mapping	P2; applicable P4 integration and P5 verification	Determine formal all-correct diagnostic outcome behavior and its dependent result communication and continuation.
OPS-01 — Guided Practice Feedback Mechanics	P3; P5	Specify response-feedback behavior for supported Partial Guided Practice.
OPS-02 — Guided Learning Completion Criteria	P3–P5	Define the operational conditions for instructional completion and the resulting applicable handoff.
OPS-03 — Validation Response Handling	P4–P5	Specify evaluation and handling of incomplete, uninterpretable, and mathematically equivalent validation responses.
OPS-04 — Curated Mathematical Items & Rubrics	P2–P5	Prepare and review the supported diagnostic items, instructional materials, validation tasks, reference answers, and mathematical assessment criteria.
OPS-05 — Cross-Capability Transition & State Design	P1–P5, as applicable	Define the implementation-level handling of learning context, eligible transitions, and consistency of applicable diagnostic and validation states.

MVP Release Boundaries

- MVP readiness requires implementation and verification of the applicable approved functional requirements, not merely completion of individual development workstreams.
- The applicable end-to-end verification scenarios must support the functional success conditions SC1 — Complete Supported Recovery Path, SC2 — Evidence-Grounded Decision Integrity, and SC3 — Appropriate Outcome-Based Continuation.
- The core recovery experience must also be assessed against SC4 — Understandable Core Recovery Experience, using the separately defined usability verification approach.
- Outstanding decisions and operational specifications must be resolved to the extent required for implementing and verifying their dependent behaviors. Unresolved rules must not be silently substituted with implementation assumptions.
- A working demonstration alone does not establish comprehensive functional readiness, learning effectiveness, or empirically validated educational impact.
- Additional mathematical concepts, diagnostic coverage, adaptive learning capabilities, and other future extensions require separate scope approval and must not be treated as MVP release requirements.
- Roadmap workstream completion, verified MVP readiness, and release authorization are distinct milestones.

This roadmap does not assign fixed delivery dates, sprint durations, staffing commitments, or production performance targets. It also does not alter the 39 approved functional requirements or the conceptual architecture principles established in Section 07.

Assumptions, Risks & Dependencies

9.1 Critical MVP Risks & Mitigation

ReThink's principal MVP risks concern the reliability of its supported diagnostic decisions, mathematical content, guided learning interactions, validation outcomes, and end-to-end product integration.

The following register identifies material risks to MVP delivery and evaluation. These risks are potential failure conditions, not confirmed implementation defects. Mitigations refer to existing product requirements, pending specifications, and verification activities rather than introducing additional capabilities.

Risk ID	Critical MVP Risk	Potential Impact	Mitigation & Verification Approach
R-01 — Diagnostic Decision Integrity	Mathematical evidence is misinterpreted or insufficient evidence produces an unsupported blocker conclusion.	Incorrect diagnostic outcomes and misleading recovery recommendations.	Verify individual evidence evaluation, cross-question synthesis, mathematical safeguards, and conservative decision behavior under FR-C3-04–08; resolve applicable OI-01 decisions.
R-02 — Mathematical Content Validity	Curated diagnostic items, worked examples, practice materials, or validation criteria contain mathematical inconsistencies or fail to assess their intended concept.	Unreliable assessment and learning experiences, even when software functions as designed.	Review mathematical content, reference solutions, and assessment criteria through OPS-04; test representative content against its intended assessment or instructional purpose.
R-03 — Guided Learning Interaction Integrity	Partial Guided Practice is implemented as passive content or its feedback and completion handling do not support the	Students may complete the intervention without the required active mathematical participation.	Verify FR-C4-05–08, including student-generated responses, instructional completion boundaries, and consistency with OPS-01/OPS-02.

	intended instructional sequence.		
R-04 — Validation Outcome Integrity	Validation responses are evaluated incorrectly or item-level results are mapped to incorrect learner states.	Misleading Understood, Procedural Only, or Needs Reinforcement outcomes.	Verify mathematical response evaluation separately from the approved three-state decision rules under FR-C5-04/05; define response-handling criteria through OPS-03.
R-05 — Cross-Capability Integration Integrity	Learning context, diagnostic outcomes, completion information, or validation states become inconsistent across product capabilities.	Invalid transitions, interrupted recovery pathways, or conflicting learner information.	Verify context continuity, outcome-gated transitions, and representation consistency under FR-X-01–03; complete applicable transition specifications through OPS-05.
R-06 — Evaluation & Claims Validity	Functional MVP completion is interpreted as sufficient proof of usability or learning effectiveness.	Product claims exceed the evidence available from implementation and testing.	Evaluate functional success conditions, core experience usability, and empirical learning outcomes as distinct evidence dimensions under Section 03; report limitations explicitly.

Risk mitigations should be assessed against their relevant evidence and outstanding dependencies. Implementation of a related requirement does not automatically establish that its associated risk has been sufficiently addressed.

The register does not assign numerical likelihood, severity, or residual risk scores without supporting evidence.

9.2 Assumption & Dependency Governance

ReThink distinguishes product assumptions, unresolved product decisions, operational specifications, and verification evidence.

Assumption Governance

The existing A1–A6 product assumptions retain their previously established definitions and validation status. They remain hypotheses unless supported by relevant research or evaluation evidence.

Assumptions must not be treated as confirmed merely because the MVP implements a corresponding product feature or because exploratory research suggests a possible need.

Evidence used to evaluate assumptions should be assessed according to its methodological limitations and relevance to the specific claim.

Dependency Governance

Outstanding product decisions and pending operational specifications remain governed by their existing registers and the roadmap dependency references in Section 08.2.

A dependency may affect the implementation or verification readiness of specific requirements without invalidating unrelated approved requirements.

When an outstanding decision or operational specification is resolved, its implications should be reviewed against the affected requirements, relevant design specifications, and verification criteria.

Risk Review & Closure

Risk treatment follows three governance principles:

1. **Evidence-Based Review:** Risks are reassessed using relevant implementation results, mathematical-content review, verification findings, usability evidence, or research outcomes.
2. **Traceable Decisions:** Material changes to assumptions, dependencies, or mitigations must remain traceable to the affected approved product baseline and supporting documentation.
3. **Explicit Residual Uncertainty:** Risks and assumptions must not be declared resolved when the evidence remains insufficient or the applicable dependent behavior has not been verified.

MVP readiness must account for unresolved dependencies that materially affect the implementation or verification of approved requirements.

This section does not redefine the existing assumptions, duplicate the outstanding decision or operational specification registers, establish additional release criteria, or expand the approved MVP scope.

Product Acceptance & PRD Governance

10.1 Product Acceptance Principles

Product acceptance determines whether the implemented ReThink MVP satisfies the applicable approved product requirements and success conditions.

Acceptance decisions must be supported by relevant verification evidence rather than inferred from document approval, implementation completion, or demonstration alone.

Acceptance Principle	Required Interpretation
Requirements Compliance	The implemented MVP must be assessed against the applicable approved functional requirements and their verification criteria.
Functional Decision Integrity	The relevant end-to-end scenarios must demonstrate appropriate diagnostic conclusions, learning transitions, and validation outcomes.
Evidence-Based Acceptance	Acceptance claims must be supported by documented verification results and explicitly identified limitations.
Dependency Resolution	Outstanding decisions and specifications affecting required acceptance scenarios must be resolved before those scenarios are considered fully verified.
Bounded Product Claims	Functional acceptance must not be presented as independent proof of learning effectiveness, long-term retention, or broader mathematical mastery.

The applicable success conditions, requirement verification bases, and verification scenario coverage remain governed by Sections 03 and 05.

This section does not introduce new numerical acceptance thresholds, assessment methods, or functional requirements.

10.2 PRD Closure & Change Governance

The Overall PRD serves as the controlled product baseline for subsequent product design, technical implementation, and verification activities.

Once the full document receives final approval, its scope boundaries, requirements, and approved architectural principles become authoritative references for the MVP.

Governance Area	Principle
Baseline Authority	Approved sections define the agreed product obligations and boundaries.
Unresolved Decisions	Open decisions remain explicitly unresolved until reviewed and approved; final PRD approval does not close them automatically.
Operational Specifications	Detailed design and implementation decisions are documented separately without silently modifying approved product behavior.
Change Control	Material changes affecting scope, required behavior, eligibility, learner states, or success conditions require explicit impact review and approval.
Traceability	Approved changes must remain traceable to affected requirements, design documents, and verification criteria.

PRD approval establishes agreement on the intended product, not proof of a completed or validated implementation.

Subsequent documentation and development must preserve the approved MVP boundaries unless changes receive explicit authorization.

ReThink Summary

TECH DOCUMENTATION

Technical Overview & MVP Boundaries

1.1 Implementation Objective

Implementasi MVP ReThink bertujuan merealisasikan satu pengalaman pemulihan pemahaman matematika yang terintegrasi, mulai dari pemilihan topik yang didukung, asesmen diagnostik berbasis bukti, pembelajaran konsep terarah, hingga validasi penerapan mandiri dan penyajian hasil belajar.

Sistem harus mempertahankan perbedaan antara bukti respons matematika, kesimpulan diagnostik, aktivitas pembelajaran, dan hasil validasi. Keputusan mengenai pemahaman siswa hanya berlaku dalam cakupan konsep dan asesmen yang didukung oleh MVP.

Implementation Objective	Expected Technical Outcome
Integrated Recovery Experience	Alur diagnosis, pembelajaran terbimbing, dan validasi dapat dijalankan secara terhubung pada supported conceptual blocker.
Evidence-Grounded Evaluation	Respons siswa dapat dievaluasi dan digunakan sebagai evidence bagi hasil diagnosis serta validasi sesuai aturan produk.
Controlled Instructional Experience	Paket pembelajaran terkurasi disajikan dalam urutan yang telah ditentukan, dengan partisipasi matematika aktif pada Guided Practice.
Consistent Learning Context	Informasi topik, konsep, diagnosis, konteks pembelajaran, dan learner state tetap konsisten sepanjang perpindahan antarkemampuan.
Verifiable Product Behavior	Perilaku implementasi dapat diperiksa berdasarkan approved functional requirements, success conditions, dan applicable verification scenarios.

Tujuan tersebut menjelaskan tanggung jawab implementasi secara logis, bukan penetapan jumlah modul, layanan, basis data, atau komponen infrastruktur.

Implementasi dianggap memenuhi kewajiban produk hanya jika perilaku terkait memiliki bukti verifikasi yang sesuai. Keberhasilan menjalankan satu demonstrasi tidak otomatis membuktikan kelengkapan MVP.

1.2 MVP Implementation Boundaries

A. Supported Implementation Scope

MVP dibatasi pada satu vertical slice matematika untuk memeriksa dan memulihkan pemahaman Distributive Property dalam konteks Sistem Persamaan Linear (SPL).

Area	Approved MVP Coverage
Mathematical Domain	Aljabar
Diagnostic Entry Topic	Sistem Persamaan Linear (SPL)
Supported Conceptual Blocker	Distributive Property
Diagnostic Assessment	Q1–Q3 sebagai asesmen inti dan maksimal satu Q4 kondisional
Diagnostic Outcomes	Confirmed, Rejected, Uncertain
Guided Learning	Visual Explanation → Worked Example → Partial Guided Practice
Optional Concept Review	Akses ulasan konsep tanpa otomatis menetapkan Confirmed
Knowledge Validation	Independent Retry dan Transfer Problem kondisional
Learner Outcomes	Understood, Procedural Only, Needs Reinforcement
Hybrid Learning Map	Explore, Focus, Progress berdasarkan konteks dan evidence yang berlaku
Product Capabilities	C1–C5 dan persyaratan lintas kemampuan yang disetujui

B. PRD-Derived Behavioral Constraints

Ketentuan berikut merupakan kewajiban perilaku yang berasal dari approved PRD. Ketentuan ini harus dipenuhi oleh desain implementasi, terlepas dari teknologi yang dipilih.

Area	Required Behavioral Constraint
Diagnostic Coverage	Sistem hanya dapat menetapkan conceptual blocker yang secara eksplisit didukung oleh cakupan diagnosis.
Diagnostic Evidence	Satu jawaban salah tidak cukup untuk mengonfirmasi blocker; evidence harus dievaluasi dan disintesis sesuai aturan diagnosis.
Conservative Outcomes	Ketika evidence tidak memadai atau bertentangan, sistem harus dapat menghasilkan Uncertain.

AI Boundaries	Interpretasi AI dibatasi oleh konteks asesmen, pengetahuan matematika terkurasi, dan safeguards yang relevan.
Instructional Sequence	Guided Learning mengikuti urutan linear yang disetujui dan tidak menggunakan adaptive scaffold selection.
Active Participation	Partial Guided Practice memerlukan respons matematika aktif dari siswa.
Completion vs. Understanding	Penyelesaian Guided Learning tidak otomatis menghasilkan status pemahaman tervalidasi.
Validation Sequence	Transfer Problem hanya diberikan setelah Independent Retry berhasil.
Validation Boundary	Kegagalan validasi tidak secara otomatis memulai ulang diagnosis atau menelusuri blocker tambahan.
Learning Map Integrity	Learning Map merepresentasikan hasil yang didukung evidence dan tidak menetapkan diagnosis atau learner state secara independen.

Detail operasional terkait feedback, completion, response handling, dan cross-capability transitions tetap mengikuti status spesifikasi masing-masing yang belum diselesaikan.

C. Technical Non-Goals

MVP tidak mencakup general-purpose AI tutoring, dynamic multi-level prerequisite tracing, diagnosis banyak conceptual blocker, autonomous instructional content generation, probabilistic mastery estimation, ataupun long-term retention tracking.

Konsep future-readiness tidak mewajibkan implementasi generalized curriculum management system atau seluruh kemampuan ekspansi produk.

D. Engineering Design Freedom

Dokumen ini tidak menetapkan backend framework, database engine, service topology, API contract, model AI tertentu, atau deployment environment.

Engineering dapat menentukan pendekatan implementasi yang sesuai setelah mempertimbangkan feasibility, kualitas evaluasi matematika, keterpeliharaan, dan batas waktu pengembangan.

Kebebasan tersebut tidak mencakup perubahan terhadap approved product behavior atau penyelesaian product decisions yang masih terbuka.

1.3 Source of Truth & Documentation Boundaries

A. Documentation Authority

Source	Role & Authority
Overall PRD v1.0 — Final Approved	Otoritas utama untuk MVP scope, product behavior, success definitions, dan product-level boundaries.
Approved Product Requirements	Otoritas atas 39 functional requirements, integrity principles, dan preliminary verification coverage.
ReThink Research Paper	Referensi ilmiah, metodologi diagnosis, rationale pembelajaran, dan technical approach yang sebelumnya diusulkan.
Existing Process & Design References	Dokumentasi pendukung untuk memahami diagnostic process, guided learning, knowledge validation, dan conceptual architecture.
Technical Implementation Blueprint	Panduan penerjemahan approved requirements menjadi logical components, workflows, data considerations, dan implementation readiness.
Engineering Design Specifications	Dokumentasi turunan untuk keputusan teknologi, implementasi, data models, API contracts, dan engineering details.

Dalam hal terdapat perbedaan antara paper, dokumen desain sebelumnya, dan approved PRD mengenai perilaku atau cakupan MVP, approved PRD menjadi acuan implementasi.

Paper tetap harus dirujuk untuk memahami metodologi dan pendekatan yang mendasari ReThink, tetapi proposed technical mechanisms di dalamnya tidak otomatis dianggap sebagai pilihan implementasi final.

B. Technical Approach & Deviation Governance

Paper ReThink mengusulkan penggunaan curated knowledge relationships, rule-based orchestration, deterministic mathematical validation, serta LLM dengan autonomy yang dibatasi.

Pendekatan tersebut harus dicatat sebagai **paper-proposed technical approach**, bukan sebagai seluruh keputusan arsitektur yang telah dikunci.

Engineering dapat mengevaluasi pendekatan alternatif dengan ketentuan:

1. Approved MVP behavior tetap terpenuhi.
2. Perbedaan dari technical approach yang diusulkan paper didokumentasikan.
3. Alasan, dampak terhadap requirements, dan implikasi verifikasi dijelaskan.
4. Perubahan yang memengaruhi product obligations dikembalikan ke controlled product review.

Pembahasan spesifik mengenai methodological references, datasets, serta technical alternatives dilakukan dalam Section 04 dan Section 05.

C. Open Product Decisions & Specifications

Reference	Status	Governance
OI-01 — AI Methodology Alignment	Open	Pembagian wewenang interpretasi AI dan keputusan diagnostik belum ditetapkan secara final.
OI-02 — Optional Review Learner-State Behavior	Open	Kelayakan validasi dan implikasi learner state setelah Optional Concept Review belum ditentukan.
OI-03 — All-Correct Diagnostic Outcome Mapping	Open	Perlakuan hasil diagnosis ketika seluruh jawaban benar masih memerlukan keputusan produk.
OPS-01–OPS-05	Pending	Mekanisme feedback, completion, validation response handling, curated mathematical assets, serta transisi antarkemampuan masih membutuhkan spesifikasi operasional.

Keputusan produk yang belum diselesaikan tidak boleh ditentukan secara implisit oleh implementasi Backend, AI, atau asumsi AI coding assistant.

Engineering dapat melanjutkan pekerjaan yang tidak bergantung pada keputusan tersebut, sedangkan dependent behavior harus mempertahankan status unresolved hingga keputusan yang relevan disetujui.

D. Implementation Governance Principle

Technical Implementation Blueprint merupakan dokumen panduan pelaksanaan, bukan pengganti Product Requirements.

Implementasi, perubahan teknik, dan hasil verifikasi harus tetap dapat ditelusuri ke approved PRD baseline.

Dokumen ini tidak menetapkan bahwa MVP telah diimplementasikan, diuji, atau diterima.

System Components & Responsibilities

2.1 Logical System Architecture

Arsitektur konseptual ReThink terdiri atas delapan komponen logis yang mendukung pengalaman pemulihan konsep secara terintegrasi.

Pemisahan komponen bertujuan memperjelas tanggung jawab pengelolaan konteks pembelajaran, evaluasi matematika, interpretasi diagnosis, penyajian intervensi, validasi, dan representasi hasil belajar.

[DIAG-01 — Logical System Architecture Diagram]

Placeholder: Diagram hubungan LC-01–LC-08, termasuk alur utama diagnosis → guided learning → knowledge validation, dukungan evaluasi matematika, koordinasi transisi, dan proyeksi hasil pada Hybrid Learning Map.

Status: Pending Diagram Development — setelah seluruh struktur Technical Blueprint disetujui.

Diagram ini nantinya merepresentasikan logical responsibilities dan information exchange, bukan microservice boundaries, deployment architecture, atau database topology.

Delapan komponen tidak harus diwujudkan sebagai delapan layanan perangkat lunak terpisah. Penggabungan atau pemisahan fisik menjadi keputusan engineering, selama kewajiban produk dan batas tanggung jawab tetap dipenuhi.

2.2 Component Responsibilities

Component	Primary Responsibility	Required Inputs	Expected Outputs	PRD Reference
LC-01 — Learning Entry & Catalog	Menyediakan katalog topik, relasi konsep, dan informasi ketersediaan dukungan.	Curated catalog, topic–concept relationships, capability availability.	Supported topic context dan diagnostic entry eligibility.	C1, F1–F2
LC-02 — Assessment Flow Controller	Mengelola urutan Q1–Q3, maksimal satu Q4 kondisional, serta pengumpulan respons.	Selected topic, assessment blueprint, question definitions, student responses.	Assessment responses beserta konteks dan metadata relevan.	C3

LC-03 — Mathematical Evaluation Support	Mendukung pemeriksaan kebenaran matematika, transformasi, dan interpretasi terhadap reference criteria.	Student responses, curated mathematical references, applicable rubrics.	Item-level mathematical observations dan evaluation results.	C3, C5
LC-04 — Diagnostic Interpretation & Decision Support	Menyintesiskan diagnostic evidence dan mendukung penentuan diagnostic outcome yang sah.	Mathematical observations, response evidence, supported diagnostic knowledge.	Confirmed, Rejected, atau Uncertain beserta evidence basis yang relevan.	C3, OI-01
LC-05 — Guided Learning Controller	Mengelola penyajian paket pembelajaran terkurasi, respons Guided Practice, dan completion.	Learning entry context, supported concept, instructional package, student interactions.	Participation evidence, completion state, dan learning handoff context.	C4
LC-06 — Knowledge Validation Controller	Mengelola Independent Retry, Transfer Problem kondisional, dan penetapan learner state.	Eligible learning context, validation items, mathematical evaluation results.	Understood, Procedural Only, atau Needs Reinforcement.	C5
LC-07 — Session & Transition Coordination	Mempertahankan konteks serta mengoordinasikan perpindahan antarkemampuan berdasarkan aturan yang berlaku.	Selected topic, diagnostic outcome, learning context, completion, validation outcomes.	Transition eligibility dan consistent workflow context.	FR-X-01–03
LC-08 — Learning Map Projection	Merepresentasikan curated concept relationships, diagnostic focus, dan evidence-based progress.	Knowledge structure, capability availability, diagnostic	Explore, Focus, dan Progress representations.	C2

		outcomes, learner states.		

Komponen di atas menjelaskan tanggung jawab konseptual. Definisi data model, API contracts, storage mechanisms, dan internal implementation tetap menjadi bagian dari engineering design.

Component Design Boundaries

- LC-03 menghasilkan evaluasi atau observasi matematika, bukan diagnostic conclusion secara independen.
- LC-04 bertanggung jawab secara logis terhadap diagnostic interpretation dan outcome coordination; alokasi final decision authority masih mengikuti OI-01.
- LC-05 mengelola aktivitas serta completion pembelajaran, bukan learner understanding status.
- LC-06 menentukan learner state berdasarkan approved validation rules dan item-level evidence yang berlaku.
- LC-07 mengelola eligibility dan perpindahan, bukan menciptakan diagnostic atau validation outcome baru.
- LC-08 merepresentasikan evidence dan status yang telah ditentukan, bukan melakukan diagnosis tersendiri.

2.3 Cross-Component Contracts & Ownership

Pertukaran informasi antarkomponen harus mempertahankan hubungan antara data input, evaluasi matematika, interpretasi, dan hasil keputusan.

Kontrak pada bagian ini merupakan logical information contracts, bukan API specification atau JSON schema.

Information / Decision	**Conceptual Owner**	**Consumers**	**Integrity Requirement**
Topic & Capability Availability	LC-01	LC-02, LC-08	Konsep yang direpresentasikan tidak otomatis menjadi supported diagnostic target.
Assessment Responses	LC-02	LC-03, LC-04	Raw student response, question context, dan interpretation harus dapat dibedakan.
Mathematical Evaluation Results	LC-03	LC-04, LC-06	Hasil item-level evaluation tidak otomatis menetapkan diagnosis atau learner state.

Diagnostic Outcome	LC-04, subject to OI-01	LC-05, LC-07, LC-08	Confirmed, Rejected, dan Uncertain hanya berlaku pada supported blocker dan evidence yang relevan.
Learning Participation & Completion	LC-05	LC-06, LC-07	Completion tidak secara otomatis membuktikan pemahaman tervalidasi.
Validation Learner State	LC-06	LC-07, LC-08	State mengikuti approved three-state mapping.
Transition Eligibility	LC-07	Workflow controllers	Transisi mengikuti outcome, konteks masuk, completion, dan applicable eligibility rules.
Learning Map Representation	LC-08	User-facing product interface	Representasi tidak boleh mengubah makna diagnostic outcome atau validation state.

Cross-Component Integrity Principles

1. Evidence and Decision Separation

Respons siswa, hasil evaluasi matematika, sintesis evidence, dan keputusan diagnosis harus dapat dibedakan secara logis.

2. Explicit Decision Ownership

Diagnostic outcomes, Guided Learning completion, dan validation learner states memiliki tanggung jawab keputusan yang berbeda. Komponen lain dapat menggunakan hasil tersebut tanpa menetapkan ulang maknanya.

3. Context Continuity

Informasi selected topic, supported conceptual blocker, learning entry context, dan applicable outcome harus tetap konsisten selama perpindahan antarkemampuan.

4. No Implicit State Promotion

Confirmed diagnosis tidak otomatis menghasilkan Understood. Penyelesaian Guided Learning juga tidak otomatis menghasilkan learner state dari Knowledge Validation.

5. Dependency-Aware Integration

Perilaku yang bergantung pada OI-01–OI-03 dan OPS-01–OPS-05 harus mempertahankan status unresolved sampai keputusan atau spesifikasi terkait disetujui.

Architectural Limitations

Section ini tidak menetapkan physical service decomposition, database architecture, endpoint definitions, transport protocols, maupun data persistence technology.

Komponen logis dapat diimplementasikan melalui pendekatan engineering yang berbeda selama behavioral obligations, evidence boundaries, dan consistency requirements dari approved PRD tetap terpenuhi.

Core Processing Workflows

Section ini menjelaskan mekanisme pemrosesan utama ReThink berdasarkan kebutuhan produk yang telah disetujui. Alur dinyatakan dalam bentuk tahapan, aturan keputusan, dan ketergantungan terbuka agar dapat digunakan sebagai dasar engineering design tanpa menentukan API, struktur database, atau detail algoritme.

3.1 Diagnostic Assessment & Outcomes

Tujuan: Memperoleh dan mengevaluasi bukti respons matematika untuk menentukan hasil diagnosis Distributive Property dalam konteks SPL.

A. Workflow Stages

Stage	Responsible Component	Processing Responsibility	Expected Output
D1 — Assessment Initialization	LC-01, LC-02	Memastikan SPL merupakan topik yang didukung dan memuat blueprint diagnosis yang relevan.	Konteks asesmen yang valid.
D2 — Core Question Administration	LC-02	Menyajikan Q1 Target Performance, Q2 Prerequisite Application, dan Q3 Conceptual Reasoning.	Respons inti beserta konteks setiap soal.
D3 — Mathematical Evidence Evaluation	LC-03	Mengevaluasi respons berdasarkan kriteria matematika dan kategori bukti yang berlaku.	Observasi matematika tingkat soal.
D4 — Cross-Question Evidence Synthesis	LC-04	Mempertimbangkan bukti pendukung, bukti yang bertentangan, dan informasi yang belum cukup.	Sintesis evidence terhadap supported blocker.
D5 — Conditional Diagnostic Probe	LC-02, LC-04	Jika diperlukan, menyajikan paling banyak satu Q4 berupa Procedural Trace atau Conceptual Probe, kemudian mengevaluasi bukti tambahan.	Additional diagnostic evidence.
D6 — Diagnostic Outcome Finalization	LC-04, subject to OI-01	Menghasilkan hasil diagnosis yang sah berdasarkan bukti dan aturan yang berlaku.	Confirmed, Rejected, atau Uncertain.

D7 — Result Communication & Handoff	LC-07, LC-08	Menyampaikan hasil, batas interpretasinya, dan opsi kelanjutan yang diperbolehkan.	Diagnostic result dan continuation context.

[DIAG-03 — Diagnostic Evidence & Conditional Probe Flow]

Placeholder untuk visualisasi alur Q1–Q3, evaluasi bukti, sintesis, kemungkinan Q4, finalisasi hasil, dan kelanjutan diagnosis.

B. PRD-Derived Decision Rules

Rule	Required Behavior
D-R01 — Supported Scope	Diagnosis hanya memeriksa Distributive Property sebagai supported conceptual blocker untuk konteks SPL.
D-R02 — Evidence Requirement	Satu kesalahan atau satu pola respons tidak otomatis mengonfirmasi blocker.
D-R03 — Cross-Question Evaluation	Keputusan mempertimbangkan bukti yang mendukung, bertentangan, atau belum memadai.
D-R04 — Conditional Probe Limit	Maksimal satu Q4 diberikan untuk mengklarifikasi kebutuhan bukti yang relevan.
D-R05 — Valid Outcomes	Hasil dibatasi menjadi Confirmed, Rejected, atau Uncertain.
D-R06 — Conservative Interpretation	Bukti ambigu, tidak memadai, atau tidak konsisten tidak boleh dipaksakan menjadi Confirmed.
D-R07 — Bounded AI Interpretation	AI tidak dapat menetapkan blocker baru di luar supported diagnostic knowledge.
D-R08 — Limited Diagnostic Claims	Hasil diagnosis tidak membuktikan penguasaan atau ketidakmampuan siswa pada seluruh SPL.

C. Open Dependencies

Reference	Dependency	Implementation Consequence
OI-01 — Open	AI Methodology Alignment	Final decision authority, pembagian evaluasi matematika dan interpretasi AI, serta safeguards memerlukan keputusan lanjutan.
OI-03 — Open	All-Correct Diagnostic Outcome Mapping	Outcome ketika seluruh respons diagnosis benar belum boleh ditentukan oleh implementasi.

OPS-04 — Pending	Curated Mathematical Items & Rubrics	Butir Q1–Q4, solusi acuan, dan kriteria evaluasi harus disiapkan serta ditinjau.

Komponen assessment, data capture, dan basic mathematical evaluation dapat dirancang tanpa menganggap dependent diagnostic behavior telah final.

3.2 Guided Learning & Optional Concept Review

Tujuan: Menyajikan intervensi terkurasi untuk Distributive Property melalui urutan pembelajaran linear yang mempertahankan partisipasi matematika aktif.

A. Workflow Stages

Stage	Responsible Component	Processing Responsibility	Expected Output
L1 — Learning Entry Context	LC-07	Mengidentifikasi apakah pembelajaran dimulai dari Confirmed Recovery atau Optional Concept Review.	Learning context yang dapat dibedakan.
L2 — Curated Package Retrieval	LC-05	Memuat paket pembelajaran Distributive Property yang didukung.	Instructional package.
L3 — Visual Explanation	LC-05	Menyajikan representasi yang menjelaskan makna konsep.	Aktivitas penjelasan konseptual.
L4 — Worked Example	LC-05	Menampilkan contoh dengan transformasi dan alasan matematika yang relevan.	Aktivitas demonstrasi prosedur dan reasoning.
L5 — Partial Guided Practice	LC-05, LC-03 bila diperlukan	Meminta siswa melengkapi langkah atau reasoning dalam solusi yang dibimbing sebagian.	Practice response evidence.
L6 — Learning Completion Evaluation	LC-05	Menentukan completion berdasarkan kriteria operasional yang berlaku.	Completion state.
L7 — Learning Continuation	LC-07	Mengarahkan pengguna sesuai konteks masuk, completion, dan eligibility.	Permitted next transition.

B. PRD-Derived Decision Rules

Rule	Required Behavior
L-R01 — Supported Package	MVP menyediakan satu curated Guided Learning package untuk Distributive Property.
L-R02 — Fixed Sequence	Urutan pembelajaran adalah Visual Explanation → Worked Example → Partial Guided Practice.
L-R03 — Active Participation	Partial Guided Practice memerlukan respons aktif dari siswa; melihat konten saja tidak memenuhi kebutuhan tersebut.
L-R04 — No Adaptive Orchestration	Sistem tidak memilih scaffold atau learning pathway baru secara dinamis.
L-R05 — Entry Context Integrity	Optional Concept Review tidak mengubah hasil Rejected atau Uncertain menjadi Confirmed.
L-R06 — Completion Boundary	Menyelesaikan pembelajaran tidak otomatis menunjukkan pemahaman yang tervalidasi.
L-R07 — Conditional Continuation	Confirmed Recovery yang memenuhi completion dapat berlanjut ke Knowledge Validation; eligibility setelah Optional Review belum diputuskan.

[DIAG-04 — Guided Learning & Knowledge Validation Transition Flow]

Placeholder untuk visualisasi dua konteks masuk, urutan pembelajaran linear, completion, dan transisi validasi sesuai eligibility.

C. Open Dependencies

Reference	Dependency	Implementation Consequence
OI-02 — Open	Optional Review Learner-State Behavior	Kelayakan validasi dan dampak learner state setelah ulasan opsional belum final.
OPS-01 — Pending	Guided Practice Feedback Mechanics	Mekanisme respons, umpan balik, dan penanganan kesalahan perlu ditetapkan.
OPS-02 — Pending	Guided Learning Completion Criteria	Completion tidak boleh ditentukan hanya berdasarkan asumsi implementasi.
OPS-04 — Pending	Curated Mathematical Items & Rubrics	Materi dan latihan harus diperiksa validitas matematikanya.

OPS-05 — Pending	Cross-Capability Transition & State Design	Handoff pembelajaran dan eligibility membutuhkan spesifikasi operasional.

3.3 Knowledge Validation & Learner States

Tujuan: Memvalidasi kemampuan siswa menggunakan Distributive Property secara mandiri serta membedakan keberhasilan prosedural dari penerapan pada representasi berbeda.

A. Workflow Stages

Stage	Responsible Component	Processing Responsibility	Expected Output
V1 — Validation Eligibility Check	LC-07	Memastikan konteks belajar dan completion memenuhi ketentuan validasi.	Eligible validation context.
V2 — Independent Retry Administration	LC-06	Menyajikan soal dengan struktur konsep yang relevan tanpa scaffolding.	Independent response.
V3 — Retry Evaluation	LC-03	Mengevaluasi respons menggunakan kriteria matematika yang berlaku.	Item-level retry outcome.
V4 — Conditional Transfer Administration	LC-06	Memberikan Transfer Problem hanya jika Independent Retry berhasil.	Transfer response.
V5 — Transfer Evaluation	LC-03	Mengevaluasi penerapan konsep pada bentuk atau konteks yang berbeda secara bermakna.	Item-level transfer outcome.
V6 — Learner-State Determination	LC-06	Menetapkan hasil belajar berdasarkan kombinasi validasi yang sah.	Understood, Procedural Only, atau Needs Reinforcement.
V7 — Outcome Communication & Continuation	LC-07, LC-08	Menyampaikan hasil dan memperbarui representasi progress jika berlaku.	Learning outcome dan next-step context.

B. PRD-Derived Decision Rules

Independent Retry	Transfer Problem	Learner State
Gagal	Tidak diberikan	**Needs Reinforcement**
Berhasil	Gagal	**Procedural Only**
Berhasil	Berhasil	**Understood**

Aturan tambahan:

Rule	Required Behavior
V-R01 — Validation Eligibility	Knowledge Validation dijalankan pada jalur yang memenuhi prasyarat pembelajaran dan konteks yang disetujui.
V-R02 — Unscaffolded Retry	Independent Retry tidak menyertakan bantuan yang menggantikan penyelesaian mandiri.
V-R03 — Meaningful Transfer	Transfer Problem menggunakan struktur konsep yang sama dengan representasi atau konteks yang berbeda secara bermakna.
V-R04 — Conditional Transfer	Transfer Problem tidak diberikan ketika Independent Retry gagal.
V-R05 — Approved State Mapping	Learner state mengikuti matriks tiga kondisi yang disetujui.
V-R06 — Reinforcement Routing	Needs Reinforcement mengarahkan siswa kembali ke Guided Learning package yang sama.
V-R07 — No Re-Diagnosis	Kegagalan validasi tidak memulai diagnosis baru atau mengonfirmasi blocker tambahan.
V-R08 — Limited Learning Claims	Understood hanya berlaku pada Distributive Property dalam cakupan validasi yang digunakan, bukan mastery seluruh SPL.

Matriks tiga kondisi berlaku jika setiap respons telah memiliki hasil evaluasi item yang sah. Respons yang kosong, tidak dapat diinterpretasikan, atau memiliki bentuk matematika ekuivalen memerlukan penanganan sesuai OPS-03.

C. Open Dependencies

Reference	Dependency	Implementation Consequence

OI-02 — Open	Optional Review Learner-State Behavior	Jalur validasi setelah Optional Review tidak dapat difinalisasi.
OPS-02 — Pending	Guided Learning Completion Criteria	Validation eligibility bergantung pada kriteria completion yang berlaku.
OPS-03 — Pending	Validation Response Handling	Penanganan respons tidak lengkap, tidak terbaca, atau ekuivalen perlu dispesifikasikan.
OPS-04 — Pending	Curated Mathematical Items & Rubrics	Independent Retry dan Transfer Problem memerlukan item serta referensi matematika yang valid.
OPS-05 — Pending	Cross-Capability Transition & State Design	Transisi menuju validasi dan sesudah hasil perlu konsisten.

3.4 Cross-Capability Transitions

Tujuan: Mempertahankan konteks pembelajaran dan mengoordinasikan perpindahan C1–C5 berdasarkan keputusan yang telah dihasilkan oleh komponen yang berwenang.

A. Transition Workflow

Transition	Trigger	Eligibility / Required Context	Owner
T1 — Learning Entry → Diagnosis	Siswa memilih memulai diagnosis SPL.	Diagnostic capability tersedia untuk topik yang dipilih.	LC-01, LC-02, LC-07
T2 — Diagnosis → Confirmed Recovery	Diagnostic outcome Confirmed.	Supported blocker dan result context sah.	LC-04, LC-07
T3 — Diagnosis → Optional Review	Rejected atau Uncertain dan siswa memilih review.	Tidak ada perubahan retroaktif menjadi Confirmed.	LC-07, LC-05
T4 — Confirmed Recovery → Validation	Guided Learning selesai.	Completion dan validation eligibility terpenuhi.	LC-05, LC-07, LC-06
T5 — Optional Review → Validation	Optional Review selesai.	**Eligibility masih OI-02 Open.**	LC-07, LC-06

T6 — Validation → Reinforcement	Independent Retry gagal.	Needs Reinforcement berdasarkan evaluasi yang sah.	LC-06, LC-07, LC-05
T7 — Outcome → Learning Map	Diagnostic outcome atau learner state tersedia.	Informasi dan evidence berasal dari komponen yang berwenang.	LC-07, LC-08

[DIAG-02 — End-to-End Processing & Outcome Branches]

Placeholder untuk visualisasi alur lintas C1–C5, percabangan hasil diagnosis, konteks Optional Review, validasi kondisional, serta pembaruan Learning Map.

B. Cross-Capability Integrity Rules

Rule	Required Behavior
T-R01 — Context Continuity	Selected topic, supported blocker, diagnostic outcome, dan learning entry context tidak boleh tertukar selama transisi.
T-R02 — Decision Ownership	Transition Coordinator tidak menetapkan hasil evaluasi matematika, diagnosis, maupun learner state baru.
T-R03 — Evidence Traceability	Hasil yang digunakan untuk perpindahan harus dapat dihubungkan dengan konteks dan sumber evidence yang relevan.
T-R04 — No Implicit State Promotion	Confirmed dan Guided Learning Completed tidak boleh diperlakukan sebagai Understood.
T-R05 — Supported Capability Boundary	Transisi tidak dapat mengaktifkan diagnostic, learning, atau validation capability yang belum didukung.
T-R06 — Dependency-Aware Routing	Transisi yang memerlukan keputusan terbuka tidak boleh dianggap memiliki aturan final.

C. Open Dependencies

Reference	Affected Transition
OI-01 — Open	T2, karena penetapan diagnostic outcome bergantung pada pembagian otoritas diagnosis yang belum final.
OI-02 — Open	T5 dan eligibility validasi setelah Optional Review.

OI-03 — Open	Kelanjutan diagnosis pada pola seluruh jawaban benar.
OPS-01–OPS-02 — Pending	Penyelesaian Guided Practice dan completion sebelum handoff.
OPS-03–OPS-04 — Pending	Evaluasi respons serta validasi matematika yang digunakan dalam transisi.
OPS-05 — Pending	Detail operasional cross-capability transition dan state consistency.

Section 03 — Implementation Boundaries

Processing workflows ini menetapkan hubungan perilaku dan tanggung jawab logis, bukan pseudocode, API contract, database schema, atau physical architecture.

Engineering bebas mengusulkan pendekatan implementasi selama approved product behavior tetap terpenuhi, hasil evaluasi dan keputusan dapat ditelusuri, serta keputusan yang belum selesai tidak diasumsikan telah disetujui.

Data, AI & Methodological Foundations

Section ini menjelaskan fondasi informasi, aset matematika, metodologi diagnosis, rancangan AI, dan sumber referensi yang diperlukan untuk mendukung implementasi MVP ReThink.

Setiap pendekatan dibedakan berdasarkan statusnya: PRD-Derived, Paper-Proposed, Research Reference, atau Engineering Consideration. Section ini tidak menetapkan physical database schema, API payload, prompt template, maupun pemilihan model AI final.

4.1 Concept, Content & Learner Data Foundations

A. Logical Information Inventory

Information Category	Required Information	Main Consumers	Status
Topic & Concept Knowledge	Subject, topic, concept, relasi prasyarat, dan capability availability.	LC-01, LC-08	PRD-Derived
Diagnostic Knowledge	Supported blocker, known error patterns, interpretive criteria, dan diagnostic coverage.	LC-03, LC-04	PRD-Derived
Assessment Assets	Q1–Q3, conditional Q4, mathematical reference answers, dan assessment metadata.	LC-02, LC-03, LC-04	PRD-Derived
Guided Learning Assets	Visual Explanation, Worked Example, Partial Guided Practice, dan associated mathematical criteria.	LC-05, LC-03	PRD-Derived
Validation Assets	Independent Retry, Transfer Problem, expected mathematical structure, dan evaluation criteria.	LC-06, LC-03	PRD-Derived
Learner Evidence	Respons asli, evaluasi matematika, hasil diagnosis, participation/completion, serta hasil validasi.	LC-03–LC-08 sesuai tanggung jawab	PRD-Derived
Session & Transition Context	Selected topic, supported blocker, learning entry context, dan applicable eligibility.	LC-02, LC-05–LC-08	PRD-Derived

Data inventory tersebut adalah kategori logis, bukan definisi entitas database atau kewajiban menyimpan seluruh kategori dalam sistem persistence yang berbeda.

B. Concept Coverage & Capability Availability

ReThink harus mempertahankan pembedaan antara konsep yang direpresentasikan, dikonfigurasi, dan benar-benar didukung.

Coverage level	Meaning	Implementation implication
Represented	Topic/concept tercatat dalam knowledge structure.	Dapat digunakan sebagai konteks katalog atau relasi.
Configured	Sebagian metadata atau aset operasional tersedia.	Tidak otomatis berarti diagnostic recovery dapat dijalankan.
Actively Supported	Capability relevan memiliki cakupan dan aset yang memenuhi ketentuan.	Dapat menjalankan proses sesuai capability yang didukung.

SPL menjadi supported diagnostic entry topic, sedangkan Distributive Property merupakan satu-satunya conceptual blocker dengan complete recovery coverage pada MVP.

C. Curated Mathematical Asset Package

Untuk membangun satu complete recovery loop, tim perlu menyiapkan aset yang saling konsisten.

Asset group	Minimum content consideration	Verification concern
Diagnostic Items	Q1–Q3 dan kandidat Q4 yang sesuai blueprint	Peran diagnostik dan kebenaran matematika
Error Pattern Knowledge	Pattern yang relevan terhadap Distributive Property	Tidak menyimpulkan blocker dari satu pola ambigu
Reference Solutions & Rubrics	Expected answers, acceptable transformations, reasoning criteria	Mathematical validity dan interpretability
Guided Learning Content	Visual explanation, worked example, partial guided practice	Konsistensi konsep dan pedagogi
Independent Retry	Item mandiri pada struktur konsep yang relevan	Tidak memberikan scaffold tersembunyi

Transfer Problem	Konsep sama dengan representasi atau konteks berbeda	Perbedaan bersifat bermakna, bukan sekadar angka

OPS-04 masih Pending. Tabel ini mengidentifikasi kebutuhan aset, bukan mengesahkan isi question bank, rubrik, atau instrumen evaluasi tertentu.

D. Learner Evidence Integrity

Implementasi harus menjaga perbedaan antara raw student response, mathematical observation, AI interpretation, diagnostic outcome, Guided Learning completion, dan validation learner state.

Hasil pada satu tingkat tidak boleh otomatis dianggap sebagai hasil pada tingkat berikutnya. Data yang tidak dikumpulkan atau belum dievaluasi juga tidak boleh diubah menjadi kesimpulan bahwa siswa memiliki conceptual gap.

Representasi learner evidence harus menjaga keterkaitan dengan konsep, tahap asesmen, serta konteks sesi yang relevan.

4.2 Diagnostic Methodological Foundation

A. Evidence-Centered Design Adaptation

Rancangan diagnosis ReThink menggunakan Evidence-Centered Design (ECD) sebagai referensi metodologis untuk menghubungkan klaim tentang siswa, bukti yang dapat diamati, dan tugas yang menghasilkan bukti.

Adaptasi konseptual yang digunakan adalah:

Claim → Evidence → Task → Interpretation → Verification

ECD-related construct	ReThink interpretation	Engineering relevance	Status
Student Model	Hipotesis mengenai Distributive Property sebagai possible blocker.	Membatasi klaim yang dapat dihasilkan sistem.	Research Reference
Task Model	Pertanyaan diagnostik dengan peran evidence berbeda.	Menentukan informasi yang diharapkan dari Q1–Q4.	Research Reference
Evidence Model	Kebenaran, transformasi, pola kesalahan, dan explanation evidence.	Menjaga hubungan observasi dengan hipotesis.	Research Reference

Evidence Synthesis	Penggabungan supporting, counterevidence, dan insufficient evidence.	Mendukung penalaran diagnosis lintas pertanyaan.	Paper/Design-Proposed
Diagnostic Outcome	Confirmed, Rejected, atau Uncertain.	Menghasilkan hasil yang dibatasi approved rules.	PRD-Derived
Verification	Evaluasi keputusan terhadap reference cases.	Membantu menguji diagnostic consistency.	Engineering Consideration

ReThink mengadaptasi prinsip ECD dan tidak mengklaim mengimplementasikan keseluruhan statistical measurement model dalam ECD.

B. Pre-Test Question Blueprint

Question	Diagnostic role	Input type	Expected evidence
Q1	Target Performance — SPL	Multiple Choice	Outcome pada tugas topik target
Q2	Prerequisite Application — Distributive Property	Short Mathematical Response	Bukti penerapan atau transformasi konsep
Q3	Conceptual Reasoning	Short Explanation	Bukti reasoning terhadap transformasi matematika
Q4	Conditional Targeted Probe	Procedural Trace atau Conceptual Probe	Klarifikasi information gap spesifik

Q1 menggunakan distractor yang dapat dirancang berdasarkan kemungkinan kesalahan matematis yang bermakna. Q2 dan Q3 memberikan tipe bukti berbeda sehingga diagnosis tidak hanya bergantung pada jawaban akhir.

Q4 tidak diwajibkan untuk semua siswa dan tidak boleh menjadi probing loop berulang.

Butir soal, contoh, accepted responses, serta aturan pemilihan probe tetap membutuhkan operational specification dan mathematical review.

C. Evidence Evaluation & Interpretation

Evaluasi diagnosis membedakan tiga tingkat informasi:

Layer	Example	Limitation
Observation	Siswa menulis $3(x + 2) = 3x + 2$.	Menunjukkan respons yang diamati.

Interpretation	Respons konsisten dengan partial distribution.	Belum membuktikan adanya miskonsepsi yang menetap.
Diagnostic Inference	Bukti lintas pertanyaan mendukung atau tidak mendukung blocker.	Hanya dapat menghasilkan keputusan dalam supported scope.

Empat dimensi evaluasi yang dapat digunakan dari rancangan diagnosis sebelumnya adalah mathematical correctness, error characteristics, conceptual explanation, dan evidence clarity.

Sistem juga perlu mempertimbangkan supporting evidence, counterevidence, dan insufficient evidence, termasuk penjelasan alternatif untuk kesalahan siswa.

D. Methodological Boundaries

ECD, DINA, cognitive diagnostic multiple-choice design, dan Rule Space merupakan sumber pemahaman metodologis. ReThink MVP tidak diwajibkan menerapkan model statistik tersebut.

Implementasi tidak boleh menghasilkan probabilitas mastery yang seolah terkalibrasi tanpa landasan pengukuran yang memadai.

OI-01 — Open: pembagian final diagnostic decision authority.
OI-03 — Open: all-correct diagnostic outcome mapping.
OPS-04 — Pending: mathematical items dan evaluation rubrics.

4.3 AI & Mathematical Evaluation Architecture

A. Paper-Proposed Technical Approach

Paper ReThink menggambarkan pendekatan hibrida berupa knowledge graph untuk relasi konsep, rule-based orchestration, deterministic mathematical validation, serta LLM terbatas sebagai supporting layer.

Pendekatan tersebut menjadi referensi awal engineering design, bukan keputusan final untuk physical architecture maupun pemilihan model.

Responsibility	**Paper-proposed approach**	**Design boundary**
Mathematical Correctness	Deterministic expression checking	Hasil matematika perlu dapat diverifikasi
Error Pattern Evaluation	Curated known error patterns	Tidak bebas menetapkan blocker baru

Reasoning Interpretation	Context-bounded LLM	Tidak mengabaikan evidence atau counterevidence
Diagnostic Outcome	Structured orchestration	Final decision allocation tetap OI-01 Open
AI Output Handling	Structured output validation	Invalid output tidak boleh dianggap hasil sah
Guided Learning	Curated instructional sequence	Tidak menggunakan autonomous adaptive teaching
Knowledge Validation	Controlled retry/transfer rules	Learner state mengikuti approved decision matrix

B. Paper-Proposed Technology Register

Technical role	**Technology mentioned in paper**	**Adoption status**
Frontend interface	Flutter	Paper-Proposed — Not Locked
Backend & orchestration	FastAPI	Paper-Proposed — Not Locked
Concept relationships	Neo4j	Paper-Proposed — Not Locked
Learner data & metadata	PostgreSQL	Paper-Proposed — Not Locked
Mathematical evaluation	SymPy	Paper-Proposed — Not Locked
AI structured-output validation	Pydantic	Paper-Proposed — Not Locked
Language model	DeepSeek-V4.1-Flash via API	Paper-Proposed — Not Locked
Mathematical visualization	JSXGraph, SVG, atau Canvas	Paper-Listed Candidates — Not Locked

Engineering boleh merekomendasikan alternatif setelah mengevaluasi feasibility, reliability, compatibility, dan cost. Perbedaan dari rancangan paper dicatat dalam Technical Deviation Register pada Section 05.

Pemilihan library atau framework alternatif tidak boleh mengubah approved product behavior.

C. AI Responsibilities & Safeguards

Area	Required consideration	Status
Input Context	AI menerima informasi yang relevan terhadap supported task dan concept.	Paper-Proposed / Engineering Consideration
Mathematical Reliability	Interpretasi atau feedback AI tidak boleh bertentangan dengan referensi matematika yang valid.	PRD-Derived boundary
Diagnostic Coverage	AI tidak boleh memperluas supported blockers secara independen.	PRD-Derived
Evidence Handling	Interpretasi harus mempertimbangkan ambiguitas dan counterevidence.	PRD-Derived
Output Validation	Output AI perlu diperiksa sebelum digunakan oleh proses berikutnya.	Paper-Proposed
Failure Handling	Kegagalan AI tidak boleh diubah menjadi kesimpulan diagnosis yang tampak valid.	Engineering Consideration
Decision Authority	Penanggung jawab finalisasi diagnosis memerlukan alignment.	OI-01 Open

D. Implementation Questions Left to Engineering

Detail yang belum ditentukan meliputi pilihan model AI final, inference strategy, prompt design, mathematical comparison approach, persistence implementation, output schema, dan API structure.

Engineering dapat mendesain alternatif yang memenuhi kewajiban PRD dan membawa proposal perubahan yang memengaruhi product behavior untuk ditinjau melalui governance.

4.4 Research References & External Datasets

A. Methodological Reference Registry

Reference	Contribution	ReThink usage status
Mislevy, Almond, dkk. — Evidence-Centered Design (2003)	Kerangka hubungan student claims, evidence, dan tasks.	Adapted methodological reference
de la Torre — Cognitively Based Multiple-Choice Options (2009)	Distractor dengan nilai diagnostik.	Question-design research reference

de la Torre — DINA Model (2009)	Cognitive diagnostic modeling dan atribut keterampilan.	Theoretical reference; not implemented
Tatsuoka — Rule Space (1983)	Pendekatan diagnosis berbasis pola respons.	Theoretical reference; not implemented
Bastani et al. (2025)	Risiko pembelajaran dari GenAI tanpa safeguards.	AI/pedagogical reference
Visualization, Worked Examples & Transfer Literature	Landasan pembelajaran melalui representasi, contoh, scaffolding, dan penerapan konsep.	Instructional methodology references

Referensi lengkap, DOI/URL, dan detail bibliografis harus dicocokkan dengan daftar pustaka paper serta sumber primer sebelum digunakan sebagai paket handoff final. Registry ini bukan klaim bahwa semua metode telah diimplementasikan.

B. External Dataset Registry

Dataset	Information referenced	Potential use	MVP status
Eedi Misconceptions Graph	Concept/misconception relationships	Eksplorasi concept–blocker knowledge	Deferred Research
Eedi MAP	Questions, student explanations, misconception labels	Contoh reasoning dan interpretive evaluation	Deferred Research
FoundationalASSIST	Student responses, interactions, skill mapping	Studi response patterns dan skill-related evidence	Deferred Research

Dataset eksternal tidak menjadi mandatory dependency untuk menjalankan MVP.

Operational diagnostic knowledge ReThink tetap memerlukan curated assets yang valid untuk Distributive Property.

C. Reference Adoption & Data Governance

Sebelum menggunakan dataset eksternal untuk training, evaluation, atau penyusunan curated assets, engineering perlu memeriksa akses, lisensi, cakupan konsep, kualitas label, representativitas, dan kesesuaian matematis terhadap supported MVP.

Referensi yang diadopsi tidak otomatis memperluas diagnostic coverage atau membuktikan ketepatan diagnosis ReThink.

Katalog sumber dan tautan bibliografis yang telah diverifikasi akan menjadi bagian dari handoff reference registry, tanpa menambah requirement implementasi baru.

Section 04 — Boundary Statement

Section ini menetapkan fondasi informasi dan referensi teknis yang diperlukan untuk memulai system design. Dokumen tidak menentukan ERD, database schema, JSON contract, prompt template, technology selection, atau AI decision algorithm final.

Semua implementasi tetap mengacu pada approved PRD serta approved Technical Blueprint Sections 01–03.

Implementation Readiness & Technical Governance

Section ini menjelaskan kesiapan implementasi, keputusan yang masih terbuka, persyaratan verifikasi, serta mekanisme pengambilan keputusan teknis selama pengembangan MVP ReThink.

Dokumen ini tidak menggantikan Engineering Backlog. Section 05 menetapkan hal-hal yang perlu diselesaikan, disiapkan, dan diperiksa, sedangkan pembagian tugas implementasi disusun dalam backlog terpisah.

5.1 Open Decisions & Specification Dependencies

A. Engineering Participation in Open Decisions

Tim Backend dan AI memiliki kewenangan untuk menganalisis, mengusulkan, dan melengkapi alternatif penyelesaian terhadap open product questions maupun pending operational specifications.

Rekomendasi tim teknis perlu mempertimbangkan:

- **PRD alignment:** kesesuaian dengan tujuan, kewajiban fungsional, dan batas MVP yang telah disetujui.
- **Product effectiveness:** kontribusi pendekatan terhadap reliabilitas diagnosis, kualitas pengalaman belajar, validitas matematika, dan ketercapaian recovery loop.
- **Technical feasibility:** kelayakan implementasi dalam batas sumber daya, waktu, dan kompleksitas.
- **Reliability & testability:** kemampuan pendekatan untuk menghasilkan perilaku konsisten dan dapat diverifikasi.
- **Engineering trade-offs:** keuntungan, keterbatasan, risiko, dan konsekuensi dari alternatif yang diajukan.

Tim teknis tidak harus menerima seluruh pendekatan awal dari paper tanpa evaluasi. Jika terdapat alternatif yang lebih layak atau lebih efektif, tim dapat mengajukannya dengan alasan yang dapat diperiksa.

Namun, rekomendasi teknis yang mengubah approved product behavior, status pembelajaran, diagnostic decision rules, atau MVP scope harus melalui product review sebelum dijadikan keputusan final.

B. Open Product Decision Register

ID	Open decision	Engineering contribution	Resolution authority

OI-01	AI Methodology Alignment	Usulkan pembagian peran LLM, mathematical evaluator, orchestration, evidence synthesis, dan decision safeguards. Sertakan trade-offs serta verification approach.	Product + AI/Math review
OI-02	Optional Review Learner-State Behavior	Evaluasi konsekuensi apabila Optional Review memberikan atau tidak memberikan akses ke validation dan learner-state update.	Product review
OI-03	All-Correct Diagnostic Outcome Mapping	Usulkan penanganan ketika seluruh respons diagnosis benar, termasuk konsekuensi terhadap hasil diagnosis dan navigasi pengguna.	Product review

Ketiga keputusan tetap berstatus **Open** sampai alternatif yang diajukan ditinjau dan disetujui.

C. Pending Operational Specification Register

ID	Pending specification	Expected technical contribution
OPS-01	Guided Practice Feedback Mechanics	Usulkan mekanisme pemeriksaan respons, feedback, dan penanganan kesalahan dalam batas pembelajaran linear.
OPS-02	Guided Learning Completion Criteria	Definisikan kandidat completion rules berdasarkan aktivitas yang diwajibkan dan bukti partisipasi siswa.
OPS-03	Validation Response Handling	Usulkan penanganan jawaban salah, kosong, ambigu, tidak valid, atau ekuivalen secara matematis.
OPS-04	Curated Mathematical Items & Rubrics	Siapkan rancangan mathematical assets, reference solutions, error patterns, dan rubrik untuk mathematical review.
OPS-05	Cross-Capability Transition & State Design	Rancang operational transitions, evidence continuity, serta aturan konsistensi state berdasarkan approved workflows.

OPS-01–OPS-05 tetap **Pending**. Tim engineering dapat menyiapkan spesifikasi usulan, tetapi spesifikasi yang memengaruhi product behavior atau evaluasi pembelajaran memerlukan review yang sesuai sebelum ditetapkan sebagai baseline.

D. Decision Proposal Format

Setiap open question yang membutuhkan penyelesaian dapat menggunakan format ringkas berikut.

Field	Expected content
Reference ID	OI/OPS atau area keputusan terkait
Problem / Decision Needed	Hal yang belum dapat dipastikan dari dokumen saat ini
Proposed Approach	Solusi yang direkomendasikan tim
Alternatives Considered	Alternatif utama yang dibandingkan
Rationale & Product Impact	Alasan pemilihan dan dampak terhadap efektivitas ReThink
PRD Alignment	Requirement yang dipenuhi atau berpotensi terdampak
Trade-offs & Risks	Risiko reliabilitas, kompleksitas, pengalaman pengguna, atau matematika
Verification Approach	Cara membuktikan bahwa solusi memenuhi kebutuhan
Decision Status	Proposed / Approved / Rejected / Deferred

Proposal tidak harus panjang. Penjelasan yang cukup untuk memahami keputusan, dampak, dan alasan teknis lebih penting daripada dokumentasi yang berlebihan.

5.2 Verification & Quality Preparation

A. Verification Objectives

Implementasi MVP perlu memiliki bukti bahwa setiap capability bekerja sesuai approved Product Requirements dan menghasilkan outcome yang benar pada skenario yang didukung.

Verification harus membedakan:

Verification area	Primary concern
Functional correctness	Komponen dan alur memenuhi functional requirements terkait.
Diagnostic reliability	Hasil diagnosis konsisten dengan evidence dan supported blocker scope.

Mathematical validity	Soal, solusi acuan, evaluasi respons, dan interpretasi matematis dapat diperiksa.
Learning flow integrity	Guided Learning mengikuti paket dan urutan yang disetujui.
Validation correctness	Retry, Transfer Problem, dan learner-state mapping mengikuti aturan yang berlaku.
Cross-capability consistency	Konteks, eligibility, outcome, dan Learning Map tidak saling bertentangan.
AI safeguards	Respons atau kegagalan AI tidak menghasilkan klaim diagnosis yang tidak didukung.

B. Required Verification Preparation

Preparation	Expected evidence	PRD reference
Functional requirement coverage	Keterkaitan implementasi dengan 39 approved FR	C1–C5, FR-X
Mathematical reference cases	Curated inputs, expected mathematical results, dan review matematika	OPS-04
Diagnostic test cases	Kasus Confirmed, Rejected, Uncertain, termasuk kondisi probing	C3
Guided Learning scenarios	Pemeriksaan urutan konten, partisipasi aktif, dan completion sesuai aturan	C4
Validation decision cases	Retry gagal, retry berhasil/transfer gagal, dan keduanya berhasil	C5
Integration scenarios	Pengujian perpindahan, kontinuitas konteks, dan representasi Learning Map	FR-X, V-X-01–08
AI reliability cases	Input ambigu, output tidak valid, unsupported claims, serta penanganan kegagalan	IP-01/IP-02, OI-01

Skenario pengujian harus menggunakan expected behavior yang berasal dari approved requirements. Kondisi yang masih bergantung pada OI/OPS tidak boleh dinyatakan telah memenuhi kriteria final sebelum dependensinya diselesaikan.

C. Readiness Principles

Sebuah capability dapat dinyatakan siap diverifikasi apabila perilaku yang relevan telah diimplementasikan, aset yang diperlukan tersedia, aturan keputusan telah disetujui, dan expected outcomes dapat diperiksa.

Keberhasilan demonstrasi end-to-end belum cukup untuk menyimpulkan bahwa seluruh functional requirements telah terpenuhi.

Demikian pula, ketepatan matematis pada satu soal tidak otomatis membuktikan reliabilitas diagnosis pada berbagai pola respons.

Penilaian keberhasilan MVP tetap mengikuti SC1–SC4 pada approved PRD, termasuk pemisahan functional completeness dan UX evaluation.

5.3 Technical Decision & Deviation Governance

A. Engineering Design Authority

Tim engineering memiliki keleluasaan memilih pendekatan implementasi untuk backend, AI runtime, mathematical evaluation, data persistence, frontend integration, serta deployment.

Keputusan tersebut dapat diambil berdasarkan feasibility, reliability, maintainability, kebutuhan produk, dan trade-offs selama memenuhi approved PRD.

Pilihan teknologi dalam paper menjadi referensi awal, bukan mandatory implementation commitment.

B. Paper-Proposed Technology Baseline

Technical area	Paper-proposed approach	Governance
Frontend	Flutter	Dapat diganti melalui engineering decision
Backend orchestration	FastAPI	Dapat diganti
Concept relationships	Neo4j	Dapat diganti selama relationship semantics terjaga
Learner data	PostgreSQL	Dapat diganti
Mathematical checking	SymPy	Dapat diganti dengan pendekatan yang dapat diverifikasi

Structured AI outputs	Pydantic	Dapat diganti dengan mekanisme validasi setara
LLM	DeepSeek-V4.1-Flash	Dapat dievaluasi berdasarkan kualitas, biaya, latency, dan reliability
Mathematical visualization	JSXGraph, SVG, Canvas	Opsi implementasi, bukan kewajiban

C. Technical Decision & Deviation Register

Setiap keputusan signifikan atau perubahan dari pendekatan yang disebutkan paper perlu dicatat.

Field	Description
Decision ID / Area	Identitas dan area keputusan
Original Reference	Pendekatan dari paper atau desain sebelumnya
Selected / Proposed Approach	Pendekatan baru yang dipilih atau direkomendasikan
Reason for Decision	Justifikasi teknis dan dampak produk
PRD Impact	Requirement atau behavioral constraint yang terdampak
Verification Impact	Pengujian dan evidence yang diperlukan
Approval / Decision Status	Proposed / Approved / Rejected / Deferred

Contoh: apabila engineering memilih relational database daripada dedicated graph database, tim harus menjelaskan bagaimana concept relationships dan supported capability boundaries tetap direpresentasikan.

Perubahan tersebut tidak otomatis memerlukan persetujuan Product jika tidak mengubah approved behavior. Sebaliknya, penggantian diagnostic decision policy atau learner-state mapping memerlukan product review meskipun dilakukan tanpa mengganti teknologi.

D. Governance & Escalation Rules

Change category	Handling

Implementation detail within PRD	Engineering dapat memutuskan dan mendokumentasikan.
Technology deviation from paper	Engineering mengevaluasi dan mencatat rationale serta dampaknya.
Pending operational specification	Engineering menyusun usulan; pihak terkait meninjau dan menyetujui sebelum dijadikan aturan final.
Open product decision	Tim teknis boleh merekomendasikan solusi, tetapi penyelesaian final melalui product review.
Change to approved requirements	Memerlukan controlled change review sebelum implementasi dianggap sesuai baseline baru.

E. Handoff Completion Principle

Technical Blueprint dapat digunakan untuk memulai engineering design meskipun sebagian OI/OPS masih terbuka.

Tim dapat melanjutkan pekerjaan independen, menyiapkan mathematical assets, membandingkan pendekatan AI, dan membuat desain teknis awal.

Namun, dependent decision logic tidak boleh dinyatakan final sebelum ketentuan yang relevan diselesaikan dan tercatat.

Engineering Backlog akan disusun secara terpisah untuk menerjemahkan approved requirements, workflow responsibilities, readiness dependencies, dan technical decisions menjadi actionable development tasks.

Section 05 tidak menetapkan bahwa MVP telah dibangun, diuji, atau berhasil mencapai learning outcomes.